

Topological Energy Condensate Theory (TECT): 14

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WHAT CAN NOW BE RIGOROUSLY CLOSED

We isolate the strongest statements that can be proved *without overclaiming* from the uploaded Math15 draft together with the previously established Math10–13 structure.

Setup

Consider the minimal complex-triplet Brazovskii wave equation

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi, \quad Z < 0, \quad Y > 0, \quad g > 0, \quad (1)$$

and a locked single-shell background

$$\Psi_0(x) = A e^{iG \cdot x} z_0, \quad |G| = q_*, \quad q_*^2 = -\frac{Z}{2Y}, \quad z_0^\dagger z_0 = 1. \quad (2)$$

Let the transverse fluctuation be

$$\delta \Psi_\perp(x, t) = A e^{iG \cdot x} \chi(x, t), \quad \chi(x, t) \in Q_{z_0} := (\mathbb{C} z_0)^\perp, \quad z_0^\dagger \chi = 0. \quad (3)$$

Theorem 1. Linearized quartic no-go

At linear order in the fluctuation, the quartic term $g(\Psi^\dagger \Psi) \Psi$ does not directly generate any enriched first-order operator of the form

$$(\partial_i \Psi_0)(-i \partial_j) \otimes s^a.$$

Expand

$$\Psi = \Psi_0 + \delta \Psi. \quad (4)$$

Then, to first order,

$$g(\Psi^\dagger \Psi) \Psi = g \left[(\Psi_0^\dagger \Psi_0) \delta \Psi + (\Psi_0^\dagger \delta \Psi) \Psi_0 + (\delta \Psi^\dagger \Psi_0) \Psi_0 \right] + O(\delta \Psi^2). \quad (5)$$

Now decompose

$$\delta \Psi = \delta \Psi_\parallel + \delta \Psi_\perp, \quad \delta \Psi_\parallel = \delta \rho z_0, \quad \delta \Psi_\perp \in Q_{z_0}. \quad (6)$$

Since

$$z_0^\dagger \delta \Psi_\perp = 0, \quad (7)$$

the mixed terms $(\Psi_0^\dagger \delta \Psi) \Psi_0$ and $(\delta \Psi^\dagger \Psi_0) \Psi_0$ depend only on the longitudinal amplitude fluctuation $\delta \rho$, while $(\Psi_0^\dagger \Psi_0) \delta \Psi = A^2 \delta \Psi$ is a local mass-type term.

Therefore the linearized quartic contribution contains only zeroth-order operators in derivatives acting on $\delta \Psi$, and in particular contains no term bilinear in

$$\partial_i \Psi_0 \quad \text{and} \quad \partial_j \chi.$$

Hence it does not directly produce an enriched first-order tensor N_{ij}^a .

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Theorem 2. Exact shell cancellation of the isotropic kinetic first-order term

For the minimal isotropic kinetic operator

$$Z\nabla^2 - Y\nabla^4,$$

the shell-projected first-order transport term in the envelope χ vanishes exactly at the Brazovskii shell $q_*^2 = -Z/(2Y)$.

Write

$$\delta\Psi_\perp = Ae^{iG\cdot x}\chi. \quad (8)$$

Introduce the shifted differential operator

$$D_G := \nabla + iG. \quad (9)$$

Then

$$\nabla(e^{iG\cdot x}\chi) = e^{iG\cdot x}D_G\chi, \quad \nabla^2(e^{iG\cdot x}\chi) = e^{iG\cdot x}D_G^2\chi, \quad (10)$$

with

$$D_G^2 = \nabla^2 + 2iG \cdot \nabla - |G|^2 = \nabla^2 + 2iG \cdot \nabla - q_*^2. \quad (11)$$

Hence

$$(Z\nabla^2 - Y\nabla^4)\delta\Psi_\perp = Ae^{iG\cdot x}\left[ZD_G^2 - Y(D_G^2)^2\right]\chi. \quad (12)$$

Retain only terms up to first order in the slow envelope derivative $\nabla\chi$. Since

$$D_G^2 = -q_*^2 + 2iG \cdot \nabla + O(\nabla^2), \quad (13)$$

we get

$$(D_G^2)^2 = (-q_*^2 + 2iG \cdot \nabla + O(\nabla^2))^2 \quad (14)$$

$$= q_*^4 - 4iq_*^2 G \cdot \nabla + O(\nabla^2). \quad (15)$$

Therefore

$$ZD_G^2 - Y(D_G^2)^2 = \left(-Zq_*^2 - Yq_*^4\right) + 2i\left(Z + 2Yq_*^2\right)G \cdot \nabla + O(\nabla^2). \quad (16)$$

Now impose the shell condition

$$q_*^2 = -\frac{Z}{2Y}. \quad (17)$$

Then

$$Z + 2Yq_*^2 = Z + 2Y\left(-\frac{Z}{2Y}\right) = 0. \quad (18)$$

Hence the entire first-order transport coefficient vanishes:

$$2i\left(Z + 2Yq_*^2\right)G \cdot \nabla = 0. \quad (19)$$

Thus the minimal isotropic kinetic operator produces no nonzero first-order shell transport term at the exact Brazovskii shell.

Theorem 3. Locked cross-shell quartic projection vanishes at zeroth order

In the exactly polarization-locked background, the zeroth-order projected cross-shell quartic term vanishes identically on Q_{z_0} .

Consider a second shell mode with internal direction u_{**} . The projected cross-shell quartic contribution has the form

$$\Pi_0 \left[(u_{**}^\dagger \chi) u_{**} \right]. \quad (20)$$

In the locked background,

$$u_{**} = z_0. \quad (21)$$

Therefore

$$\Pi_0 \left[(z_0^\dagger \chi) z_0 \right] = (z_0^\dagger \chi) \Pi_0 z_0 = 0, \quad (22)$$

because $\chi \in Q_{z_0}$ implies $z_0^\dagger \chi = 0$, and also $\Pi_0 z_0 = 0$.

Hence the locked zeroth-order cross-shell quartic projection is exactly zero.

Corollary. What is actually closed now

For the minimal locked isotropic complex-triplet Brazovskii equation, the following statement is rigorously established:

the enriched Pauli-sector shell tensor N_{ij}^a is *not yet derived* from the minimal locked single-mode calculation alone.

(23)

More precisely:

1. the linear quartic term gives no direct enriched first-order source;
2. the isotropic kinetic first-order shell term cancels exactly at q_* ;
3. the locked zeroth-order cross-shell quartic projection vanishes.

Therefore any genuine nonzero shell-enriched tensor must come from *additional structure*, such as:

1. a full 12-mode BCC gradient-expansion calculation beyond the locked single-mode truncation,
2. polarization unlocking,
3. or an explicit Class II mediator / enriched UV channel.

What remains honestly open

The remaining microscopic witness problem is therefore not solved by the present Math15 derivation. What remains open is the explicit production of the three shell witnesses

$$g_{\parallel}, \quad g_I, \quad g_J$$

(or equivalently $\lambda_{\parallel}, \alpha, \beta$) from the *actual* full microscopic TECT linearization.

Thus the corrected honest status is:

minimal locked isotropic PDE alone is insufficient,

(24)

whereas

the Class II-enriched route remains a mathematically consistent explicit mechanism

(25)

for generating $\delta N_{ij}^a \neq 0$ at shell-reduced level.

A. Summary

Corrected closure after Math15

Closed rigorously:

1. The linearized quartic term does not directly generate

$$(\partial_i \Psi_0)(-i\partial_j) \otimes s^a.$$

2. For the minimal isotropic Brazovskii operator,

$$Z\nabla^2 - Y\nabla^4,$$

the first-order shell transport coefficient is

$$2i(Z + 2Yq_*^2)G \cdot \nabla,$$

hence vanishes exactly at

$$q_*^2 = -\frac{Z}{2Y}.$$

3. In the exactly locked background, the zeroth-order cross-shell quartic projection onto

$$Q_{z_0}$$

vanishes identically.

Therefore:

the minimal locked isotropic PDE does not yet by itself close the N_{ij}^a witness problem.

What remains open is the explicit microscopic generation of

$$g_{\parallel}, \quad g_I, \quad g_J$$

from one of the following:

full 12-mode BCC Bloch linearization, polarization unlocking, or explicit Class II mediation.

Hence the honest current status is:

Dirac emergence remains theorem-level / shell-reduced closed,

but

full microscopic closure in complete TECT is still open.

NEXT RIGOROUS STEP IN THE TECT PROOF CHAIN

a. Principle. The next step is *not* to re-prove the reduced Weyl/Dirac theorem. That part is already closed once the shell witnesses

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

are nonzero in the required pattern. The actual remaining microscopic task is to prove that the *full microscopic TECT transport space* contains the required seed directions.

b. Microscopic transport space. Let

$$V_{\text{mic}}$$

denote the symmetry-allowed microscopic transport tensor space obtained by linearizing the full TECT dynamics around the relevant defect/BCC background.

Let the explicit shell-adapted seed subspace be

$$V_{\text{seed}} = \text{span}\left\{N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)}\right\}.$$

For a general seed combination

$$N = c_{\parallel} N_{\text{seed}}^{(\parallel)} + c_E N_{\text{seed}}^{(\perp, E)} + c_O N_{\text{seed}}^{(\perp, O)},$$

the reduced witnesses are

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}. \quad (26)$$

Assuming

$$I_{\theta} \neq 0, \quad I_{\theta'^3} \neq 0,$$

one immediately gets the exact nondegeneracy criterion

$$\boxed{c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0)} \iff \lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (27)$$

c. Immediate consequence. Therefore the reduced Weyl witness problem is already closed *inside the explicit seed ansatz space*. So the remaining task is not to prove that such seeds *could* work. The remaining task is to prove that the actual microscopic transport space contains them:

$$\boxed{V_{\text{seed}} \subset V_{\text{mic}} \quad \text{or at least} \quad V_{\text{mic}} \text{ contains one longitudinal and one transverse seed direction.}} \quad (28)$$

d. Carrier-audit completion criterion. Equivalently, let

$$T^{(\parallel)}, \quad T^{(\perp, E)}, \quad T^{(\perp, O)}$$

be the corresponding target covectors/witness channels. Then the carrier audit succeeds iff

$$\boxed{P_{V_{\text{mic}}} T^{(\parallel)} \neq 0 \quad \text{and} \quad (P_{V_{\text{mic}}} T^{(\perp, E)}, P_{V_{\text{mic}}} T^{(\perp, O)}) \neq (0, 0).} \quad (29)$$

e. Why the longitudinal channel is the true bottleneck. Suppose the transverse odd-gradient extraction has already been shown to survive microscopically. Then the transverse condition

$$(\alpha, \beta) \neq (0, 0)$$

is no longer the open part of the problem. The only remaining microscopic carrier question becomes the survival of the primitive longitudinal channel producing

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3} \neq 0.$$

Hence the sharp next theorem to prove is:

[Longitudinal primitive-channel survival theorem: target form] Let V_{mic} be the full microscopic transport tensor space obtained from the true TECT linearization around the physically relevant background. Assume:

1. the transverse carrier channel survives in V_{mic} ;
2. the microscopic transport sector contains one admissible primitive longitudinal direction whose reduced projection onto the seed covector $T^{(\parallel)}$ is nonzero.

Then

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),$$

and therefore the local shell-reduced Weyl symbol is nondegenerate. Consequently the local Dirac reduction theorem applies.

f. After this step. Once the carrier audit is closed, the next microscopic tasks are:

1. remote-label mismatch audit implying $V(0) = 0$,
2. remote-gap positivity audit giving $\Delta > 0$,
3. exact residual valley symmetry audit giving mass protection,
4. only afterwards the full derivative-anisotropy/FRG closure.

Thus the correct proof order is

$$\boxed{\text{carrier audit} \rightarrow V(0) = 0 \rightarrow \Delta > 0 \rightarrow \text{mass protection} \rightarrow M_Z/\text{FRG}.}$$

B. Summary

Correct next step in the TECT proof chain

The reduced Weyl/Dirac witness theorem is already closed inside the explicit Class II seed space:

$$\lambda_{\parallel} = \frac{1}{2}c_{\parallel}I_{\theta^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta},$$

so nondegeneracy holds iff

$$c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).$$

Therefore the true remaining microscopic task is not to re-prove the seed theorem, but to prove that the full microscopic TECT transport space contains the required seed directions:

$$V_{\text{seed}} \subset V_{\text{mic}} \quad \text{or at least one longitudinal and one transverse seed direction.}$$

Because the transverse carrier channel is already largely solved, the main bottleneck is the survival of the primitive longitudinal channel:

$$\theta'(\xi)^3 \sigma_3 \implies \lambda_{\parallel} \neq 0.$$

Hence the correct proof order is

$$\boxed{\text{longitudinal carrier audit} \rightarrow V(0) = 0 \rightarrow \Delta > 0 \rightarrow \text{mass protection} \rightarrow M_Z/\text{FRG}.}$$

LONGITUDINAL PRIMITIVE-CHANNEL THEOREM: EXACT SYMMETRY FORM AND NONZERO WITNESS CRITERION

a. Goal. We now close the part of the longitudinal problem that *can* be closed rigorously at the present stage:

1. identify the exact symmetry type of the longitudinal primitive channel;
2. prove that, once this channel survives with a nonzero microscopic coefficient, the reduced witness satisfies

$$\lambda_{\parallel} \neq 0.$$

This does *not* yet prove that the full complete microscopic TECT linearization actually generates the channel. It proves the sharp theorem-level statement:

if the longitudinal primitive carrier survives microscopically, then the longitudinal witness is automatically nonzero.

A. Residual singlet $\oplus$ doublet decomposition

For the explicit local defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

one has

$$\partial_\xi z_B(\xi) = \theta'(\xi) e_1(\xi).$$

Therefore the rank-two fiber splits as

$$Q_B = \mathbb{C}e_1 \oplus \mathbb{C}e_2,$$

and the traceless Hermitian sector decomposes under the residual axial $U(1)$ as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}.$$

At the same time, the worldvolume momentum side decomposes as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} p_{\parallel} \oplus \text{span}\{p_1, p_2\}.$$

Hence both sides carry the same residual

$$\text{singlet} \oplus \text{doublet}$$

pattern:

$$p_{\parallel} \leftrightarrow \sigma_3, \quad (p_1, p_2) \leftrightarrow (\sigma_1, \sigma_2).$$

b. Consequence. The longitudinal channel must transform as the singlet-to-singlet intertwiner. Because both source and target singlet spaces are one-dimensional,

$$\dim \text{Hom}_{U(1)_{\text{res}}}(\mathbb{R} p_{\parallel}, \mathbb{R} \sigma_3) = 1.$$

Therefore the symmetry-allowed longitudinal primitive channel is unique up to an overall scalar coefficient.

Any symmetry-allowed primitive longitudinal transport channel is proportional to $n^\mu \sigma_3$.

(30)

B. Canonical geometric realization of the singlet channel

Let

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B].$$

Inside the explicit seed representative one has

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3. \quad (31)$$

Hence the canonical odd first-order longitudinal transport seed is

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (32)$$

This proves that the explicit primitive longitudinal seed is exactly the unique residual-singlet channel picked out by the defect geometry.

C. General microscopic longitudinal sector and projection criterion

Let V_{mic} be the true microscopic first-order transport space obtained from the full TECT linearization around the relevant defect background.

By the residual singlet/doublet decomposition, any microscopic transport field splits uniquely as

$$V^\mu(\xi) = V_{\text{sing}}^\mu(\xi) + V_{\text{dbl}}^\mu(\xi),$$

where

$$V_{\text{sing}}^\mu(\xi) \in \text{span}\{n^\mu \sigma_3\}, \quad V_{\text{dbl}}^\mu(\xi) \in \text{span}\{E^\mu{}_a \sigma_a, J^\mu{}_a \sigma_a\}.$$

Because the longitudinal intertwiner space is one-dimensional, every admissible microscopic longitudinal primitive term has the form

$$V_{\text{long}}^\mu(\xi) = c_{\parallel} \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu + V_{\text{null}}^\mu(\xi), \quad (33)$$

where V_{null}^μ is any component orthogonal to the longitudinal target covector under the reduced projection.

Using (31),

$$V_{\text{long}}^\mu(\xi) = \frac{1}{2} c_{\parallel} \theta'(\xi)^3 n^\mu \sigma_3 + V_{\text{null}}^\mu(\xi). \quad (34)$$

D. Reduced witness evaluation

Inside the explicit seed space, the reduced witness coefficients are

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}.$$

For the explicit longitudinal seed one has

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3}, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \neq 0. \quad (35)$$

Therefore, for the general microscopic longitudinal component (33),

$$\boxed{\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}.} \quad (36)$$

Since $I_{\theta'^3} \neq 0$, it follows immediately that

$$\boxed{c_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (37)$$

E. Sharp theorem

[Longitudinal primitive-channel theorem] Consider the explicit CP^2 defect representative with residual axial splitting

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} p_{\parallel} \oplus \mathbb{R}_{\perp}^2, \quad \mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}.$$

Then:

1. the symmetry-allowed primitive longitudinal channel is unique up to an overall scalar and is represented by the singlet intertwiner

$$n^\mu \sigma_3;$$

2. geometrically, the canonical seed representative is

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3;$$

3. if the true microscopic TECT transport space contains this primitive channel with coefficient $c_{\parallel} \neq 0$, then the reduced longitudinal witness is automatically nonzero:

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3} \neq 0.$$

Part (1) follows from the residual $U(1)$ singlet/doublet decomposition and the one-dimensionality of the singlet intertwiner space. Part (2) follows from the explicit identity

$$(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

Part (3) follows by substitution into the reduced witness formula together with

$$I_{\theta'^3} \neq 0.$$

F. Honest closure statement

Thus the longitudinal frontier is no longer ambiguous. It is reduced to exactly one finite microscopic coefficient test:

$$\boxed{P_{V_{\text{mic}}} T^{(\parallel)} \neq 0 \iff c_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (38)$$

What remains open is only the first implication:

does the complete microscopic TECT linearization actually produce $c_{\parallel} \neq 0$?

The rest of the longitudinal witness chain is now closed.

C. Summary

Closed part of the longitudinal problem

Residual axial symmetry gives

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} p_{\parallel} \oplus \mathbb{R}_{\perp}^2, \quad \mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}.$$

Hence the primitive longitudinal channel is the unique singlet-to-singlet intertwiner:

$$\boxed{\text{longitudinal primitive channel} \propto n^{\mu} \sigma_3.}$$

In the explicit defect representative,

$$(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3,$$

so the canonical longitudinal seed is

$$\boxed{V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3.}$$

Its reduced witness is

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0.$$

Therefore:

$$\boxed{c_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0.}$$

So the longitudinal frontier is reduced to one finite microscopic test:

$$\boxed{\text{Does the complete microscopic TECT transport space actually generate } c_{\parallel} \neq 0?}$$

Everything after that implication is now closed.

I. LONGITUDINAL CARRIER CLASSIFICATION IN THE TECT DIRAC PROGRAM

a. Goal. The remaining carrier frontier is no longer the full witness problem. The transverse carrier has already been fixed explicitly, while the only unresolved part is the microscopic survival of the primitive longitudinal channel:

$$\lambda_{\parallel} \neq 0.$$

Our task is therefore to classify which microscopic operator classes can, and which cannot, generate the required longitudinal carrier.

b. Target longitudinal structure. The shell-reduced local symbol has the form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

Hence the longitudinal sector requires a singlet-to-singlet carrier between

$$p_{\parallel} \quad \text{and} \quad \sigma_3.$$

In the explicit defect representative, the canonical primitive internal channel is

$$(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

Therefore the actual longitudinal primitive seed must be of the form

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3.$$

Its witness is

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0.$$

So any microscopic class can close the longitudinal sector only if it can generate a nonzero coefficient

$$c_{\parallel}$$

in front of this primitive odd singlet channel.

II. NECESSARY STRUCTURAL CONDITIONS

a. Condition 1: an (i, a) -carrier is mandatory. The shell-enriched target coupling has schematic form

$$\delta L_{\text{enr}} = N_{ij}^a (\partial_i \phi_0) (-i \partial_j) \otimes s^a.$$

Therefore any UV/microscopic source able to induce the longitudinal witness must simultaneously carry:

1. one spatial derivative index i or j ,
2. one internal shell-relevant label a ,
3. one derivative acting on the test/low sector,
4. one shell/background derivative factor.

Equivalently, it must contain an (i, a) -type carrier.

b. Condition 2: the longitudinal singlet must be odd. A naive singlet built only from

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} \propto \theta'(\xi)^2 \sigma_3$$

is insufficient. Indeed, the corresponding overlap would be

$$\int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^2,$$

and because

$$f_+ \text{ is even, } \quad f_- \text{ is odd, } \quad \theta'(\xi)^2 \text{ is even,}$$

the integrand is odd, hence

$$\int d\xi f_+ f_- \theta'^2 = 0.$$

Therefore the longitudinal carrier must include one additional odd prefactor, giving precisely

$$\theta'(\xi)^3 \sigma_3.$$

III. EXCLUSION THEOREM FOR INSUFFICIENT CLASSES

[Structural exclusion of non-carriers] The following microscopic correction types cannot generate the required longitudinal primitive witness:

1. pure scalar potential corrections,
2. pure projector derivative corrections,
3. amplitude-only derivative corrections,
4. the primitive projector channel $3P - \mathbf{1}_3$ by itself,
5. the naive even singlet channel proportional only to $\theta'(\xi)^2 \sigma_3$.

We check each case.

a. (i) Pure scalar potentials. Terms such as

$$(\Psi^\dagger \Psi)^3, \quad (\Psi^\dagger \Psi)(\partial_i P \partial_i P), \quad ((\partial_i P \partial_i P))^2$$

carry no shell-relevant internal tangent label a , or produce no linear shell-projector mixing around the locked background. Hence they do not generate an (i, a) -carrier.

b. (ii) Pure projector derivative terms. Terms built only from P and its derivatives remain internal/projector-sector terms. They do not couple linearly to the shell/test derivative sector, so they cannot induce the required first-order enriched mixed symbol.

c. (iii) Amplitude-only derivative terms. If a term depends only on ρ and its derivatives, it has no internal Pauli/triplet label and therefore cannot produce the traceless shell-relevant σ_3 carrier.

d. (iv) Primitive projector tensor $3P - \mathbf{1}_3$. On the rank-two fiber Q_B , this reduces to the identity and is therefore too coarse to generate the shell Pauli triplet. In particular, by itself it does not isolate the required primitive longitudinal witness as a nontrivial traceless channel.

e. (v) Naive quadratic commutator singlet. The commutator-square channel gives

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

But without an extra odd factor $\theta'(\xi)$, its low-mode overlap vanishes by parity:

$$\int d\xi f_+ f_- \theta'^2 = 0.$$

So the naive even singlet does not contribute to $\lambda_{\parallel}$.

Thus none of these classes can close the longitudinal carrier problem.

IV. ADMISSIBLE MICROSCOPIC CLASSES

We now classify the three minimal viable local classes.

Class I: direct shell–projector derivative coupling

A representative form is

$$L_{\text{UV}}^{(I)} = \Lambda_{ij}^a (\partial_i \Psi^\dagger) T_a \Psi (T_a \partial_j P) + \text{h.c.}$$

This class can work in principle because it explicitly inserts the needed (i, a) -carrier tensor. Hence Class I is *admissible*. However, it is less explanatory microscopically because the carrier tensor

$$\Lambda_{ij}^a$$

is inserted directly by hand rather than generated dynamically.

Class II: heavy auxiliary internal vector mediator

Introduce an auxiliary mediator

$$X_i^a$$

carrying simultaneously a spatial index i and an internal index a . This is precisely the minimal type of field able to couple the shell-derivative structure to the internal tangent/projector sector and then, after elimination, induce the required first-order enriched tensor.

Inside the seed decomposition, Class II naturally splits into:

$$X_i^3 \leftrightarrow \text{longitudinal singlet seed,}$$

$$X_i^1 \leftrightarrow \text{transverse-even seed,} \quad X_i^2 \leftrightarrow \text{transverse-odd seed.}$$

Hence the explicit seed basis

$$V_{\text{seed}} = \text{span}\{N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)}\}$$

is exactly the local realization of the Class II carrier decomposition.

Therefore Class II is not merely allowed. It is the minimal UV carrier structure matching the final reduced witness problem.

Class III: connection/gauge-like mediator

A connection-like mediator A_i^a is also structurally plausible, since it carries the same index type (i, a) . Therefore it is not excluded by symmetry.

However, in the present derivation chain it is less explicit than Class II. So it remains a secondary candidate rather than the current closure route.

V. PRIORITY THEOREM FOR THE LONGITUDINAL CARRIER PROBLEM

[Current priority ordering] For the present TECT longitudinal carrier problem, the viable microscopic classes are ordered by explanatory strength as

$$\boxed{\text{Class II} > \text{Class I} > \text{Class III.}}$$

Class II generates the required (i, a) -carrier dynamically through the auxiliary mediator and matches exactly the explicit shell-level seed decomposition. Class I can also realize the carrier, but only by inserting the carrier tensor directly at the microscopic level. Class III is structurally admissible but currently less explicit in the established derivation chain.

Therefore Class II is the strongest current candidate for closing the longitudinal carrier sector.

VI. SHARP HONEST CONCLUSION

The longitudinal primitive witness problem is now reduced to the following finite microscopic yes/no test:

Does the true complete TECT microscopic transport space contain a nonzero Class II-type longitudinal seed?

More explicitly, it is sufficient to find

$$N_{\parallel} \in V_{\text{micro}}$$

such that

$$\langle T^{(\parallel)}, N_{\parallel} \rangle \neq 0.$$

Equivalently, one needs

$$c_{\parallel} \neq 0.$$

Once this happens,

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3} \neq 0,$$

because

$$I_{\theta'^3} \neq 0.$$

Thus the current theorem-level closure is:

all insufficient carrier classes are excluded, and Class II is the minimal natural surviving candidate.

What remains open is not the reduced witness logic, but only the final microscopic realization question:

does complete TECT actually realize this nontrivial Class II longitudinal sector?

VII. FINAL SUMMARY OF THE LONGITUDINAL CARRIER CLASSIFICATION STEP

a. What is now closed. The longitudinal witness requires the primitive odd singlet channel

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3.$$

Hence

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0.$$

Therefore

$$c_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

b. What is excluded. The following cannot generate the required longitudinal carrier:

pure scalar corrections, pure projector corrections, amplitude-only derivative corrections,

because they do not provide the required (i, a) -carrier. Also,

$$3P - \mathbf{1}_3$$

is too coarse on Q_B , and the naive even singlet

$$\theta'(\xi)^2 \sigma_3$$

fails by parity:

$$\int d\xi f_+ f_- \theta'^2 = 0.$$

c. What remains viable. Among the minimal viable local microscopic classes:

$$\boxed{\text{Class II} > \text{Class I} > \text{Class III.}}$$

Class II is the minimal natural carrier because it dynamically realizes the same three seed directions:

$$X_i^3 \leftrightarrow \text{longitudinal seed}, \quad X_i^1, X_i^2 \leftrightarrow \text{transverse seeds.}$$

d. Exact current frontier. The remaining longitudinal question is reduced to one finite microscopic test:

$$\boxed{\exists N_{\parallel} \in V_{\text{micro}} \text{ such that } \langle T^{(\parallel)}, N_{\parallel} \rangle \neq 0.}$$

Equivalently,

$$\boxed{c_{\parallel} \neq 0.}$$

Once this is true, the longitudinal witness is automatically nonzero:

$$\lambda_{\parallel} \neq 0.$$

e. Honest status. So the current proof chain has established:

$$\boxed{\text{insufficient classes are excluded, and Class II is the minimal surviving longitudinal carrier candidate.}}$$

What is still open is only:

$$\boxed{\text{whether the complete microscopic TECT dynamics actually realizes that nonzero Class II longitudinal seed.}}$$

VIII. GENERIC NONVANISHING OF THE LONGITUDINAL CLASS II CARRIER

a. Goal. We now prove the strongest honest statement that can be closed at this stage: once the actual microscopic TECT transport space contains one admissible longitudinal Class II seed, the longitudinal witness is generically nonzero. This does *not* yet prove that complete microscopic TECT realizes the Class II sector. It proves that, *if such a sector is realized*, then the surviving longitudinal coefficient is not a fine-tuned accident.

IX. AMBIENT SETUP AND THE LONGITUDINAL FUNCTIONAL

Let

$$W = \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}$$

be the ambient enriched tensor space, and let

$$V_{\text{micro}} \subset W$$

be the actual symmetry-allowed microscopic transport tensor space extracted from the full TECT linearization around the relevant defect background.

Let

$$T^{(\parallel)} \in W^*$$

be the longitudinal witness covector. Define the longitudinal linear functional

$$L_{\parallel}(N) := \langle T^{(\parallel)}, N \rangle, \quad N \in V_{\text{micro}}.$$

By construction,

$$L_{\parallel}(N) = 0 \iff \text{the reduced projection of } N \text{ has zero longitudinal component.}$$

Inside the explicit shell-reduced seed basis,

$$N = c_{\parallel} B_{\parallel} + c_I B_I + c_J B_J,$$

one has

$$L_{\parallel}(N) = \mu_{\parallel} c_{\parallel}, \quad \mu_{\parallel} \neq 0,$$

and therefore

$$g_c \propto c_{\parallel}.$$

Equivalently,

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0.$$

Hence the longitudinal problem is exactly the problem of whether

$$L_{\parallel}|_{V_{\text{micro}}}$$

is identically zero or not.

X. SEED EXISTENCE IMPLIES GENERIC NONVANISHING

[Longitudinal seed criterion] Assume that the actual microscopic transport space V_{micro} contains one admissible longitudinal seed

$$N_{\text{seed}}^{(\parallel)} \in V_{\text{micro}}$$

such that

$$L_{\parallel}(N_{\text{seed}}^{(\parallel)}) \neq 0.$$

Then:

1. $L_{\parallel}|_{V_{\text{micro}}} \not\equiv 0$;
2. the bad set

$$Z_{\parallel} := \{N \in V_{\text{micro}} : L_{\parallel}(N) = 0\}$$

is a proper hyperplane in V_{micro} ;

3. its complement

$$U_{\parallel} := V_{\text{micro}} \setminus Z_{\parallel}$$

is open, dense, and full measure;

4. therefore the longitudinal witness is generically nonzero:

$$L_{\parallel}(N) \neq 0 \quad \text{for generic } N \in V_{\text{micro}}.$$

Since $L_{\parallel}$ is linear, its zero set is either the entire space or a hyperplane. Because there exists $N_{\text{seed}}^{(\parallel)} \in V_{\text{micro}}$ with

$$L_{\parallel}(N_{\text{seed}}^{(\parallel)}) \neq 0,$$

the functional $L_{\parallel}|_{V_{\text{micro}}}$ is not identically zero. Therefore its zero locus

$$Z_{\parallel} = \ker(L_{\parallel}|_{V_{\text{micro}}})$$

is a proper linear hyperplane in V_{micro} .

A proper hyperplane is closed, has empty interior relative to V_{micro} , and Lebesgue measure zero in any finite-dimensional affine coordinate system. Hence its complement is open, dense, and full measure. This proves the claim.

a. Immediate corollary. Because

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0,$$

the same theorem implies:

if one admissible longitudinal seed exists, then $\lambda_{\parallel} \neq 0$ generically.

XI. CANONICAL MINIMAL CLASS II BASIS

Inside the canonical Class II parametrization, choose

$$m^a(P_0) = m \delta^{a3},$$

together with

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j,$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}.$$

Then the induced shell-level coefficients satisfy

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

In this basis, the witness functionals are diagonal:

$$L_{\parallel}(N) = \mu_{\parallel} c_{\parallel}, \quad L_I(N) = \mu_I c_I, \quad L_J(N) = \mu_J c_J,$$

with

$$\mu_{\parallel} \neq 0, \quad \mu_I \neq 0, \quad \mu_J \neq 0.$$

Therefore the exact longitudinal completion criterion is simply

$$c_{\parallel} \neq 0.$$

XII. ONE-PARAMETER CANONICAL FAMILY

For the smallest concrete Class II realization, define

$$\mu := \frac{\lambda g_X}{M_X^2} m.$$

Then the induced shifts are

$$\delta A = -\mu c_{\parallel}, \quad \delta B = -\mu c_I, \quad \delta C = -\mu c_J,$$

and the effective coefficients are

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J.$$

The shell Weyl determinant becomes

$$\det M = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2).$$

Assume the full-support condition

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0).$$

Then:

1. $A_{\text{eff}} = 0$ is one linear equation in μ , hence yields at most one isolated bad value;
2. $B_{\text{eff}} = 0$ and $C_{\text{eff}} = 0$ simultaneously is at worst a finite condition in μ , hence again gives at most finitely many isolated bad values.

Therefore the bad set in the one-parameter μ -space is finite, and the good Weyl region is open, dense, and full measure.

[Canonical one-parameter genericity] Assume

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0).$$

Then in the canonical one-parameter Class II family, the set

$$U_{\text{Weyl}} = \{\mu : A_{\text{eff}} \neq 0, (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0)\}$$

is the complement of at most finitely many isolated values of μ . In particular, it is open, dense, and full measure in μ -space.

Directly from the affine formulas for $A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}$. The equation $A_{\text{eff}} = 0$ has at most one solution. The simultaneous vanishing of $(B_{\text{eff}}, C_{\text{eff}})$ gives at most finitely many isolated solutions in one real parameter. Hence the bad set is finite.

XIII. PURELY INDUCED REGIME

A particularly important special case is

$$A_0 = B_0 = C_0 = 0.$$

Then

$$A_{\text{eff}} = -\mu c_{\parallel}, \quad B_{\text{eff}} = -\mu c_I, \quad C_{\text{eff}} = -\mu c_J.$$

Therefore, if

$$\mu \neq 0, \quad c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0),$$

one immediately gets

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0), \quad \det M \neq 0.$$

So in the purely induced regime, once the three Class II seed directions are present, any nonzero μ already gives a Weyl-good local symbol.

XIV. SHARP HONEST CONCLUSION

We have now closed the following theorem-level statement:

If the actual microscopic TECT transport space contains one admissible longitudinal Class II seed,

(39)

then the longitudinal witness is generically nonzero, and its bad set is a proper hyperplane.

(40)

Moreover, in the canonical one-parameter Class II family:

the bad parameter set is at most finite.

(41)

What remains open is only the embedding question:

does complete microscopic TECT actually realize this Class II longitudinal seed in the physical defect background?

That is the only place where the present proof chain is still conditional.

XV. FINAL SUMMARY OF THE GENERIC LONGITUDINAL CLASS II STEP

a. What is now rigorously closed. Let

$$L_{\parallel}(N) = \langle T^{(\parallel)}, N \rangle$$

be the longitudinal witness functional on the actual microscopic transport space

$$V_{\text{micro}}.$$

If there exists one admissible longitudinal seed

$$N_{\text{seed}}^{(\parallel)} \in V_{\text{micro}}$$

such that

$$L_{\parallel}(N_{\text{seed}}^{(\parallel)}) \neq 0,$$

then

$$L_{\parallel}|_{V_{\text{micro}}} \not\equiv 0.$$

Hence the bad set

$$Z_{\parallel} = \{N \in V_{\text{micro}} : L_{\parallel}(N) = 0\}$$

is a proper hyperplane, and therefore its complement is open, dense, and full measure.

b. Equivalent witness statement. Since

$$\lambda_{\parallel} = \frac{1}{2}c_{\parallel}I_{\theta^3}, \quad I_{\theta^3} \neq 0,$$

this is equivalent to:

$\text{one admissible longitudinal seed} \implies \lambda_{\parallel} \neq 0 \text{ generically.}$

c. Canonical Class II basis. In the canonical Class II coordinates,

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J,$$

so the exact longitudinal condition is simply

$c_{\parallel} \neq 0.$

d. One-parameter canonical family. For the smallest Class II realization with

$$\mu = \frac{\lambda g_X}{M_X^2} m,$$

the effective coefficients are

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J.$$

Thus, under

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0),$$

the bad set in μ -space is at most finite. Hence the Weyl-good set is open, dense, and full measure.

e. Honest current frontier. So the proof chain has now reduced the longitudinal issue to a single finite embedding question:

$\text{Does complete microscopic TECT actually contain one admissible longitudinal Class II seed?}$

Everything after that implication is now closed.

XVI. WHY THE LONGITUDINAL SEED $B_{\parallel}$ IS NOT SYMMETRY-FORBIDDEN

a. Goal. The next rigorous step is to separate two logically distinct statements:

$$\boxed{\text{(A) } B_{\parallel} \text{ is not forbidden by the residual shell symmetry;}} \quad (42)$$

$$\boxed{\text{(B) the actual complete microscopic TECT space } V_{\text{TECT}} \text{ really contains } B_{\parallel}.} \quad (43)$$

At the present stage, statement (A) can be proved. Statement (B) still requires the finite microscopic carrier audit.

XVII. RESIDUAL SYMMETRY AND THE UNIQUE LONGITUDINAL INTERTWINER

Let the shell direction be

$$\hat{n} = \frac{G_*}{|G_*|},$$

and choose an orthonormal frame

$$\{e_1, e_2, \hat{n}\}.$$

Under the residual shell symmetry

$$H_{\text{sh}} \cong SO(2)_{\perp},$$

the worldvolume momentum sector decomposes as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} p_{\parallel} \oplus \text{span}\{p_1, p_2\},$$

while the traceless internal fiber sector decomposes as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}.$$

Thus both sides carry the same

$$\text{singlet} \oplus \text{doublet}$$

splitting. The most general local symmetry-compatible reduced symbol is therefore

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

This is exactly the natural local S^2 -branch symbol. In particular, the longitudinal channel is the singlet-to-singlet intertwiner

$$p_{\parallel} \longleftrightarrow \sigma_3,$$

and this intertwiner is one-dimensional. Therefore the longitudinal primitive tensor is unique up to scale.

Hence:

$$\boxed{\text{the residual symmetry does not eliminate the longitudinal channel;}}$$

it selects it uniquely.

XVIII. MINIMAL ALLOWED TENSOR SPACE AND EXPLICIT NON-FORBIDDANCE

Inside the minimal shell-enriched projector ansatz, the allowed shell-relevant tensor space is exactly

$$V_{\min} = \text{span}\{B_{\parallel}, B_I, B_J\}.$$

Equivalently, in component form the explicit invariant basis is

$$N_{ij}^3 = g_{\parallel} \hat{n}_i \hat{n}_j,$$

$$N_{ij}^1 = g_I \hat{n}_i e_{1j} - g_J \hat{n}_i e_{2j},$$

$$N_{ij}^2 = g_I \hat{n}_i e_{2j} + g_J \hat{n}_i e_{1j}.$$

The associated exact witness functionals satisfy

$$W_{\parallel}[N] = g_{\parallel}, \quad W_I[N] = g_I, \quad W_J[N] = g_J.$$

Therefore, at the level of the minimal shell-allowed symmetry class,

$$\boxed{P_V T^{(\parallel)} \neq 0, \quad P_V T^{(I)} \neq 0, \quad P_V T^{(J)} \neq 0.}$$

So none of the three reduced shell channels is symmetry-forbidden.

In particular,

$$\boxed{B_{\parallel} \text{ is an allowed invariant tensor direction.}}$$

XIX. EXACT DICHOTOMY: FORBIDDEN VS. GENERICALLY NONVANISHING

Let

$$W = \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}$$

be the ambient tensor space, and let

$$V_{\text{TECT}} \subset W$$

be the actual symmetry-allowed microscopic transport space extracted from the complete TECT linearization around the chosen defect background.

For the longitudinal channel define the target covector

$$T_{ija}^{(\parallel)} = e_{\parallel i} G_{*j} u_a^{(\parallel)},$$

and the linear witness functional

$$L_{\parallel}(N) := \langle T^{(\parallel)}, N \rangle.$$

Then the exact finite-dimensional dichotomy is

$$L_{\parallel}|_{V_{\text{TECT}}} \equiv 0 \quad \Longleftrightarrow \quad P_V T^{(\parallel)} = 0.$$

Hence exactly one of the following holds:

1.

$$P_V T^{(\parallel)} = 0,$$

so the longitudinal channel is symmetry-forbidden in the actual microscopic space;

2.

$$P_V T^{(\parallel)} \neq 0,$$

so the longitudinal channel is generically nonvanishing on an open dense full-measure subset of V_{TECT} .

Thus the real remaining question is finite:

compute $P_V T^{(\parallel)}$ or find one admissible seed with nonzero overlap.

XX. MINIMAL SUFFICIENT CONDITION FOR ACTUAL SURVIVAL

A practical sufficient condition is the existence of one explicit admissible microscopic tensor

$$N_{\text{seed}}^{(\parallel)} \in V_{\text{TECT}}$$

such that

$$\langle T^{(\parallel)}, N_{\text{seed}}^{(\parallel)} \rangle \neq 0.$$

Then automatically

$$P_V T^{(\parallel)} \neq 0,$$

hence the channel survives generically.

Inside the minimal Class II parametrization, the canonical longitudinal seed is represented by

$$m^a(P_0) = m \delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j,$$

which gives

$$g_c \propto c_{\parallel}.$$

Therefore, once such a seed belongs to the actual microscopic space,

$$c_{\parallel} \neq 0 \implies g_c \neq 0$$

and hence the longitudinal witness survives generically.

XXI. WHY VANISHING WOULD REQUIRE MORE THAN RESIDUAL SYMMETRY

The above proves that residual shell symmetry alone does not force

$$P_V T^{(\parallel)} = 0.$$

So if the longitudinal channel vanishes in the actual microscopic realization, the reason cannot be the basic

$$SO(2)_{\perp}$$

residual shell symmetry by itself.

Instead, vanishing would require at least one of the following:

1. an additional microscopic symmetry not contained in the minimal shell residual symmetry;
2. a dynamical cancellation among the actual basis tensors of V_{TECT} ;
3. the absence of any genuine (i, a) -carrier sector in the full microscopic linearization.

This is structurally analogous to the mirror-lifting scalar-overlap problem, where the established stabilizer symmetry by itself does not force vanishing, and any vanishing requires extra dynamical or involutive cancellation structure.

XXII. SHARP HONEST CONCLUSION

We can now state the strongest honest theorem of this step:

[Non-forbiddance of the longitudinal shell channel] At the shell-reduced symmetry level, the longitudinal channel $B_{\parallel}$ is not symmetry-forbidden. More precisely, in the minimal shell-allowed tensor space

$$V_{\text{min}} = \text{span}\{B_{\parallel}, B_I, B_J\},$$

one has

$$P_V T^{(\parallel)} \neq 0.$$

Therefore residual shell symmetry selects $B_{\parallel}$ as the unique singlet longitudinal intertwiner rather than excluding it.

If the actual complete microscopic TECT transport space V_{TECT} contains one admissible longitudinal seed with nonzero overlap against $T^{(\parallel)}$, then the longitudinal witness is generically nonzero.

a. What remains open. What is still not automatic is the actual embedding statement

$$B_{\parallel} \in V_{\text{TECT}} \quad \text{or at least} \quad P_V T^{(\parallel)} \neq 0$$

for the full complete microscopic TECT linearization. That remains a finite overlap / basis audit.

XXIII. FINAL SUMMARY OF THE NON-FORBIDDANCE STEP

a. What is now closed. At the shell-reduced symmetry level, the residual symmetry

$$H_{\text{sh}} \cong SO(2)_{\perp}$$

does not forbid the longitudinal channel. Instead, it selects a unique singlet-to-singlet intertwiner

$$p_{\parallel} \longleftrightarrow \sigma_3.$$

In the minimal shell-allowed tensor space,

$$V_{\text{min}} = \text{span}\{B_{\parallel}, B_I, B_J\},$$

all three channels survive:

$$P_V T^{(\parallel)} \neq 0, \quad P_V T^{(I)} \neq 0, \quad P_V T^{(J)} \neq 0.$$

Hence

$$\boxed{B_{\parallel} \text{ is not symmetry-forbidden.}}$$

b. Exact logical meaning. For the actual complete microscopic TECT transport space V_{TECT} ,

$$L_{\parallel}|_{V_{\text{TECT}}} \equiv 0 \iff P_V T^{(\parallel)} = 0.$$

So the real remaining question is finite-dimensional:

$$\boxed{\text{compute } P_V T^{(\parallel)} \text{ or exhibit one admissible seed with nonzero overlap.}}$$

c. Minimal sufficient condition for actual survival. If there exists

$$N_{\text{seed}}^{(\parallel)} \in V_{\text{TECT}}$$

such that

$$\langle T^{(\parallel)}, N_{\text{seed}}^{(\parallel)} \rangle \neq 0,$$

then automatically

$$P_V T^{(\parallel)} \neq 0,$$

so the longitudinal witness is generically nonzero.

d. Honest current frontier. Thus the present proof chain has established:

$$\boxed{\text{non-forbiddance is closed,}}$$

but still leaves open:

$$\boxed{\text{actual embedding of the longitudinal Class II seed into the complete microscopic TECT space.}}$$

e. Interpretation. Therefore any failure of the longitudinal channel would have to come from

additional microscopic symmetry, or dynamical cancellation, or absence of a genuine (i, a) -carrier sector, not from the basic residual shell symmetry itself.

XXIV. FINITE BASIS-LEVEL AUDIT THEOREM FOR THE ACTUAL MICROSCOPIC TECT SPACE

a. Goal. We now reduce the remaining microscopic carrier problem to an explicitly computable finite-dimensional basis audit. The objective is not yet to prove that the complete TECT linearization already contains the required seed space. Rather, we prove that once a basis of the actual microscopic transport space is chosen, the carrier audit becomes a finite overlap test.

XXV. AMBIENT FINITE-DIMENSIONAL FORMULATION

Let

$$W = \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}$$

be the ambient shell-enriched tensor space, and let

$$V_{\text{TECT}} \subset W$$

be the actual symmetry-allowed microscopic transport tensor space obtained by linearizing the complete TECT dynamics around the chosen defect/BCC background.

Fix shell data

$$G_* \neq 0, \quad \hat{n} := \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\} \text{ orthonormal.}$$

For the three required reduced channels define the shell target covectors

$$T_{ija}^{(\parallel)} = e_{\parallel i} G_{*j} u_a^{(\parallel)}, \quad T_{ija}^{(I)} = e_{1i} G_{*j} u_a^{(1)} + e_{2i} G_{*j} u_a^{(2)}, \quad T_{ija}^{(J)} = e_{2i} G_{*j} u_a^{(1)} - e_{1i} G_{*j} u_a^{(2)},$$

and the corresponding linear witness functionals

$$L_\alpha(N) := \langle T^{(\alpha)}, N \rangle, \quad \alpha \in \{\parallel, I, J\}.$$

Then the exact carrier criterion is

$$L_\alpha|_{V_{\text{TECT}}} \equiv 0 \iff P_V T^{(\alpha)} = 0,$$

where $P_V : W \rightarrow V_{\text{TECT}}$ is the orthogonal projector. Equivalently,

$$P_V T^{(\alpha)} \neq 0$$

means that the actual microscopic TECT linearization contains a shell-relevant nontrivial carrier direction in channel α .

XXVI. BASIS-LEVEL OPERATIONAL AUDIT

Choose any basis

$$\{E_A\}_{A=1}^d$$

of V_{TECT} , where $d = \dim V_{\text{TECT}} < \infty$. Write

$$N = \sum_{A=1}^d x_A E_A.$$

Then

$$L_\alpha(N) = \sum_{A=1}^d x_A \ell_{\alpha A}, \quad \ell_{\alpha A} := \langle T^{(\alpha)}, E_A \rangle.$$

Hence the entire carrier audit becomes the computation of the three overlap vectors

$$\ell_{\parallel} = (\ell_{\parallel 1}, \dots, \ell_{\parallel d}), \quad \ell_I = (\ell_{I1}, \dots, \ell_{Id}), \quad \ell_J = (\ell_{J1}, \dots, \ell_{Jd}).$$

The exact dichotomy is:

$$P_V T^{(\alpha)} = 0 \iff \ell_{\alpha A} = 0 \text{ for all } A,$$

and

$$P_V T^{(\alpha)} \neq 0 \iff \exists A \text{ such that } \ell_{\alpha A} \neq 0.$$

Therefore the surviving-carrier problem is finite and explicit.

XXVII. SHELL-ADAPTED DECOMPOSITION OF BASIS ELEMENTS

Let

$$V_{\min} = \text{span}\{B_{\parallel}, B_I, B_J\}$$

be the minimal shell-adapted witness subspace. Each basis element $E_A \in V_{\text{TECT}}$ can be decomposed as

$$E_A = a_A B_{\parallel} + b_A B_I + c_A B_J + R_A,$$

where

$$R_A \perp V_{\min}.$$

Because the witness functionals are linear and vanish on the orthogonal complement R_A , we get

$$\ell_{\parallel A} = \langle T^{(\parallel)}, E_A \rangle = \mu_{\parallel} a_A,$$

$$\ell_{IA} = \langle T^{(I)}, E_A \rangle = \mu_I b_A, \quad \ell_{JA} = \langle T^{(J)}, E_A \rangle = \mu_J c_A,$$

with nonzero constants

$$\mu_{\parallel} \neq 0, \quad \mu_I \neq 0, \quad \mu_J \neq 0$$

in the shell-adapted normalization.

Hence:

$$\ell_{\parallel A} \neq 0 \iff a_A \neq 0,$$

and

$$(\ell_{IA}, \ell_{JA}) \neq (0, 0) \iff (b_A, c_A) \neq (0, 0).$$

So the actual microscopic test becomes:

$$\boxed{\exists A : a_A \neq 0 \quad \text{and} \quad \exists B : (b_B, c_B) \neq (0, 0).} \quad (44)$$

XXVIII. EXACT BASIS-LEVEL COMPLETION THEOREM

[Basis-level carrier-audit theorem] Let $V_{\text{TECT}} \subset W$ be the actual microscopic transport space and let $\{E_A\}_{A=1}^d$ be any basis. For each basis element define the overlap coefficients

$$\ell_{\parallel A} = \langle T^{(\parallel)}, E_A \rangle, \quad \ell_{IA} = \langle T^{(I)}, E_A \rangle, \quad \ell_{JA} = \langle T^{(J)}, E_A \rangle.$$

Then the microscopic carrier audit succeeds if and only if:

1. there exists at least one basis element E_A with

$$\ell_{\parallel A} \neq 0;$$

2. there exists at least one basis element E_B with

$$(\ell_{IB}, \ell_{JB}) \neq (0, 0).$$

Equivalently, the carrier audit succeeds if and only if

$$P_V T^{(\parallel)} \neq 0 \quad \text{and} \quad (P_V T^{(I)}, P_V T^{(J)}) \neq (0, 0).$$

By the basis expansion,

$$L_\alpha(N) = \sum_A x_A \ell_{\alpha A}.$$

Thus $L_\alpha|_{V_{\text{TECT}}} \equiv 0$ if and only if every coefficient $\ell_{\alpha A}$ vanishes. Therefore

$$P_V T^{(\alpha)} \neq 0$$

is equivalent to the existence of at least one basis vector with nonzero overlap in channel α .

For the full Weyl/Dirac witness problem, the longitudinal channel and at least one transverse channel must survive. So the stated two conditions are necessary and sufficient.

XXIX. IMMEDIATE REDUCED DIRAC CONSEQUENCE

Once the basis-level carrier audit succeeds, the shell-reduced witness coefficients satisfy

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta},$$

with

$$I_{\theta} \neq 0, \quad I_{\theta'^3} \neq 0.$$

Therefore the reduced symbol is nondegenerate exactly when

$$c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).$$

Equivalently, in the shell tensor form

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j},$$

the local Weyl condition is

$$A \neq 0, \quad (B, C) \neq (0, 0).$$

So after the basis audit, the PDE problem is completely reduced to a finite coefficient check.

XXX. PRACTICAL AUDIT ALGORITHM

A concrete operational version is as follows.

a. Step 1. Linearize the complete microscopic TECT equation around the chosen defect/BCC background and extract the first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}.$$

b. Step 2. Choose a basis

$$\{E_A\}_{A=1}^d$$

for the resulting finite-dimensional symmetry-allowed transport space V_{TECT} .

c. Step 3. For each basis element, compute the overlaps

$$\ell_{\parallel A}, \quad \ell_{IA}, \quad \ell_{JA}.$$

d. Step 4. If there exists at least one A with

$$\ell_{\parallel A} \neq 0,$$

then the longitudinal carrier survives generically.

e. Step 5. If there exists at least one B with

$$(\ell_{IB}, \ell_{JB}) \neq (0, 0),$$

then the transverse carrier survives generically.

f. Step 6. If both checks pass, then the local Weyl witness audit is complete.

XXXI. SHARP HONEST CONCLUSION

We have now reduced the remaining microscopic problem to the finite-dimensional statement:

$$\boxed{\exists A : \ell_{\parallel A} \neq 0, \quad \exists B : (\ell_{IB}, \ell_{JB}) \neq (0, 0).}$$

This is the exact basis-level completion criterion.

Thus the strongest honest status is:

$$\boxed{\text{the actual microscopic carrier problem is no longer an uncontrolled PDE problem;}}$$

$$\boxed{\text{it is a finite basis-overlap audit on } V_{\text{TECT}}.}$$

What remains open is the execution of this audit for the final complete microscopic TECT linearization.

XXXII. FINAL SUMMARY OF THE BASIS-LEVEL CARRIER AUDIT STEP

a. What is now closed. Let

$$V_{\text{TECT}} \subset W$$

be the actual microscopic transport space and choose any basis

$$\{E_A\}_{A=1}^d.$$

For the shell target covectors $T^{(\parallel)}, T^{(I)}, T^{(J)}$, define

$$\ell_{\parallel A} = \langle T^{(\parallel)}, E_A \rangle, \quad \ell_{IA} = \langle T^{(I)}, E_A \rangle, \quad \ell_{JA} = \langle T^{(J)}, E_A \rangle.$$

Then:

$$P_V T^{(\parallel)} \neq 0 \iff \exists A : \ell_{\parallel A} \neq 0,$$

and

$$(P_V T^{(I)}, P_V T^{(J)}) \neq (0, 0) \iff \exists B : (\ell_{IB}, \ell_{JB}) \neq (0, 0).$$

Therefore the exact basis-level completion criterion is

$$\boxed{\exists A : \ell_{\parallel A} \neq 0, \quad \exists B : (\ell_{IB}, \ell_{JB}) \neq (0, 0).}$$

b. Equivalent shell-adapted statement. If

$$E_A = a_A B_{\parallel} + b_A B_I + c_A B_J + R_A, \quad R_A \perp \text{span}\{B_{\parallel}, B_I, B_J\},$$

then

$$\ell_{\parallel A} = \mu_{\parallel} a_A, \quad \ell_{IA} = \mu_I b_A, \quad \ell_{JA} = \mu_J c_A,$$

with

$$\mu_{\parallel}, \mu_I, \mu_J \neq 0.$$

So the audit is equivalent to

$$\boxed{\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0).}$$

c. Dirac consequence. Once this basis audit succeeds, the reduced witness coefficients satisfy

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta},$$

hence the local Weyl condition becomes

$$\boxed{c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).}$$

Equivalently, in shell tensor form,

$$\boxed{A \neq 0, \quad (B, C) \neq (0, 0).}$$

d. Honest status. So the remaining microscopic frontier is no longer conceptual:

$$\boxed{\text{it is the explicit execution of a finite basis-overlap audit for the final complete TECT linearization.}}$$

XXXIII. RESIDUAL AXIAL REPRESENTATION DECOMPOSITION OF ACTUAL MICROSCOPIC BASIS ELEMENTS

a. Goal. We now turn the carrier audit into a canonical representation-theoretic decomposition. The aim is to prove that every shell-projected microscopic first-order basis element admits a unique invariant decomposition into one longitudinal singlet coefficient and one transverse doublet pair:

$$E_A \mapsto (a_A, b_A, c_A).$$

This is the exact structural step needed before the final explicit microscopic overlap audit.

XXXIV. RESIDUAL AXIAL SYMMETRY AND THE INVARIANT COUPLING SPACE

Let the selected shell direction be

$$\hat{n} = \frac{G_*}{|G_*|},$$

and choose an orthonormal frame

$$\{e_1, e_2, \hat{n}\}.$$

Under the residual axial symmetry $H_\perp$ (the connected $SO(2)_\perp$ piece, or equivalently the $O(2)_\perp$ invariant classification), the worldvolume momentum sector decomposes as

$$V_p = \mathbb{R} p_\parallel \oplus E_\perp, \quad E_\perp = \text{span}\{p_1, p_2\},$$

while the shell-relevant traceless internal sector decomposes as

$$V_{\text{int}} = \mathbb{R} \sigma_3 \oplus \Sigma_\perp, \quad \Sigma_\perp = \text{span}\{\sigma_1, \sigma_2\}.$$

Thus both source and target carry the same

$$1 \oplus 2$$

decomposition.

Define the invariant first-order shell coupling space

$$\text{Inv} := \text{Hom}_{H_\perp}(V_p, V_{\text{int}}).$$

[Dimension and canonical basis of the invariant space] The invariant coupling space is exactly three-dimensional:

$$\dim \text{Inv} = 3.$$

A canonical basis is given by

$$B_\parallel : p_\parallel \mapsto \sigma_3,$$

$$B_I : (p_1, p_2) \mapsto (\sigma_1, \sigma_2),$$

$$B_J : (p_1, p_2) \mapsto (-\sigma_2, \sigma_1).$$

Equivalently, the most general symmetry-compatible reduced symbol is

$$A_1(p) = a p_\parallel \sigma_3 + b (p_1 \sigma_1 + p_2 \sigma_2) + c (-p_2 \sigma_1 + p_1 \sigma_2).$$

Because

$$V_p \cong 1 \oplus 2, \quad V_{\text{int}} \cong 1 \oplus 2,$$

one has

$$\text{Hom}_{H_\perp}(V_p, V_{\text{int}}) \cong \text{Hom}(1, 1) \oplus \text{Hom}(2, 2),$$

since the mixed blocks

$$\text{Hom}(1, 2), \quad \text{Hom}(2, 1)$$

vanish by inequivalence of irreducible sectors.

Now

$$\dim \text{Hom}(1, 1) = 1.$$

For the real transverse doublet representation,

$$\text{End}_{H_\perp}(2) = \text{span}\{\mathbf{1}_2, J\},$$

where J is the planar complex structure

$$J(p_1, p_2) = (-p_2, p_1).$$

Hence

$$\dim \text{Hom}(2, 2) = 2.$$

Therefore

$$\dim \text{Inv} = 1 + 2 = 3,$$

and the stated basis follows.

XXXV. PROJECTION OF ACTUAL MICROSCOPIC BASIS ELEMENTS ONTO THE INVARIANT SECTOR

Let

$$V_{\text{TECT}} \subset W$$

be the actual microscopic first-order transport space produced by the complete TECT linearization, and choose any basis

$$\{E_A\}_{A=1}^d.$$

Not every E_A is itself already written in shell-adapted invariant form. However, only its invariant shell-reduced component contributes to the reduced Weyl witness problem. Therefore define the invariant projection

$$P_{\text{inv}} : W \rightarrow \text{Inv}.$$

For each basis element, set

$$E_A^{\text{inv}} := P_{\text{inv}} E_A.$$

[Unique shell-adapted decomposition] For every microscopic basis element E_A , there exists a unique triple

$$(a_A, b_A, c_A) \in \mathbb{R}^3$$

such that

$$E_A^{\text{inv}} = a_A B_{\parallel} + b_A B_I + c_A B_J.$$

By the previous theorem,

$$\text{Inv} = \text{span}\{B_{\parallel}, B_I, B_J\}$$

is a three-dimensional vector space with the displayed basis. Therefore every element of Inv , and in particular E_A^{inv} , admits a unique expansion in that basis.

So the actual microscopic carrier audit is reduced to the coefficient map

$$E_A \longmapsto (a_A, b_A, c_A).$$

XXXVI. DUAL WITNESS EXTRACTION OF THE COEFFICIENTS

Let

$$W_{\parallel}, \quad W_I, \quad W_J$$

be the dual witness functionals normalized by

$$\begin{aligned} W_{\parallel}(B_{\parallel}) &= 1, & W_{\parallel}(B_I) &= W_{\parallel}(B_J) = 0, \\ W_I(B_I) &= 1, & W_I(B_{\parallel}) &= W_I(B_J) = 0, \\ W_J(B_J) &= 1, & W_J(B_{\parallel}) &= W_J(B_I) = 0. \end{aligned}$$

Then for

$$E_A^{\text{inv}} = a_A B_{\parallel} + b_A B_I + c_A B_J,$$

one gets immediately

$$a_A = W_{\parallel}(E_A^{\text{inv}}), \quad b_A = W_I(E_A^{\text{inv}}), \quad c_A = W_J(E_A^{\text{inv}}).$$

Equivalently, in shell target-covector form,

$$\ell_{\parallel A} := \langle T^{(\parallel)}, E_A \rangle, \quad \ell_{IA} := \langle T^{(I)}, E_A \rangle, \quad \ell_{JA} := \langle T^{(J)}, E_A \rangle,$$

there exist fixed nonzero normalization constants

$$\mu_{\parallel}, \mu_I, \mu_J \neq 0$$

such that

$$\ell_{\parallel A} = \mu_{\parallel} a_A, \quad \ell_{IA} = \mu_I b_A, \quad \ell_{JA} = \mu_J c_A.$$

Hence:

$$a_A \neq 0 \iff \ell_{\parallel A} \neq 0,$$

and

$$(b_A, c_A) \neq (0, 0) \iff (\ell_{IA}, \ell_{JA}) \neq (0, 0).$$

XXXVII. REPRESENTATION-THEORETIC AUDIT CRITERION

We may now state the exact executable criterion.

[Representation decomposition audit theorem] Let $\{E_A\}_{A=1}^d$ be any basis of the actual microscopic TECT transport space V_{TECT} . For each A , let

$$E_A^{\text{inv}} = a_A B_{\parallel} + b_A B_I + c_A B_J$$

be its unique invariant shell-reduced decomposition. Then:

1. the longitudinal carrier survives if and only if

$$\exists A \text{ such that } a_A \neq 0;$$

2. the transverse carrier survives if and only if

$$\exists A \text{ such that } (b_A, c_A) \neq (0, 0).$$

Equivalently, the full local Weyl witness audit succeeds if and only if

$$\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0).$$

This follows immediately from the dual-witness extraction formulas above. If all $a_A = 0$, then every invariant basis component lacks the longitudinal singlet and thus

$$W_{\parallel}(E_A^{\text{inv}}) = 0 \quad \forall A,$$

so the longitudinal witness vanishes identically on V_{TECT} . Conversely, if at least one $a_A \neq 0$, then the longitudinal witness is not identically zero.

The transverse statement is identical, using (b_A, c_A) and the two transverse dual witnesses.

XXXVIII. RELATION TO THE REDUCED DIRAC CONDITIONS

Inside the shell-adapted reduced witness space, the exact nondegeneracy condition is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

With the seed parametrization,

$$\lambda_{\parallel} = \frac{1}{2}c_{\parallel}I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta},$$

where

$$I_{\theta'^3} \neq 0, \quad I_{\theta} \neq 0.$$

Thus, once the basis-level representation audit finds

$$\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0),$$

the reduced Weyl symbol is nondegenerate generically.

So the remaining microscopic problem is no longer abstract: it is exactly the computation of the invariant coordinates

$$(a_A, b_A, c_A)$$

for the actual complete TECT basis elements.

XXXIX. SHARP HONEST STATUS

What has been closed here is the representation-theoretic reduction:

$$E_A \mapsto (a_A, b_A, c_A) \text{ is a canonical and unique shell-reduced decomposition.}$$

What remains open is only the final execution:

$$\text{for the actual complete TECT basis, exhibit at least one } A \text{ with } a_A \neq 0 \text{ and at least one } B \text{ with } (b_B, c_B) \neq (0, 0).$$

That is the last microscopic carrier audit in executable finite-dimensional form.

XL. FINAL SUMMARY OF THE REPRESENTATION DECOMPOSITION STEP

a. Closed result. Under the residual axial shell symmetry, both the worldvolume momentum sector and the shell-relevant internal sector decompose as

$$1 \oplus 2.$$

Hence the invariant first-order shell-coupling space is

$$\text{Inv} = \text{Hom}_{H_{\perp}}(V_p, V_{\text{int}}), \quad \dim \text{Inv} = 3.$$

A canonical basis is

$$B_{\parallel}, \quad B_I, \quad B_J.$$

b. Canonical decomposition of actual basis elements. For every actual microscopic basis element $E_A \in V_{\text{TECT}}$, its invariant shell-reduced part satisfies

$$E_A^{\text{inv}} = P_{\text{inv}} E_A = a_A B_{\parallel} + b_A B_I + c_A B_J,$$

with a unique coefficient triple

$$(a_A, b_A, c_A) \in \mathbb{R}^3.$$

c. *Exact extraction.* The dual witnesses recover these coefficients exactly:

$$a_A = W_{\parallel}(E_A^{\text{inv}}), \quad b_A = W_I(E_A^{\text{inv}}), \quad c_A = W_J(E_A^{\text{inv}}).$$

Equivalently,

$$\ell_{\parallel A} = \mu_{\parallel} a_A, \quad \ell_{IA} = \mu_I b_A, \quad \ell_{JA} = \mu_J c_A,$$

with

$$\mu_{\parallel}, \mu_I, \mu_J \neq 0.$$

d. *Executable carrier criterion.* Therefore the actual microscopic audit succeeds if and only if

$$\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0).$$

e. *Dirac consequence.* Once these two conditions hold, the reduced shell witness satisfies

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),$$

hence the local shell-reduced Weyl symbol is nondegenerate generically.

f. *Honest frontier.* So the remaining microscopic problem is no longer conceptual. It is exactly this finite audit:

$$\text{compute } (a_A, b_A, c_A) \text{ for the actual complete TECT basis elements.}$$

XLI. CLASS-BY-CLASS COEFFICIENT AUDIT THEOREM

a. *Goal.* We now classify the symmetry-allowed microscopic operator classes by the invariant coefficient pattern they can contribute to the shell-reduced basis

$$\{B_{\parallel}, B_I, B_J\}.$$

For each microscopic basis element E_A , recall the unique invariant decomposition

$$E_A^{\text{inv}} = a_A B_{\parallel} + b_A B_I + c_A B_J.$$

The purpose of this section is to determine, class by class, which of the three coordinates

$$a_A, \quad b_A, \quad c_A$$

must vanish identically, and which may survive.

XLII. REFERENCE INVARIANT BASIS AND WITNESS DICTIONARY

The shell-reduced invariant first-order operator space is

$$\text{Inv} = \text{span}\{B_{\parallel}, B_I, B_J\},$$

with canonical basis

$$B_{\parallel} : \quad p_{\parallel} \sigma_3,$$

$$B_I : \quad p_1 \sigma_1 + p_2 \sigma_2,$$

$$B_J : \quad -p_2 \sigma_1 + p_1 \sigma_2.$$

Equivalently, in tensor form,

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j},$$

and the shell witnesses read

$$\lambda_{\parallel} \propto A, \quad \alpha \propto B, \quad \beta \propto C.$$

Thus the coefficient audit is exactly the problem of deciding, for each microscopic class, whether it can produce:

$$a_A \neq 0, \quad (b_A, c_A) \neq (0, 0).$$

XLIII. EXCLUDED CLASSES: NO (i, a) -CARRIER

[Type 0 exclusion theorem] Suppose E_A belongs to one of the following classes:

1. Type 0A: pure scalar potential corrections;
2. Type 0B: pure projector derivative corrections;
3. Type 0C: amplitude-only derivative corrections.

Then

$$E_A^{\text{inv}} = 0, \quad \text{hence} \quad (a_A, b_A, c_A) = (0, 0, 0).$$

Any shell-relevant first-order enriched tensor must carry simultaneously:

1. one spatial derivative index i or j ,
2. one internal reduced-shell/tangent index a ,
3. one derivative acting on the test mode sector,
4. one shell/background derivative factor.

Pure scalar potentials are polarization-blind or start only at quadratic order in the tangent fluctuation. Pure projector derivative terms remain internal/projector-sector terms and do not produce a linear shell-test mixed source. Amplitude-only derivative terms contain no internal Pauli/tangent label a .

Therefore none of these classes can contribute to the first-order shell-enriched mixed tensor, so their projection to Inv vanishes identically.

XLIV. EVEN SINGLET NO-GO

[Naive even singlet no-go] Consider the naive longitudinal singlet source

$$V_{\parallel, \text{naive}}^\mu(\xi) = \lambda_c \theta'(\xi)^2 n^\mu \sigma_3.$$

Then its projected longitudinal witness vanishes:

$$a_A = 0.$$

More precisely, its mixed low-mode overlap is zero by parity.

For a reflection-symmetric defect,

$$\theta(-\xi) = \theta(\xi), \quad \theta'(-\xi) = -\theta'(\xi),$$

so $\theta'(\xi)^2$ is even. The relevant mixed overlap is proportional to

$$\int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^2,$$

where f_+ is even and f_- is odd. Hence the integrand is odd, so the integral vanishes:

$$\int d\xi f_+ f_- \theta'^2 = 0.$$

Therefore the naive even singlet does not contribute to the first-order mixed symbol, and thus gives no longitudinal coefficient.

a. Consequence. A longitudinal singlet source must carry one additional odd prefactor. So the minimal viable longitudinal class is not $\theta'^2 \sigma_3$, but an odd-upgraded primitive channel.

XLV. PURE TRANSVERSE DOUBLET CLASS

[Pure transverse class] Consider the canonical transverse doublet source

$$V_{\perp}^{\mu}(\xi) = \theta'(\xi) \left[\Lambda_E E^{\mu}_a \sigma_a + \Lambda_O J^{\mu}_a \sigma_a \right], \quad a = 1, 2.$$

Then its invariant coefficient pattern is

$$a_A = 0, \quad (b_A, c_A) \propto (\Lambda_E, \Lambda_O).$$

In particular, this class can generate the transverse channels B_I, B_J , but not the longitudinal singlet $B_{\parallel}$.

The source takes values only in the transverse doublet sector (σ_1, σ_2) , and its worldvolume factor is likewise purely transverse. Therefore it transforms in the doublet-to-doublet intertwiner sector $\text{Hom}(2, 2)$, not in the singlet-to-singlet sector $\text{Hom}(1, 1)$.

Hence no $B_{\parallel}$ component is present, while the two transverse invariant coordinates are exactly the even/odd coefficients Λ_E, Λ_O , up to fixed normalization.

XLVI. PRIMITIVE LONGITUDINAL ODD SINGLET CLASS

Let

$$\mathcal{T}_{\xi} := i[P_B, \partial_{\xi} P_B], \quad S_{\xi} := (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}.$$

Write its Pauli decomposition as

$$S_{\xi} = s_1(\xi) \sigma_1 + s_2(\xi) \sigma_2 + s_3(\xi) \sigma_3.$$

[Primitive longitudinal class] Consider the odd-upgraded primitive transport kernel

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) S_{\xi} n^{\mu}.$$

Then:

1. if $s_3(\xi) \equiv 0$, then $a_A = 0$;
2. if $s_3(\xi) \not\equiv 0$ and the projected overlap

$$J_{\parallel} := \int d\xi f_{+}(\xi) f_{-}(\xi) \theta'(\xi) s_3(\xi)$$

is nonzero, then

$$a_A \neq 0;$$

3. in the pure longitudinal adapted frame, if $S_{\xi} \propto \sigma_3$, then

$$b_A = c_A = 0.$$

The first statement is immediate: the longitudinal witness probes the σ_3 -singlet component. If $s_3 \equiv 0$, there is no singlet projection.

For the second statement, $\theta'(\xi)$ is odd and S_{ξ} is even when built from quadratic response data. Hence the transport kernel is odd, exactly as required to couple even and odd low modes. If the scalar overlap $J_{\parallel}$ is nonzero, then the projected longitudinal coefficient is nonzero.

The third statement follows because in the adapted singlet frame the primitive class lies entirely in the σ_3 -sector and therefore has no transverse doublet projection.

a. Explicit seed representative. In the explicit representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3,$$

so

$$V_{(\parallel)}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3,$$

and

$$a_A \neq 0 \iff I_{\theta'^3} \neq 0.$$

This is the canonical longitudinal seed.

XLVII. CLASS I, II, III: ADMISSIBLE (i, a) -CARRIER CLASSES

[Admissible carrier classes] At lowest derivative order, the minimal viable local microscopic classes are:

1. Class I: direct shell–projector derivative couplings;
2. Class II: heavy auxiliary internal vector mediators X_i^a ;
3. Class III: connection-like mediators A_i^a .

Each is symmetry-admissible because each introduces an effective (i, a) -carrier. However, their audit status differs.

Any microscopic class capable of generating the reduced shell tensor must introduce an object carrying both a spatial derivative index and an internal tangent index. Classes I, II, III all satisfy this structural requirement, so none is excluded by symmetry.

Their difference is not admissibility but explicitness: Class I inserts the carrier tensor directly by hand, Class II generates it dynamically through mediator elimination, and Class III is structurally plausible but less explicit in the present derivation chain.

XLVIII. CLASS II AS THE FULLY EXPLICIT THREE-COORDINATE CLASS

[Class II coordinate theorem] In the Class II interpretation, the three mediator directions correspond exactly to the three shell-reduced invariant coordinates:

$$X_i^3 \leftrightarrow B_{\parallel}, \quad X_i^1 \leftrightarrow B_I, \quad X_i^2 \leftrightarrow B_J.$$

Equivalently, in the explicit seed basis

$$V_{\text{seed}} = \text{span}\{N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)}\},$$

one has

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

Thus Class II is the unique currently explicit microscopic class that realizes all three coefficient directions (a, b, c) at once.

The longitudinal seed is represented by the X_i^3 -channel, while the transverse even/odd seeds are represented by the X_i^1 - and X_i^2 -channels respectively. After projection to the reduced defect/shell sector, these induce the three independent coefficients g_c, g_E, g_O , which are exactly the shell witness coordinates. Therefore the Class II mediator decomposition is isomorphic to the invariant basis decomposition.

XLIX. CLASS-BY-CLASS AUDIT TABLE

The result of the classification can be summarized as follows.

Microscopic class	Possible invariant coordinates	Audit status
Type 0A / 0B / 0C	$(0, 0, 0)$	excluded
Naive even singlet $\theta'^2 \sigma_3$	$a = 0$	excluded by parity
Pure transverse doublet	$a = 0, (b, c) \neq (0, 0)$ possible	explicitly allowed
Primitive odd longitudinal singlet	$a \neq 0$ possible, $b = c = 0$	minimal longitudinal mechanism
Class I	a, b, c all possible in principle	admissible, less explicit
Class II	a, b, c explicitly realizable	strongest current class
Class III	a, b, c possible in principle	plausible, less explicit

L. SHARP HONEST CONCLUSION

The final coefficient audit problem is now sharply reduced.

1. Pure scalar/projector/amplitude classes are removed completely.
2. A naive even singlet is removed by parity.
3. The transverse doublet class is already explicitly realized.
4. The longitudinal primitive odd singlet is the minimal surviving longitudinal mechanism.
5. Class II is the only currently explicit microscopic class that packages all three invariant coordinates in one unified carrier structure.

Therefore the remaining microscopic frontier is no longer broad. It is the explicit yes/no audit:

Does the complete TECT linearization realize at least one nonzero primitive longitudinal seed,

together with the already available transverse seed, inside the actual microscopic transport space?

Equivalently, the final executable target is:

$$\boxed{\exists A : a_A \neq 0 \quad \text{and} \quad \exists B : (b_B, c_B) \neq (0, 0).}$$

LI. FINAL SUMMARY OF THE CLASS-BY-CLASS COEFFICIENT AUDIT

a. Closed classification. For each actual microscopic basis element E_A , its invariant shell-reduced part

$$E_A^{\text{inv}} = a_A B_{\parallel} + b_A B_I + c_A B_J$$

falls into one of the following coefficient patterns:

Class	Coefficient pattern
Type 0A / 0B / 0C	$(a_A, b_A, c_A) = (0, 0, 0)$
Naive even singlet $\theta'^2 \sigma_3$	$a_A = 0$
Pure transverse doublet	$a_A = 0, (b_A, c_A) \neq (0, 0)$ possible
Primitive odd longitudinal singlet	$a_A \neq 0$ possible, $b_A = c_A = 0$
Class I	a_A, b_A, c_A admissible in principle
Class II	a_A, b_A, c_A explicitly realizable
Class III	a_A, b_A, c_A admissible in principle

b. Key structural conclusions.

1. Pure scalar/projector/amplitude-only classes are completely excluded.
2. A naive even longitudinal singlet is excluded by parity.
3. The transverse witness sector is already explicitly realized by the canonical odd transverse doublet source.
4. The minimal surviving longitudinal mechanism is the primitive odd singlet

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu, \quad S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}.$$

5. Class II is the strongest explicit microscopic carrier class because

$$X_i^3 \leftrightarrow B_\parallel, \quad X_i^1 \leftrightarrow B_I, \quad X_i^2 \leftrightarrow B_J.$$

c. Exact remaining frontier. Therefore the final microscopic audit is reduced to:

$$\boxed{\exists A : a_A \neq 0 \quad \text{and} \quad \exists B : (b_B, c_B) \neq (0, 0).}$$

Since the transverse half is already explicitly available, the only genuinely unresolved microscopic question is:

$$\boxed{\text{does the complete TECT linearization realize at least one nonzero primitive longitudinal seed?}}$$

LII. LONGITUDINAL PRIMITIVE 3-STEP AUDIT THEOREM

a. Goal. We now convert the remaining longitudinal carrier question into an explicit finite audit. The objective is to isolate a practical sufficient-and-necessary chain for the survival of the primitive longitudinal channel in the shell-reduced Dirac witness problem.

LIII. SETUP

Fix the defect/projector data

$$P_B(\xi), \quad Q_B(\xi) = \mathbf{1} - P_B(\xi), \quad e_1(\xi), e_2(\xi), \quad f_\pm(\xi), \quad \mathcal{T}_\xi := i[P_B, \partial_\xi P_B].$$

The longitudinal primitive channel is expected to come from the traceless quadratic Q_B -response:

$$S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}.$$

Equivalently, in the Q_B -frame,

$$(T_\xi)_{ab} := \langle e_a, \mathcal{T}_\xi e_b \rangle, \quad S_\xi = T_\xi^2 - \frac{1}{2} \text{Tr}(T_\xi^2) \mathbf{1}_2.$$

Write its Pauli decomposition as

$$S_\xi = s_1(\xi) \sigma_1 + s_2(\xi) \sigma_2 + s_3(\xi) \sigma_3.$$

The primitive longitudinal transport kernel is then

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu.$$

LIV. THE THREE AUDIT STEPS

a. Step 1: Nontrivial quadratic traceless response. Compute

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}.$$

If

$$S_\xi \equiv 0,$$

then there is no primitive longitudinal channel at all.

b. Step 2: Nonvanishing singlet component. Decompose

$$S_\xi = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3.$$

If

$$s_3(\xi) \equiv 0,$$

then the quadratic response has no longitudinal singlet projection, so the primitive longitudinal witness vanishes.

c. Step 3: Low-mode overlap test. Define the longitudinal overlap functional

$$J_{\parallel}^{(3)} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi).$$

If

$$J_{\parallel}^{(3)} \neq 0,$$

then the projected longitudinal witness is nonzero:

$$\lambda_{\parallel} \neq 0.$$

IV. LONGITUDINAL PRIMITIVE AUDIT THEOREM

[3-step longitudinal audit] Let

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3,$$

and define

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi).$$

Then the following implications hold:

1. If $S_\xi \equiv 0$, then the primitive longitudinal channel is absent.
2. If $S_\xi \neq 0$ but $s_3(\xi) \equiv 0$, then the primitive longitudinal channel has no singlet projection, hence

$$\lambda_{\parallel} = 0.$$

3. If $s_3(\xi) \neq 0$ but $J_{\parallel}^{(3)} = 0$, then the longitudinal singlet exists algebraically but is killed by the graded low-mode overlap.

4. If

$$J_{\parallel}^{(3)} \neq 0,$$

then

$$\lambda_{\parallel} \neq 0.$$

Therefore the survival of the longitudinal primitive channel is reduced to the finite chain

$$\boxed{T_\xi \longrightarrow S_\xi \longrightarrow s_3(\xi) \longrightarrow J_{\parallel}^{(3)}}.$$

The first statement is immediate from the definition of the primitive channel:

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu.$$

If $S_\xi \equiv 0$, then $V_{(\parallel)}^\mu \equiv 0$.

For the second statement, the longitudinal witness couples only to the singlet σ_3 -component of the traceless Q_B -response. Hence if $s_3(\xi) \equiv 0$, the primitive kernel has no longitudinal singlet content.

For the third statement, even when the singlet component exists, the actual projected witness is determined by its overlap with the graded low-mode pair f_+, f_- . Thus the projected coefficient is exactly

$$\lambda_{\parallel} \propto J_{\parallel}^{(3)}.$$

If this overlap vanishes, then the projected longitudinal witness vanishes.

For the fourth statement, the same formula shows that any nonzero value of $J_{\parallel}^{(3)}$ implies

$$\lambda_{\parallel} \neq 0.$$

This proves the chain.

LVI. PARITY STRUCTURE AND THE NECESSITY OF THE ODD UPGRADE

A naive singlet source would be

$$V_{\parallel, \text{naive}}^\mu(\xi) = \theta'(\xi)^2 n^\mu \sigma_3.$$

But then the relevant overlap would be

$$\int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^2.$$

Since

$$f_+ \text{ is even, } \quad f_- \text{ is odd, } \quad \theta'(\xi)^2 \text{ is even,}$$

the integrand is odd, and therefore

$$\int d\xi f_+ f_- \theta'^2 = 0.$$

So the naive even singlet cannot contribute. This proves that the primitive longitudinal channel must carry one additional odd factor, namely the odd upgrade

$$\theta'(\xi) S_\xi.$$

LVII. EXPLICIT TOY REPRESENTATIVE

In the explicit seed representative, one has

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

Therefore

$$s_1(\xi) = 0, \quad s_2(\xi) = 0, \quad s_3(\xi) = \frac{1}{2} \theta'(\xi)^2.$$

The primitive transport kernel becomes

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3.$$

Hence

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi)^3 = \frac{1}{2} I_{\theta'^3}.$$

Using the explicit seed evaluation,

$$I_{\theta^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

we obtain

$$J_{\parallel}^{(3)} = -\frac{3\sqrt{3}\pi}{512} \theta_0^3 \kappa^3.$$

Thus for a nontrivial defect profile $\theta_0 \kappa \neq 0$,

$$J_{\parallel}^{(3)} \neq 0, \quad \lambda_{\parallel} \neq 0.$$

LVIII. SHARP HONEST STATUS

The 3-step audit theorem closes the longitudinal problem up to one final microscopic question.

1. Inside the explicit seed representative, the primitive longitudinal channel survives completely:

$$S_{\xi} \neq 0, \quad s_3(\xi) \neq 0, \quad J_{\parallel}^{(3)} \neq 0.$$

2. Therefore longitudinal witness existence is no longer an abstract issue.

3. What remains open is only whether the *actual complete microscopic TECT linearization* produces a nonzero traceless quadratic Q_B -response with

$$J_{\parallel}^{(3)} \neq 0$$

in the physically relevant defect background.

Equivalently, the remaining frontier is the executable finite test:

$$\boxed{S_{\xi} \stackrel{?}{\neq} 0, \quad s_3(\xi) \stackrel{?}{\neq} 0, \quad J_{\parallel}^{(3)} \stackrel{?}{\neq} 0.}$$

LIX. FINAL SUMMARY OF THE LONGITUDINAL 3-STEP AUDIT

- Closed result.* The primitive longitudinal carrier problem is reduced to a finite audit chain:

$$\boxed{T_{\xi} \longrightarrow S_{\xi} = (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} \longrightarrow s_3(\xi) \longrightarrow J_{\parallel}^{(3)}.$$

- Definitions.* Write

$$S_{\xi} = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3,$$

and define

$$J_{\parallel}^{(3)} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi).$$

- Exact implications.* Then:

$$S_{\xi} \equiv 0 \implies \text{no primitive longitudinal channel,}$$

$$s_3(\xi) \equiv 0 \implies \text{no longitudinal singlet projection,}$$

$$J_{\parallel}^{(3)} = 0 \implies \text{the graded low-mode overlap kills the longitudinal witness,}$$

$$\boxed{J_{\parallel}^{(3)} \neq 0 \implies \lambda_{\parallel} \neq 0.}$$

d. Toy representative. In the explicit seed representative,

$$S_\xi = \frac{1}{2}\theta'(\xi)^2\sigma_3, \quad s_3(\xi) = \frac{1}{2}\theta'(\xi)^2,$$

so

$$J_{\parallel}^{(3)} = \frac{1}{2}I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{512}\theta_0^3\kappa^3 \neq 0 \quad (\theta_0\kappa \neq 0).$$

Hence

$$\lambda_{\parallel} \neq 0$$

in the toy representative.

e. Honest frontier. Therefore the only remaining microscopic question is:

does the complete microscopic TECT linearization produce a nonzero traceless quadratic Q_B -response with $J_{\parallel}^{(3)} \neq 0$?

So the longitudinal frontier is no longer abstract. It is exactly the executable 3-step audit:

$$S_\xi \neq 0, \quad s_3(\xi) \neq 0, \quad J_{\parallel}^{(3)} \neq 0.$$

LX. QUADRATIC-RESPONSE NON-FORBIDDANCE THEOREM FOR THE LONGITUDINAL SINGLET

a. Goal. We now prove the strongest honest statement available at this stage: the σ_3 -component of the traceless quadratic Q_B -response is *not* forbidden by the minimal residual symmetry of the defect/shell configuration. Equivalently, the vanishing of the longitudinal singlet coefficient

$$s_3(\xi)$$

cannot be attributed to the basic shell residual symmetry alone; it would require either an additional exchange symmetry or an accidental/dynamical cancellation.

LXI. GENERAL QUADRATIC Q_B -RESPONSE

Let

$$P_B(\xi)$$

be the local rank-one projector of the defect background, and let

$$Q_B(\xi) = \mathbf{1} - P_B(\xi)$$

be the rank-two orthogonal complement bundle. Choose the adapted local Q_B -frame

$$(e_1(\xi), e_2(\xi)),$$

where e_1 is the tangent direction singled out by the defect gradient and e_2 is the orthogonal spectator direction.

Consider any Hermitian quadratic response operator restricted to Q_B :

$$R_\xi := \Pi_Q \mathcal{R}_\xi \Pi_Q \in \text{End}(Q_B),$$

where $\mathcal{R}_\xi$ is built quadratically from the local background response data. Its traceless part is

$$S_\xi := R_\xi - \frac{1}{2}\text{Tr}(R_\xi)\mathbf{1}_2.$$

In the adapted frame,

$$R_\xi = \begin{pmatrix} r_{11}(\xi) & r_{12}(\xi) \\ r_{12}(\xi) & r_{22}(\xi) \end{pmatrix}, \quad r_{11}, r_{22} \in \mathbb{R}.$$

Therefore

$$S_\xi = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3,$$

with

$$s_1(\xi) = \Re r_{12}(\xi), \quad (45)$$

$$s_2(\xi) = -\Im r_{12}(\xi), \quad (46)$$

$$s_3(\xi) = \frac{1}{2}(r_{11}(\xi) - r_{22}(\xi)). \quad (47)$$

Hence the longitudinal singlet coefficient is exactly the diagonal anisotropy:

$$s_3(\xi) = \frac{1}{2}(\langle e_1, R_\xi e_1 \rangle - \langle e_2, R_\xi e_2 \rangle). \quad (48)$$

LXII. RESIDUAL SYMMETRY DOES NOT FORCE $s_3 = 0$

The shell-relevant local symmetry splits both the worldvolume and internal traceless sectors into

$$1 \oplus 2.$$

The unique longitudinal singlet channel is

$$p_{\parallel} \longleftrightarrow \sigma_3,$$

while the transverse channels are

$$(p_1, p_2) \longleftrightarrow (\sigma_1, \sigma_2).$$

Therefore σ_3 is not an illegal structure. It is the unique singlet generator selected by the same residual symmetry that defines the Weyl template.

This immediately implies the following.

[Non-forbiddance of the quadratic longitudinal singlet] Let S_ξ be the traceless part of any shell-relevant Hermitian quadratic Q_B -response in the adapted frame (e_1, e_2) . Then the minimal residual shell symmetry does not force

$$s_3(\xi) \equiv 0.$$

Equivalently, the longitudinal singlet σ_3 is symmetry-allowed.

The residual symmetry distinguishes one longitudinal singlet and one transverse doublet. The traceless fiber sector decomposes exactly as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R}\sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}.$$

Therefore σ_3 is part of the allowed invariant structure. A vanishing theorem

$$s_3(\xi) \equiv 0$$

could follow only if there were an additional symmetry exchanging e_1 and e_2 or forcing the quadratic response to act isotropically on Q_B . But such an exchange symmetry is not contained in the minimal shell residual symmetry itself, because the defect gradient already singles out e_1 . Hence the basic residual symmetry alone does not imply $s_3 \equiv 0$.

LXIII. ANISOTROPY CRITERION

Because $s_3(\xi)$ is exactly the diagonal anisotropy (48), we get the sharp criterion:
 [Diagonal-anisotropy criterion] For a shell-relevant Hermitian quadratic response R_ξ on Q_B ,

$$s_3(\xi) \equiv 0 \iff \langle e_1, R_\xi e_1 \rangle = \langle e_2, R_\xi e_2 \rangle \text{ for all } \xi.$$

Therefore a sufficient condition for longitudinal-singlet survival is:

$$\exists \xi \text{ such that } \langle e_1, R_\xi e_1 \rangle \neq \langle e_2, R_\xi e_2 \rangle.$$

This is immediate from

$$s_3(\xi) = \frac{1}{2}(r_{11}(\xi) - r_{22}(\xi)).$$

So the actual microscopic audit is sharply reduced: to produce a longitudinal primitive seed, it is enough to show that the quadratic Q_B -response is not isotropic between the defect-selected tangent direction e_1 and the spectator direction e_2 .

LXIV. WHY DEFECT GEOMETRY NATURALLY ALLOWS SUCH ANISOTROPY

The defect gradient already breaks the symmetry between e_1 and e_2 .
 Indeed, the basic commutator channel

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B]$$

is built from the derivative of the defect projector. It therefore measures variation along the distinguished defect direction. In the explicit representative, $\mathcal{T}_\xi$ exchanges the condensate line $\mathbb{C}z_B$ with $\mathbb{C}e_1$, while annihilating e_2 . Thus its quadratic action need not be isotropic on Q_B .

This is the structural reason why a σ_3 -type splitting is natural rather than exceptional.

LXV. EXPLICIT SEED REPRESENTATIVE

In the explicit representative one has

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0,$$

but

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \sigma_3).$$

Hence

$$S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

Therefore

$$s_1(\xi) = 0, \quad s_2(\xi) = 0, \quad s_3(\xi) = \frac{1}{2} \theta'(\xi)^2.$$

So for any nontrivial defect profile,

$$\theta'(\xi) \neq 0 \implies s_3(\xi) \neq 0.$$

This explicit model saturates the general non-forbiddance theorem: the longitudinal singlet is not only allowed, it is actually realized.

LXVI. PARITY-ALLOWED LONGITUDINAL PRIMITIVE CHANNEL

The quadratic response $S_\xi \sim \theta'(\xi)^2 \sigma_3$ is even in ξ . To survive the graded low-mode overlap, one must multiply by one additional odd factor $\theta'(\xi)$. Thus the primitive longitudinal transport kernel is

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu.$$

In the explicit seed representative,

$$V_{(\parallel)}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3.$$

Its projected witness is

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) = \frac{1}{2} I_{\theta'^3}.$$

Since

$$I_{\theta'^3} \neq 0$$

for a nontrivial defect profile, the longitudinal witness survives:

$$J_{\parallel}^{(3)} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

LXVII. SHARP HONEST CONCLUSION

The actual logical position is now clear.

1. The coefficient $s_3(\xi)$ is not forbidden by the minimal shell residual symmetry.
2. Its vanishing would require either:
 - (a) an additional $e_1 \leftrightarrow e_2$ exchange symmetry,
 - (b) an exact isotropy of the quadratic Q_B -response,
 - (c) or an accidental/dynamical cancellation.
3. The explicit seed representative realizes

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3,$$

hence

$$s_3(\xi) \not\equiv 0$$

and

$$J_{\parallel}^{(3)} \neq 0.$$

Therefore the remaining microscopic frontier is not a symmetry question. It is the coefficient-identification question:

does the complete microscopic TECT linearization produce a non-isotropic quadratic Q_B -response,

equivalently $s_3(\xi) \not\equiv 0$ with $J_{\parallel}^{(3)} \neq 0$?

LXVIII. FINAL SUMMARY OF THE QUADRATIC-RESPONSE NON-FORBIDDANCE STEP

a. Closed result. For any shell-relevant Hermitian quadratic response on Q_B ,

$$R_\xi = \Pi_Q \mathcal{R}_\xi \Pi_Q, \quad S_\xi = R_\xi - \frac{1}{2} \text{Tr}(R_\xi) \mathbf{1}_2,$$

the traceless part has the Pauli decomposition

$$S_\xi = s_1(\xi) \sigma_1 + s_2(\xi) \sigma_2 + s_3(\xi) \sigma_3,$$

with

$$s_3(\xi) = \frac{1}{2} \left(\langle e_1, R_\xi e_1 \rangle - \langle e_2, R_\xi e_2 \rangle \right).$$

b. Non-forbiddance theorem. The minimal shell residual symmetry does not force

$$s_3(\xi) \equiv 0.$$

The reason is that the defect gradient already singles out e_1 , so the longitudinal singlet

$$\sigma_3$$

is symmetry-allowed rather than forbidden.

c. Sharp criterion. Thus

$$s_3(\xi) \equiv 0 \iff \langle e_1, R_\xi e_1 \rangle = \langle e_2, R_\xi e_2 \rangle \text{ for all } \xi.$$

So a sufficient condition for longitudinal survival is simply:

$$\exists \xi : \langle e_1, R_\xi e_1 \rangle \neq \langle e_2, R_\xi e_2 \rangle.$$

d. Explicit seed representative. In the explicit representative,

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2} (1 + \sigma_3),$$

hence

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

Therefore

$$s_3(\xi) = \frac{1}{2} \theta'(\xi)^2 \neq 0$$

for a nontrivial defect.

Multiplying by the required odd factor $\theta'(\xi)$ gives

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3,$$

and the projected overlap becomes

$$J_{\parallel}^{(3)} = \frac{1}{2} I_{\theta'^3} \neq 0.$$

e. Honest frontier. Therefore the remaining longitudinal microscopic question is no longer:

“is σ_3 symmetry-forbidden?”

That question is closed.

The actual remaining question is:

does the complete microscopic TECT linearization produce a non-isotropic quadratic Q_B -response

so that

$$s_3(\xi) \neq 0 \quad \text{and} \quad J_{\parallel}^{(3)} \neq 0?$$

LXIX. DIAGONAL-ANISOTROPY SUFFICIENT CRITERION FOR $s_3(\xi) \neq 0$

a. Goal. We now isolate the sharpest practical sufficient condition for the survival of the longitudinal singlet component in the actual quadratic Q_B -response. The point is to replace the vague question

“does the complete microscopic TECT linearization produce $s_3(\xi) \neq 0$?”

by a concrete diagonal-anisotropy test.

LXX. GENERAL HERMITIAN QUADRATIC RESPONSE ON Q_B

Let

$$R_\xi^{\text{act}} = \Pi_Q \mathcal{R}_\xi^{\text{act}} \Pi_Q \in \text{End}(Q_B)$$

be the actual shell-relevant Hermitian quadratic response extracted from the complete microscopic TECT linearization around the chosen defect background.

Choose the adapted Q_B -frame

$$(e_1(\xi), e_2(\xi)),$$

where e_1 is the defect-selected tangent direction and e_2 is the orthogonal spectator direction. Then

$$R_\xi^{\text{act}} = \begin{pmatrix} r_{11}(\xi) & r_{12}(\xi) \\ r_{12}(\xi) & r_{22}(\xi) \end{pmatrix}, \quad r_{11}, r_{22} \in \mathbb{R}.$$

Its traceless part is

$$S_\xi^{\text{act}} = R_\xi^{\text{act}} - \frac{1}{2} \text{Tr}(R_\xi^{\text{act}}) \mathbf{1}_2 = s_1(\xi) \sigma_1 + s_2(\xi) \sigma_2 + s_3(\xi) \sigma_3,$$

with

$$s_3(\xi) = \frac{1}{2} (r_{11}(\xi) - r_{22}(\xi)).$$

Equivalently,

$$\boxed{s_3(\xi) = \frac{1}{2} (\langle e_1, R_\xi^{\text{act}} e_1 \rangle - \langle e_2, R_\xi^{\text{act}} e_2 \rangle)}. \quad (49)$$

LXXI. MINIMAL DIAGONAL-ANISOTROPY CRITERION

[Diagonal-anisotropy sufficient criterion] Let R_ξ^{act} be the actual Hermitian quadratic Q_B -response. If there exists at least one ξ_0 such that

$$\langle e_1(\xi_0), R_{\xi_0}^{\text{act}} e_1(\xi_0) \rangle \neq \langle e_2(\xi_0), R_{\xi_0}^{\text{act}} e_2(\xi_0) \rangle, \quad (50)$$

then

$$\boxed{s_3(\xi) \neq 0}. \quad (51)$$

By (49),

$$s_3(\xi_0) = \frac{1}{2} (\langle e_1, R_{\xi_0}^{\text{act}} e_1 \rangle - \langle e_2, R_{\xi_0}^{\text{act}} e_2 \rangle).$$

If the two diagonal matrix elements differ at ξ_0 , then $s_3(\xi_0) \neq 0$. Hence $s_3(\xi)$ cannot vanish identically.

a. Interpretation. So the longitudinal singlet survives whenever the actual quadratic Q_B -response is not isotropic between the defect-selected tangent channel e_1 and the spectator channel e_2 .

LXXII. CANONICAL DECOMPOSITION OF THE ACTUAL RESPONSE

Every Hermitian 2×2 response may be written as

$$R_\xi^{\text{act}} = u_0(\xi)\mathbf{1}_2 + u_1(\xi)\sigma_1 + u_2(\xi)\sigma_2 + u_3(\xi)\sigma_3.$$

Then

$$S_\xi^{\text{act}} = u_1(\xi)\sigma_1 + u_2(\xi)\sigma_2 + u_3(\xi)\sigma_3,$$

so

$$s_3(\xi) = u_3(\xi).$$

Therefore the practical criterion becomes:

$$\boxed{u_3(\xi) \neq 0 \iff R_\xi^{\text{act}} \text{ contains a genuine } \sigma_3\text{-anisotropic component.}} \quad (52)$$

This makes the final audit especially simple: one does not need the whole response, only the coefficient of the diagonal anisotropy operator

$$|e_1\rangle\langle e_1| - |e_2\rangle\langle e_2| \cong \sigma_3.$$

LXXIII. STRUCTURAL ORIGIN OF DIAGONAL ANISOTROPY

The crucial geometric fact is that the defect already distinguishes the two Q_B -channels. Indeed,

$$e_1(\xi) \propto \partial_\xi z_B(\xi)$$

is the tangent direction selected by the defect gradient, whereas e_2 is the orthogonal spectator. So an exact equality

$$\langle e_1, R_\xi^{\text{act}} e_1 \rangle = \langle e_2, R_\xi^{\text{act}} e_2 \rangle$$

for all ξ would require more than the minimal shell residual symmetry. It would require either:

1. an additional exchange symmetry $e_1 \leftrightarrow e_2$, or
2. an exact dynamical isotropy of the quadratic Q_B -response.

Neither is forced by the basic defect/shell setup.

Hence the natural expectation is that a generic actual quadratic response should contain a nontrivial σ_3 -component unless some extra cancellation is present.

LXXIV. COMPARISON WITH EXPLICIT SEED REPRESENTATIVE

In the explicit seed representative,

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

Thus

$$s_3(\xi) = \frac{1}{2} \theta'(\xi)^2.$$

This is the model example of the general criterion: the diagonal matrix element in the e_1 -channel differs from that in the e_2 -channel, so s_3 survives.

Moreover, after multiplying by the required odd prefactor $\theta'(\xi)$,

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3,$$

and the low-mode overlap gives

$$J_{\parallel}^{(3)} = \frac{1}{2} I_{\theta'^3} \neq 0.$$

So the explicit representative saturates the criterion completely.

LXXV. INDEPENDENT SUPPORT FROM THE CHANNEL HESSIAN

The same structural mechanism also appears in the improved effective projector-Hessian model. In the adapted (e_1, e_2) -basis, the tangent-space Hessian is

$$W_B(\xi) = \begin{pmatrix} (\lambda + \mu) \cos^2 2\theta(\xi) & 0 \\ 0 & \lambda + 2\mu \cos^2 \theta(\xi) \end{pmatrix}.$$

Thus the two diagonal channels are generically unequal:

$$W_{11}(\xi) \neq W_{22}(\xi)$$

unless special parameter/profile identities happen to hold. Therefore the defect geometry naturally generates diagonal anisotropy on Q_B .

This does not by itself prove the final complete TECT coefficient extraction, but it strongly supports the claim that $s_3(\xi) \neq 0$ is a natural and stable possibility rather than an exceptional fine tuning.

LXXVI. SHARP HONEST CONCLUSION

We can now state the strongest honest theorem of this step:

[Practical sufficient criterion for longitudinal-singlet survival] Let R_ξ^{act} be the actual shell-relevant Hermitian quadratic Q_B -response of the complete microscopic TECT linearization. If

$$\exists \xi_0 \text{ such that } \langle e_1(\xi_0), R_{\xi_0}^{\text{act}} e_1(\xi_0) \rangle \neq \langle e_2(\xi_0), R_{\xi_0}^{\text{act}} e_2(\xi_0) \rangle,$$

then

$$s_3(\xi) \neq 0.$$

Hence the only further step needed for longitudinal survival is the overlap test

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi).$$

If this is also nonzero, then

$$\lambda_{\parallel} \neq 0.$$

a. Meaning. So the longitudinal frontier is no longer:

“derive the whole PDE and solve everything.”

It is now exactly the finite two-part audit:

(i) show diagonal anisotropy on Q_B ,	(ii) show nonzero graded overlap $J_{\parallel}^{(3)}$.
---	--

LXXVII. FINAL SUMMARY OF THE DIAGONAL-ANISOTROPY SUFFICIENT CRITERION

a. Closed result. For the actual Hermitian quadratic Q_B -response

$$R_\xi^{\text{act}} = \Pi_Q \mathcal{R}_\xi^{\text{act}} \Pi_Q,$$

its traceless part is

$$S_\xi^{\text{act}} = s_1(\xi) \sigma_1 + s_2(\xi) \sigma_2 + s_3(\xi) \sigma_3,$$

with

$s_3(\xi) = \frac{1}{2} \left(\langle e_1, R_\xi^{\text{act}} e_1 \rangle - \langle e_2, R_\xi^{\text{act}} e_2 \rangle \right).$
--

b. Practical sufficient criterion. Therefore,

$$\boxed{\exists \xi_0 : \langle e_1(\xi_0), R_{\xi_0}^{\text{act}} e_1(\xi_0) \rangle \neq \langle e_2(\xi_0), R_{\xi_0}^{\text{act}} e_2(\xi_0) \rangle \implies s_3(\xi) \neq 0.}$$

So longitudinal-singlet survival is guaranteed once the quadratic Q_B -response is not isotropic between the defect-selected tangent channel e_1 and the spectator channel e_2 .

c. Explicit representative. In the explicit seed representative,

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3,$$

hence

$$s_3(\xi) = \frac{1}{2} \theta'(\xi)^2 \neq 0.$$

After the required odd upgrade,

$$V_{(\parallel)}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3,$$

and

$$J_{\parallel}^{(3)} = \frac{1}{2} I_{\theta'^3} \neq 0.$$

d. Independent support. The improved projector-Hessian model already exhibits diagonal channel anisotropy:

$$W_B(\xi) = \begin{pmatrix} (\lambda + \mu) \cos^2 2\theta(\xi) & 0 \\ 0 & \lambda + 2\mu \cos^2 \theta(\xi) \end{pmatrix},$$

so e_1 and e_2 are generically not equivalent dynamically.

e. Honest frontier. Hence the longitudinal frontier is now exactly the finite two-step check:

$$\boxed{\text{(i) verify diagonal anisotropy } s_3(\xi) \neq 0, \quad \text{(ii) verify } J_{\parallel}^{(3)} \neq 0.}$$

This is a coefficient-identification problem, not a symmetry problem.

LXXVIII. GENERIC NONVANISHING THEOREM FOR THE LONGITUDINAL SINGLET COEFFICIENT $s_3(\xi)$

a. Goal. We now sharpen the previous diagonal-anisotropy criterion to a genericity theorem. The question is no longer merely whether

$$s_3(\xi) \neq 0$$

is allowed. The sharper question is:

$$\text{when is } s_3(\xi) \equiv 0 \text{ forced, and when is it nongeneric?}$$

LXXIX. ACTUAL QUADRATIC RESPONSE FAMILY

Let the actual shell-relevant Hermitian quadratic Q_B -response be a family

$$R_\xi(c) = u_0(\xi; c) \mathbf{1}_2 + u_1(\xi; c) \sigma_1 + u_2(\xi; c) \sigma_2 + u_3(\xi; c) \sigma_3,$$

depending on microscopic coefficients

$$c = (c_1, \dots, c_m) \in \mathcal{P}$$

in some finite-dimensional local parameter space $\mathcal{P}$ of symmetry-allowed quadratic response data.

Its traceless part is

$$S_\xi(c) = u_1(\xi; c)\sigma_1 + u_2(\xi; c)\sigma_2 + u_3(\xi; c)\sigma_3,$$

and the longitudinal singlet coefficient is

$$s_3(\xi; c) = u_3(\xi; c).$$

Equivalently, in the adapted frame (e_1, e_2) ,

$$u_3(\xi; c) = \frac{1}{2} \left(\langle e_1, R_\xi(c)e_1 \rangle - \langle e_2, R_\xi(c)e_2 \rangle \right).$$

Thus $u_3 \equiv 0$ means exact diagonal isotropy of the quadratic response on Q_B .

LXXX. WHAT CAN FORCE $u_3(\xi) \equiv 0$

[Forcing mechanisms for $u_3 \equiv 0$] The identity

$$u_3(\xi; c) \equiv 0$$

can be structurally forced only if at least one of the following occurs:

1. an additional symmetry exchanges e_1 and e_2 ;
2. the quadratic response is exactly isotropic on Q_B :

$$R_\xi(c) = u_0(\xi; c)\mathbf{1}_2 + u_1(\xi; c)\sigma_1 + u_2(\xi; c)\sigma_2 \quad \text{with} \quad u_3(\xi; c) \equiv 0;$$

3. the microscopic coefficients are fine-tuned to a cancellation locus.

In particular, the minimal defect/shell residual symmetry alone does not force $u_3 \equiv 0$.

The longitudinal singlet σ_3 is part of the symmetry-allowed shell-reduced structure:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

Hence the basic residual symmetry allows a singlet channel rather than excluding it.

Therefore a vanishing theorem

$$u_3(\xi; c) \equiv 0$$

cannot follow from the minimal symmetry alone. It can only come from a stronger condition that makes the two Q_B -directions e_1, e_2 exactly equivalent, or from a coefficient cancellation.

But the defect gradient already singles out e_1 , so such an equivalence is additional, not intrinsic.

LXXXI. GENERIC NONVANISHING IN A FINITE-DIMENSIONAL MICROSCOPIC FAMILY

We now formulate the precise genericity statement.

Fix one point ξ_0 . Assume that

$$c \mapsto u_3(\xi_0; c)$$

is a polynomial, affine, or real-analytic function on $\mathcal{P}$. Define the bad set

$$\mathcal{Z}_{\xi_0} := \{c \in \mathcal{P} : u_3(\xi_0; c) = 0\}.$$

[Generic nonvanishing theorem for u_3] Assume there exists at least one admissible benchmark parameter

$$c^{\text{bench}} \in \mathcal{P}$$

such that

$$u_3(\xi_0; c^{\text{bench}}) \neq 0$$

for some ξ_0 . Then:

1. $u_3(\xi_0; \cdot)$ is not identically zero on $\mathcal{P}$;
2. the bad set $\mathcal{Z}_{\xi_0}$ is a proper algebraic/analytic subset of $\mathcal{P}$;
3. therefore its complement

$$\mathcal{U}_{\xi_0} := \mathcal{P} \setminus \mathcal{Z}_{\xi_0}$$

is open and dense, and full measure whenever $\mathcal{P}$ is equipped with the natural Lebesgue-class measure;

4. hence $u_3(\xi) \not\equiv 0$ is generic in the actual microscopic family.

Since $u_3(\xi_0; \cdot)$ is polynomial/analytic and nonzero at c^{bench} , it is not the zero function. Therefore its zero locus cannot contain any open set. Hence

$$\mathcal{Z}_{\xi_0}$$

is a proper algebraic/analytic subset, and its complement is open dense. The last statement follows because if $u_3(\xi_0; c) \neq 0$ at one point ξ_0 , then $u_3(\xi; c)$ cannot vanish identically in ξ .

a. Meaning. So once one admissible anisotropic benchmark exists inside the same microscopic operator class,

$$u_3(\xi) \equiv 0$$

is no longer a natural outcome. It becomes a nongeneric cancellation condition.

LXXXII. EXPLICIT BENCHMARK

The explicit seed representative provides exactly such a benchmark. There,

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3,$$

hence

$$u_3(\xi) = s_3(\xi) = \frac{1}{2} \theta'(\xi)^2.$$

Therefore, for a nontrivial defect profile,

$$u_3(\xi) \not\equiv 0.$$

Moreover, the parity-correct longitudinal kernel is

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3,$$

and the overlap is

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) u_3(\xi) = \frac{1}{2} I_{\theta'^3} \neq 0.$$

Thus the benchmark is not merely anisotropic. It already lies in the longitudinal-good region:

$$u_3(\xi) \not\equiv 0, \quad J_{\parallel}^{(3)} \neq 0.$$

LXXXIII. INDEPENDENT STRUCTURAL SUPPORT

The same diagonal anisotropy pattern also appears in the explicit local projector-Hessian framework. In the adapted frame, the tangent-space static Hessian is channel-diagonal and distinguishes the two internal directions:

$$H_B^{\text{stat}} = -\rho \partial_\xi^2 + \begin{pmatrix} \lambda \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2 & 0 \\ 0 & \lambda - \rho \theta'(\xi)^2 \end{pmatrix}.$$

So the two channels are already dynamically inequivalent in the explicit effective local model. This strongly supports the claim that diagonal anisotropy on Q_B is natural rather than exceptional.

LXXXIV. SHARP HONEST CONCLUSION

We can now state the strongest honest result of this step.

[Generic nonvanishing of the longitudinal singlet coefficient] Let $R_\xi(c)$ be a finite-dimensional symmetry-allowed family of Hermitian quadratic Q_B -responses. Assume:

1. no additional residual symmetry exchanges e_1 and e_2 ;
2. the family contains at least one admissible benchmark c^{bench} with

$$u_3(\xi_0; c^{\text{bench}}) \neq 0$$

for some ξ_0 .

Then

$$u_3(\xi) \equiv 0$$

is nongeneric: it defines at most a proper algebraic/analytic subset of the microscopic coefficient space. Equivalently,

$$\boxed{u_3(\xi) \neq 0 \text{ is the generic situation.}}$$

a. Honest frontier. What still remains open is not the genericity statement itself. The remaining microscopic task is the identification of the actual complete TECT coefficient point:

$$c_{\text{TECT}} \in \mathcal{P}.$$

One must still check that the actual full quadratic response of complete TECT lies outside the bad subset and also satisfies

$$J_{\parallel}^{(3)} \neq 0.$$

So the exact remaining longitudinal audit is:

$$\boxed{c_{\text{TECT}} \notin \{u_3 \equiv 0\} \quad \text{and} \quad J_{\parallel}^{(3)}(c_{\text{TECT}}) \neq 0.}$$

LXXXV. FINAL SUMMARY OF THE GENERIC NONVANISHING STEP FOR $s_3(\xi)$

a. Closed result. Let

$$R_\xi(c) = u_0(\xi; c)\mathbf{1}_2 + u_1(\xi; c)\sigma_1 + u_2(\xi; c)\sigma_2 + u_3(\xi; c)\sigma_3$$

be a finite-dimensional symmetry-allowed family of quadratic Q_B -responses. Then

$$u_3(\xi; c) = \frac{1}{2} \left(\langle e_1, R_\xi(c)e_1 \rangle - \langle e_2, R_\xi(c)e_2 \rangle \right).$$

So

$$u_3(\xi) \equiv 0$$

means exact diagonal isotropy between the defect-selected channel e_1 and the spectator channel e_2 .

b. What can force $u_3 \equiv 0$. Such a vanishing can be forced only by:

- (i) an extra $e_1 \leftrightarrow e_2$ exchange symmetry, (ii) exact isotropy, or (iii) fine tuning.

It is *not* forced by the basic defect/shell residual symmetry.

c. Genericity theorem. If the microscopic family contains one admissible benchmark with

$$u_3(\xi_0) \neq 0$$

for some ξ_0 , then the bad set

$$\{c : u_3(\xi_0; c) = 0\}$$

is a proper algebraic/analytic subset. Hence

$$\boxed{u_3(\xi) \neq 0 \text{ is generic.}}$$

d. Explicit benchmark. In the explicit seed representative,

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3,$$

so

$$u_3(\xi) = \frac{1}{2} \theta'(\xi)^2 \neq 0.$$

After the odd upgrade,

$$V_{(\parallel)}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3,$$

and

$$J_{\parallel}^{(3)} = \frac{1}{2} I_{\theta'^3} \neq 0.$$

e. Independent support. The explicit projector-Hessian model already distinguishes the two Q_B -channels dynamically, so diagonal anisotropy is natural rather than exceptional.

f. Honest frontier. Therefore the remaining microscopic question is not whether $u_3 \equiv 0$ is generically possible. That is already settled.

The actual remaining task is to identify the true complete TECT coefficient point

$$c_{\text{TECT}}$$

and verify that

$$\boxed{u_3(\xi; c_{\text{TECT}}) \neq 0 \quad \text{and} \quad J_{\parallel}^{(3)}(c_{\text{TECT}}) \neq 0.}$$

LXXXVI. GENERIC NONVANISHING THEOREM FOR THE LONGITUDINAL OVERLAP FUNCTIONAL $J_{\parallel}^{(3)}$

a. Goal. We now close the last general genericity step on the longitudinal side. Previously we reduced the primitive longitudinal survival problem to the finite test

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} \neq 0 \quad \text{and} \quad J_{\parallel}^{(3)} \neq 0.$$

The remaining question is therefore: when is

$$J_{\parallel}^{(3)}(c) = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) u_3(\xi; c)$$

generically nonzero in a symmetry-allowed microscopic family?

LXXXVII. DEFINITION OF THE LONGITUDINAL OVERLAP FUNCTIONAL

Let

$$R_\xi(c) = u_0(\xi; c)\mathbf{1}_2 + u_1(\xi; c)\sigma_1 + u_2(\xi; c)\sigma_2 + u_3(\xi; c)\sigma_3$$

be a finite-dimensional symmetry-allowed family of Hermitian quadratic Q_B -responses, depending on microscopic coefficients

$$c = (c_1, \dots, c_m) \in \mathcal{P}.$$

The parity-correct primitive longitudinal transport kernel is

$$V_{(\parallel)}^\mu(\xi; c) = \theta'(\xi) u_3(\xi; c) n^\mu \sigma_3.$$

Define the longitudinal overlap functional

$$J_{\parallel}^{(3)}(c) := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) u_3(\xi; c). \quad (53)$$

This is exactly the projected longitudinal witness up to a fixed nonzero normalization factor. Equivalently,

$$\lambda_{\parallel}(c) \propto J_{\parallel}^{(3)}(c).$$

LXXXVIII. LINEARITY / ANALYTICITY OF $J_{\parallel}^{(3)}$

[Functional character of $J_{\parallel}^{(3)}$] Assume $u_3(\xi; c)$ depends linearly on the microscopic tensor/response coefficients c , or more generally real-analytically on c . Then $J_{\parallel}^{(3)}(c)$ is respectively:

1. a linear functional on the finite-dimensional microscopic coefficient space $\mathcal{P}$, or
2. a real-analytic scalar function on $\mathcal{P}$.

The map

$$u_3(\cdot; c) \mapsto \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) u_3(\xi; c)$$

is linear in u_3 . Therefore, if u_3 depends linearly on c , then so does $J_{\parallel}^{(3)}$.

More generally, if $u_3(\xi; c)$ is real-analytic in c for each ξ , then the weighted integral against the fixed profile

$$f_+(\xi) f_-(\xi) \theta'(\xi)$$

preserves real-analytic dependence. Hence $J_{\parallel}^{(3)}(c)$ is real-analytic.

a. Interpretation. So $J_{\parallel}^{(3)}$ is not a mysterious nonlocal quantity. It is a finite-dimensional witness functional of exactly the same kind as the shell-level longitudinal witness $v_{\parallel}$ in the earlier genericity theorems.

LXXXIX. GENERIC NONVANISHING THEOREM

Fix the bad set

$$\mathcal{Z}_J := \{c \in \mathcal{P} : J_{\parallel}^{(3)}(c) = 0\}.$$

[Generic nonvanishing of the longitudinal overlap] Assume there exists at least one admissible benchmark point

$$c^{\text{bench}} \in \mathcal{P}$$

such that

$$J_{\parallel}^{(3)}(c^{\text{bench}}) \neq 0.$$

Then:

1. $J_{\parallel}^{(3)}$ is not identically zero on $\mathcal{P}$;
2. if $J_{\parallel}^{(3)}$ is linear, then

$$\mathcal{Z}_J = \ker J_{\parallel}^{(3)}$$

is a proper hyperplane in $\mathcal{P}$;

3. if $J_{\parallel}^{(3)}$ is real-analytic, then $\mathcal{Z}_J$ is a proper analytic subset of $\mathcal{P}$;
4. in either case, the good set

$$\mathcal{U}_J := \mathcal{P} \setminus \mathcal{Z}_J$$

is open and dense, and full measure in the linear case (and also Lebesgue-null complement in the standard analytic finite-dimensional setting).

Therefore

$$\boxed{J_{\parallel}^{(3)}(c) \neq 0 \text{ is generic once one nonzero benchmark exists.}} \quad (54)$$

Since

$$J_{\parallel}^{(3)}(c^{\text{bench}}) \neq 0,$$

the function $J_{\parallel}^{(3)}$ is not identically zero.

If it is linear, then its zero set is the kernel of a nonzero linear functional on a finite-dimensional space, hence a proper hyperplane.

If it is real-analytic, then the zero set of a nontrivial analytic function is a proper analytic subset and cannot contain an open set.

Thus in either case the complement is open dense. In the linear case the bad set has codimension one and measure zero. This proves the claim.

XC. EXPLICIT BENCHMARK FROM THE SEED REPRESENTATIVE

The explicit seed representative already provides a benchmark point in the good set. Indeed,

$$S_{\xi} = (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3,$$

so

$$u_3(\xi) = \frac{1}{2} \theta'(\xi)^2.$$

Hence

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) u_3(\xi) = \frac{1}{2} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3 = \frac{1}{2} I_{\theta'^3}. \quad (55)$$

Using the explicit seed evaluation,

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

we obtain

$$J_{\parallel}^{(3)} = -\frac{3\sqrt{3}\pi}{512} \theta_0^3 \kappa^3.$$

Therefore for a nontrivial defect profile,

$$\boxed{J_{\parallel}^{(3)} \neq 0.}$$

So the benchmark hypothesis of the generic nonvanishing theorem is satisfied inside the explicit seed class.

XCI. RELATION TO THE FINITE WITNESS PRINCIPLE

The earlier finite witness principle states:

1. if a linear witness vanishes identically on the allowed microscopic space, then the channel is symmetry-forbidden;
2. if it is nonzero on one admissible seed, then it is nonzero on an open dense full-measure subset.

The present longitudinal overlap theorem is exactly the same logic applied to

$$J_{\parallel}^{(3)}.$$

Thus the microscopic content of the longitudinal problem is reduced to one yes/no statement:

$$\boxed{J_{\parallel}^{(3)}|_{\mathcal{P}} \neq 0 ?} \tag{56}$$

If the answer is yes, then longitudinal overlap nonvanishing is generic. If the answer is no, then the obstruction is an exact symmetry selection rule or an exact structural cancellation in the microscopic family.

XCII. SHARP HONEST CONCLUSION

We can now state the strongest honest theorem of this step.

[Generic longitudinal-overlap theorem] Let $\mathcal{P}$ be any finite-dimensional symmetry-allowed microscopic parameter space for the actual quadratic Q_B -response, and define

$$J_{\parallel}^{(3)}(c) = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) u_3(\xi; c).$$

Assume:

1. $u_3(\xi; c)$ depends linearly or real-analytically on c ;
2. there exists one admissible benchmark $c^{\text{bench}} \in \mathcal{P}$ such that

$$J_{\parallel}^{(3)}(c^{\text{bench}}) \neq 0.$$

Then

$$J_{\parallel}^{(3)}(c) \neq 0$$

holds on an open dense subset of $\mathcal{P}$, and in the linear case on a full-measure subset. Equivalently,

$$\boxed{J_{\parallel}^{(3)} = 0 \text{ is nongeneric once one nonzero benchmark exists.}}$$

a. What remains open. The remaining microscopic task is no longer a genericity question. It is the identification of the actual complete TECT coefficient point

$$c_{\text{TECT}} \in \mathcal{P}$$

and the verification that it lies in the good set:

$$\boxed{u_3(\xi; c_{\text{TECT}}) \neq 0 \quad \text{and} \quad J_{\parallel}^{(3)}(c_{\text{TECT}}) \neq 0.}$$

So the longitudinal frontier has now been reduced to a concrete coefficient-identification problem for the actual full microscopic TECT linearization.

XCIII. FINAL SUMMARY OF THE GENERIC NONVANISHING THEOREM FOR $J_{\parallel}^{(3)}$

a. Closed result. Define the longitudinal overlap functional

$$J_{\parallel}^{(3)}(c) = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) u_3(\xi; c),$$

where $u_3(\xi; c)$ is the σ_3 -coefficient of the actual quadratic Q_B -response.

If $u_3(\xi; c)$ depends linearly (or more generally real-analytically) on the microscopic coefficients c , then $J_{\parallel}^{(3)}(c)$ is a linear (respectively analytic) witness functional on the finite-dimensional microscopic parameter space.

b. Genericity theorem. If there exists one admissible benchmark point c^{bench} such that

$$J_{\parallel}^{(3)}(c^{\text{bench}}) \neq 0,$$

then

$$J_{\parallel}^{(3)} \not\equiv 0.$$

Hence its zero set is:

a proper hyperplane, if $J_{\parallel}^{(3)}$ is linear,

or

a proper analytic subset, if $J_{\parallel}^{(3)}$ is analytic.

Therefore

$$\boxed{J_{\parallel}^{(3)}(c) \neq 0 \text{ is generic.}}$$

c. Explicit benchmark. In the explicit seed representative,

$$S_{\xi} = (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3,$$

so

$$u_3(\xi) = \frac{1}{2} \theta'(\xi)^2$$

and therefore

$$J_{\parallel}^{(3)} = \frac{1}{2} I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{512} \theta_0^3 \kappa^3 \neq 0$$

for a nontrivial defect profile.

d. Meaning. So once one nonzero benchmark exists, vanishing of the longitudinal overlap is no longer the natural outcome. It becomes a nongeneric cancellation condition.

e. Honest frontier. The remaining microscopic task is now reduced to the identification of the actual full TECT coefficient point

$$c_{\text{TECT}}$$

and the verification that

$$\boxed{u_3(\xi; c_{\text{TECT}}) \not\equiv 0 \quad \text{and} \quad J_{\parallel}^{(3)}(c_{\text{TECT}}) \neq 0.}$$

XCIV. ACTUAL COEFFICIENT-IDENTIFICATION REDUCTION THEOREM

a. Goal. We now leave the genericity layer and isolate the exact microscopic quantity that must be extracted from the complete TECT linearization in order to determine the actual longitudinal coefficient. The aim is not to overclaim that the coefficient has already been computed. Rather, the goal is to prove that the full coefficient-identification problem reduces to the extraction of one scalar profile function $h_{\parallel}(\theta)$.

XCV. ACTUAL LOCAL OVERLAP FORMULATION

The strongest honest local statement already established in the project is that the first-order projected transport kernel may be written in the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

with

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i.$$

Projecting onto the even/odd low-mode pair (f_+, f_-) gives

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi).$$

In the singlet $\oplus$ doublet $S2$ decomposition, the shell-compatible reduced symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

Accordingly, the actual local overlap formulas are

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)), \quad (57)$$

$$\alpha = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))), \quad (58)$$

$$\beta = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))). \quad (59)$$

Thus the longitudinal coefficient in the actual microscopic theory is controlled by a *single scalar profile function*

$$h_{\parallel}(\theta).$$

XCVI. REDUCTION TO A SINGLE LONGITUDINAL PROFILE FUNCTION

[Actual longitudinal coefficient-identification reduction] Assume the complete microscopic TECT linearization admits the local first-order transport form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

with shell-compatible singlet $\oplus$ doublet decomposition. Then the actual longitudinal witness coefficient is completely determined by the single scalar profile

$$h_{\parallel}(\theta),$$

through the overlap formula

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)).$$

In particular, the full microscopic longitudinal coefficient-identification problem is reduced to extracting $h_{\parallel}(\theta)$ from the complete TECT linearized transport operator.

By shell symmetry, the reduced symbol has exactly one longitudinal singlet channel:

$$p_{\parallel} \sigma_3.$$

Therefore the projected first-order operator can contribute to the longitudinal witness only through the coefficient multiplying this unique singlet channel. By the established actual local overlap formulas, that coefficient is

$$\lambda_{\parallel} = \int d\xi f_+ f_- \theta'(\xi) h_{\parallel}(\theta(\xi)).$$

Hence once $h_{\parallel}(\theta)$ is known, $\lambda_{\parallel}$ is known. No additional microscopic data are needed for the longitudinal coefficient itself.

a. Interpretation. So the longitudinal microscopic frontier is not:

“solve the whole PDE and diagonalize everything.”

It is:

derive $h_{\parallel}(\theta)$ from the actual full TECT linearization.

XCVII. MATCHING TO THE CONCRETE SEED NORMALIZATION

Inside the concrete seed transport ansatz, one has the explicit witness normalization

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3.$$

Matching this with the actual local formula (57) gives the exact coefficient-identification relation

$$g_c^{\text{act}} = -\frac{512}{3\sqrt{3}\pi \theta_0^3 \kappa^3} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)). \quad (60)$$

This is the actual longitudinal coefficient extraction formula.

[Exact actual-longitudinal matching formula] Once the function $h_{\parallel}(\theta)$ is extracted from the complete microscopic TECT linearization, the actual longitudinal shell coefficient is obtained uniquely by (60).

XCVIII. EQUIVALENT TENSOR-CONTRACTION FORM

At the shell-tensor level, the local Weyl coefficient formulas read

$$\lambda_{\parallel} = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3,$$

$$\alpha = \frac{\kappa_* |G_*|}{2} (e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2),$$

$$\beta = \frac{\kappa_* |G_*|}{2} (e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2).$$

Therefore the same longitudinal coefficient-identification problem may also be phrased as:

extract the actual tensor N_{ij}^3 from the full TECT linearization and contract it with $\hat{n}_i \hat{n}_j$.

The $h_{\parallel}(\theta)$ -formulation and the N_{ij}^3 -contraction formulation are equivalent descriptions of the same actual microscopic coefficient-identification step.

XCIX. WHAT IS AND IS NOT CLOSED HERE

[Sharp honest status of actual coefficient-identification] At the present stage, the following statement is rigorously closed:

The actual longitudinal coefficient-identification problem is reduced to the extraction of $h_{\parallel}(\theta)$, (61)

or equivalently the extraction of the shell tensor contraction

$$\hat{n}_i \hat{n}_j N_{ij}^3.$$

However, the following statement is *not yet* closed:

the actual full microscopic TECT linearization has already yielded the explicit function $h_{\parallel}(\theta)$.

The reduction statement follows from the established actual local overlap formulas and the shell-compatible singlet $\oplus$ doublet decomposition. The non-closure statement follows because the existing project documents explicitly state that the actual microscopic values of

$$g_c, \quad g_E, \quad g_O$$

and the final full microscopic projector/condensate PDE remain open. Thus $h_{\parallel}(\theta)$ has not yet been derived from the final complete TECT linearized projector equation.

C. PRACTICAL EXTRACTION PROTOCOL

The actual longitudinal coefficient-identification step is therefore the following finite protocol:

- a. (C1) Write the full microscopic TECT linearization in the local projector/defect frame.
- b. (C2) Extract the first-order transport operator $V^{\mu}(\xi)$.
- c. (C3) Decompose it as

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)).$$

- d. (C4) Project $\mathcal{M}^{\mu}(\theta)$ onto the shell-compatible singlet channel to identify $h_{\parallel}(\theta)$.
- e. (C5) Insert $h_{\parallel}(\theta)$ into

$$\lambda_{\parallel} = \int d\xi f_+ f_- \theta'(\xi) h_{\parallel}(\theta(\xi)),$$

and then into

$$g_c^{\text{act}} = -\frac{512}{3\sqrt{3}\pi\theta_0^3\kappa^3} \int d\xi f_+ f_- \theta'(\xi) h_{\parallel}(\theta(\xi)).$$

Thus the actual coefficient-identification problem has become an executable finite extraction protocol.

CI. SHARP HONEST CONCLUSION

The strongest honest conclusion of this step is:

Actual coefficient-identification is no longer an infinite-dimensional mystery.

It is exactly the extraction of $h_{\parallel}(\theta)$ (or equivalently $\hat{n}_i \hat{n}_j N_{ij}^3$).

What remains open is not the logical reduction, but only the actual derivation of $h_{\parallel}(\theta)$ from the final complete TECT microscopic linearization.

CII. FINAL SUMMARY OF THE ACTUAL COEFFICIENT-IDENTIFICATION STEP

- a. *Closed result.* The actual longitudinal coefficient-identification problem is reduced to one scalar profile function:

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)).$$

Equivalently, at shell-tensor level,

$$\lambda_{\parallel} = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3.$$

So the full microscopic longitudinal problem is no longer:

“solve the whole PDE completely.”

It is:

$$\text{extract } h_{\parallel}(\theta) \text{ or equivalently } \hat{n}_i \hat{n}_j N_{ij}^3 \text{ from the actual full TECT linearization.}$$

b. Exact matching to the concrete seed normalization. Using

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3,$$

the actual longitudinal coefficient is

$$g_c^{\text{act}} = -\frac{512}{3\sqrt{3}\pi \theta_0^3 \kappa^3} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)).$$

c. What is not yet closed. It is still *not* proved that the final complete microscopic TECT linearization has already yielded the explicit function $h_{\parallel}(\theta)$. So the current status is:

logical reduction is closed,

actual coefficient extraction remains open.

d. Executable protocol. The remaining actual microscopic task is:

$$\text{full TECT linearization} \rightarrow V^{\mu}(\xi) \rightarrow \mathcal{M}^{\mu}(\theta) \rightarrow h_{\parallel}(\theta) \rightarrow \lambda_{\parallel} \rightarrow g_c^{\text{act}}.$$

CIII. LOCAL-STRUCTURE CLASSIFICATION THEOREM FOR THE ACTUAL LONGITUDINAL COEFFICIENT $h_{\parallel}(\theta)$

a. Goal. We now identify exactly which local tensor structures inside the actual first-order transport matrix

$$\mathcal{M}^{\mu}(\theta)$$

can contribute to the longitudinal shell-reduced coefficient

$$h_{\parallel}(\theta).$$

This is the last structural reduction before any explicit full-TECT coefficient extraction.

CIV. ACTUAL LOCAL FIRST-ORDER FORM

Assume the actual complete TECT linearization admits the local first-order transport structure

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)), \quad (62)$$

where $\mathcal{M}^{\mu}(\theta)$ is a 2×2 Hermitian matrix-valued worldvolume vector in the shell-reduced low sector.

In the Pauli basis, write

$$\mathcal{M}^{\mu}(\theta) = a_0^{\mu}(\theta) \mathbf{1}_2 + a_1^{\mu}(\theta) \sigma_1 + a_2^{\mu}(\theta) \sigma_2 + a_3^{\mu}(\theta) \sigma_3. \quad (63)$$

Choose the shell-adapted worldvolume frame

$$\{n^{\mu}, e_1^{\mu}, e_2^{\mu}\},$$

and decompose the coefficient vectors as

$$a_0^{\mu}(\theta) = u_0^{\parallel}(\theta) n^{\mu} + u_0^1(\theta) e_1^{\mu} + u_0^2(\theta) e_2^{\mu}, \quad (64)$$

$$a_3^{\mu}(\theta) = u_3^{\parallel}(\theta) n^{\mu} + u_3^1(\theta) e_1^{\mu} + u_3^2(\theta) e_2^{\mu}, \quad (65)$$

$$(a_1^{\mu}(\theta), a_2^{\mu}(\theta)) = \text{general transverse-doublet data.} \quad (66)$$

CV. INVARIANT SHELL-REDUCED DECOMPOSITION

Under the residual shell symmetry $H_\perp \cong SO(2)_\perp$, the worldvolume sector and the traceless internal sector both split as

$$1 \oplus 2.$$

Hence the shell-compatible invariant first-order structures are exactly:

1. the identity singlet channel

$$n^\mu \mathbf{1}_2;$$

2. the longitudinal Pauli singlet channel

$$n^\mu \sigma_3;$$

3. the transverse-even channel

$$E^\mu{}_a \sigma_a;$$

4. the transverse-odd channel

$$J^\mu{}_a \sigma_a.$$

Therefore the invariant part of $\mathcal{M}^\mu(\theta)$ has the unique decomposition

$$\boxed{\mathcal{M}_{\text{inv}}^\mu(\theta) = h_0(\theta) n^\mu \mathbf{1}_2 + h_\parallel(\theta) n^\mu \sigma_3 + h_E(\theta) E^\mu{}_a \sigma_a + h_O(\theta) J^\mu{}_a \sigma_a.} \quad (67)$$

All other local structures are non-invariant with respect to the shell-compatible $S2$ branch and project out of the reduced witness problem.

CVI. WHICH STRUCTURES CONTRIBUTE TO $h_\parallel(\theta)$

[Local-structure classification theorem] Let $\mathcal{M}^\mu(\theta)$ be decomposed as in (67). Then:

1. the longitudinal coefficient is exactly the coefficient of the unique invariant singlet-Pauli structure:

$$\boxed{h_\parallel(\theta) = \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(\theta)];}$$

2. the identity singlet term $h_0(\theta) n^\mu \mathbf{1}_2$ does not contribute to $h_\parallel$;
3. the transverse-even and transverse-odd terms contribute only to α, β , not to $h_\parallel$;
4. any mixed non-invariant structures such as

$$n^\mu \sigma_1, \quad n^\mu \sigma_2, \quad e_a^\mu \sigma_3, \quad e_a^\mu \mathbf{1}_2$$

have zero projection onto the shell-compatible longitudinal witness.

From (67),

$$n_\mu \mathcal{M}_{\text{inv}}^\mu(\theta) = h_0(\theta) \mathbf{1}_2 + h_\parallel(\theta) \sigma_3,$$

because

$$n_\mu E^\mu{}_a = 0, \quad n_\mu J^\mu{}_a = 0.$$

Multiply by σ_3 and take the normalized trace:

$$\frac{1}{2}\text{Tr}[\sigma_3 n_\mu \mathcal{M}_{\text{inv}}^\mu(\theta)] = \frac{1}{2}\text{Tr}[h_0(\theta)\sigma_3 + h_\parallel(\theta)\sigma_3^2].$$

Since

$$\text{Tr}(\sigma_3) = 0, \quad \frac{1}{2}\text{Tr}(\sigma_3^2) = 1,$$

this becomes

$$\frac{1}{2}\text{Tr}[\sigma_3 n_\mu \mathcal{M}_{\text{inv}}^\mu(\theta)] = h_\parallel(\theta).$$

This proves item (1).

Item (2) follows from $\text{Tr}(\sigma_3 \mathbf{1}_2) = 0$.

Item (3) follows because the transverse structures are orthogonal to n^μ , so they vanish under contraction with n_μ .

Item (4) follows from the representation decomposition: they lie in $\text{Hom}(1, 2)$ or $\text{Hom}(2, 1)$, or in the wrong $H_\perp$ -channel, and therefore project out of the shell-compatible longitudinal invariant branch.

CVII. EXPLICIT COEFFICIENT DICTIONARY

From the previous theorem, the actual local coefficient dictionary is:

$$\boxed{h_\parallel(\theta) = \frac{1}{2}\text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(\theta)]}, \quad (68)$$

$$\boxed{h_0(\theta) = \frac{1}{2}\text{Tr}[n_\mu \mathcal{M}^\mu(\theta)]}, \quad (69)$$

and the transverse coefficients are extracted by contraction with the even/odd doublet projectors:

$$\boxed{h_E(\theta) = \frac{1}{2}E_{\mu a} \text{Tr}[\sigma_a \mathcal{M}^\mu(\theta)], \quad h_O(\theta) = \frac{1}{2}J_{\mu a} \text{Tr}[\sigma_a \mathcal{M}^\mu(\theta)]}. \quad (70)$$

Thus the actual shell witness formulas become

$$\lambda_\parallel = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_\parallel(\theta(\xi)), \quad (71)$$

$$\alpha = \int d\xi f_+ f_- \theta'(\xi) h_E(\theta(\xi)), \quad \beta = \int d\xi f_+ f_- \theta'(\xi) h_O(\theta(\xi)). \quad (72)$$

CVIII. CLASSIFICATION TABLE

The classification of local structures is therefore:

local structure in $\mathcal{M}^\mu(\theta)$	contribution
$n^\mu \mathbf{1}_2$	h_0 only (scalar drift)
$n^\mu \sigma_3$	$h_\parallel$ only
$E_a^\mu \sigma_a$	h_E only
$J_a^\mu \sigma_a$	h_O only
$n^\mu \sigma_{1,2}$	0 in longitudinal S2 witness
$e_a^\mu \sigma_3$	0 in longitudinal S2 witness
$e_a^\mu \mathbf{1}_2$	0 in longitudinal S2 witness

So only one local structure matters for the actual longitudinal coefficient:

$$\boxed{n^\mu \sigma_3}.$$

CIX. MATCHING TO THE CONCRETE g_c -NORMALIZATION

Inside the concrete seed/Class II normalization,

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3.$$

Matching with

$$\lambda_{\parallel} = \int d\xi f_+ f_- \theta'(\xi) h_{\parallel}(\theta(\xi))$$

gives

$$g_c^{\text{act}} = -\frac{512}{3\sqrt{3}\pi \theta_0^3 \kappa^3} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)). \quad (73)$$

Therefore the entire actual longitudinal coefficient-identification problem is reduced to reading off the coefficient of $n^{\mu} \sigma_3$ in $\mathcal{M}^{\mu}(\theta)$.

CX. SHARP HONEST CONCLUSION

We can now state the strongest honest theorem of this step.

[Sharp local classification for the actual longitudinal coefficient] In the actual shell-compatible first-order local transport operator

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)),$$

the actual longitudinal coefficient $h_{\parallel}(\theta)$ is contributed by one and only one invariant local tensor structure:

$$n^{\mu} \sigma_3.$$

Equivalently,

$$h_{\parallel}(\theta) = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(\theta)].$$

All other local structures either contribute to the scalar drift channel, to the transverse (α, β) channels, or project out of the shell-compatible S_2 longitudinal witness.

a. Meaning. The remaining actual microscopic problem is therefore no longer vague. It is precisely:

$$\boxed{\text{extract the coefficient of } n^{\mu} \sigma_3 \text{ from the full TECT local transport matrix } \mathcal{M}^{\mu}(\theta).}$$

CXI. FINAL SUMMARY OF THE LOCAL-STRUCTURE CLASSIFICATION FOR $h_{\parallel}(\theta)$

a. Closed result. In the actual shell-compatible local transport matrix

$$\mathcal{M}^{\mu}(\theta),$$

the invariant decomposition is

$$\mathcal{M}_{\text{inv}}^{\mu}(\theta) = h_0(\theta) n^{\mu} \mathbf{1}_2 + h_{\parallel}(\theta) n^{\mu} \sigma_3 + h_E(\theta) E^{\mu}_a \sigma_a + h_O(\theta) J^{\mu}_a \sigma_a.$$

b. Key theorem. The actual longitudinal coefficient is contributed by one and only one local structure:

$$\boxed{n^{\mu} \sigma_3.}$$

Equivalently,

$$\boxed{h_{\parallel}(\theta) = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(\theta)].}$$

c. *Non-contributing structures.* The following do *not* contribute to $h_{\parallel}$:

$$n^{\mu} \mathbf{1}_2, \quad E_a^{\mu} \sigma_a, \quad J_a^{\mu} \sigma_a, \quad n^{\mu} \sigma_{1,2}, \quad e_a^{\mu} \sigma_3, \quad e_a^{\mu} \mathbf{1}_2.$$

They either belong to the scalar drift channel, the transverse (α, β) sector, or project out of the shell-compatible longitudinal witness.

d. *Actual extraction formula.* Hence

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)),$$

and matching to the concrete seed normalization gives

$$g_c^{\text{act}} = -\frac{512}{3\sqrt{3}\pi \theta_0^3 \kappa^3} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)).$$

e. *Honest frontier.* So the remaining actual microscopic problem is now completely localized:

$$\text{extract the coefficient of } n^{\mu} \sigma_3 \text{ from the full TECT local transport matrix } \mathcal{M}^{\mu}(\theta).$$

Nothing else contributes to the longitudinal coefficient.

CXII. MICROSCOPIC SOURCE CLASSIFICATION THEOREM FOR THE ACTUAL LONGITUDINAL COEFFICIENT $h_{\parallel}$

a. *Goal.* We classify, source by source, which microscopic structures can contribute to the actual local longitudinal coefficient

$$h_{\parallel}(\theta) = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(\theta)].$$

The purpose is to separate:

1. source classes that are rigorously excluded from contributing to $h_{\parallel}$,
2. source classes that can contribute only indirectly,
3. source classes that provide an explicit longitudinal prototype,
4. source classes that are admissible in principle but not yet extracted from the full complete TECT linearization.

CXIII. LOCAL SOURCE DECOMPOSITION

Let the actual shell-compatible local transport matrix be decomposed as

$$\mathcal{M}^{\mu} = \mathcal{M}_{\text{drift}}^{\mu} + \mathcal{M}_{\text{trans}}^{\mu} + \mathcal{M}_{\text{comm}^2}^{\mu} + \mathcal{M}_{\text{stiff,ind}}^{\mu} + \mathcal{M}_{\text{med}}^{\mu} + \mathcal{M}_{\text{unclosed}}^{\mu},$$

where:

1. $\mathcal{M}_{\text{drift}}^{\mu}$ is the identity-valued scalar drift sector;
2. $\mathcal{M}_{\text{trans}}^{\mu}$ is the explicit transverse odd-gradient doublet source;
3. $\mathcal{M}_{\text{comm}^2}^{\mu}$ is the primitive commutator-square longitudinal source;
4. $\mathcal{M}_{\text{stiff,ind}}^{\mu}$ is the source induced only after integrating out heavy projector modes;
5. $\mathcal{M}_{\text{med}}^{\mu}$ denotes admissible mixed (i, a) -carrier sources of Class I/II/III;
6. $\mathcal{M}_{\text{unclosed}}^{\mu}$ denotes possible additional kinetic/quartic full-TECT contributions not yet theorem-level extracted.

Then

$$h_{\parallel} = h_{\parallel}^{\text{drift}} + h_{\parallel}^{\text{trans}} + h_{\parallel}^{\text{comm}^2} + h_{\parallel}^{\text{stiff,ind}} + h_{\parallel}^{\text{med}} + h_{\parallel}^{\text{unclosed}},$$

with

$$h_{\parallel}^{(S)} := \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}_{(S)}^{\mu}].$$

CXIV. CLASSES THAT CANNOT CONTRIBUTE DIRECTLY

[Direct exclusion theorem] The following source classes do not contribute directly to $h_{\parallel}$:

Type 0A (pure scalar potential), Type 0B (pure projector derivative), Type 0C (amplitude-only derivative).

Equivalently,

$$h_{\parallel}^{0A} = h_{\parallel}^{0B} = h_{\parallel}^{0C} = 0.$$

Any direct contribution to the shell-enriched longitudinal witness must contain:

1. one shell/background derivative factor,
2. one derivative on the test mode sector,
3. one internal triplet/tangent label,
4. and one structure projecting to the longitudinal invariant tensor $n^{\mu}\sigma_3$.

Type 0A, 0B, 0C terms fail this requirement: they do not produce a linear shell-projector/test mixing source with the required (i, a) -carrier. Hence they cannot contribute directly to the longitudinal coefficient.

a. Immediate corollary. Any direct microscopic source for $h_{\parallel}$ must belong to a genuinely mixed shell-internal transport class.

CXV. SCALAR DRIFT AND PURE TRANSVERSE SOURCES

[Drift/transverse decoupling] The following structures do not contribute to $h_{\parallel}$:

1. scalar drift terms of the form $n^{\mu}\mathbf{1}_2$;
2. pure transverse-even terms $E^{\mu}{}_a\sigma_a$;
3. pure transverse-odd terms $J^{\mu}{}_a\sigma_a$;
4. non-invariant mixed terms such as $n^{\mu}\sigma_{1,2}$, $e_a^{\mu}\sigma_3$, $e_a^{\mu}\mathbf{1}_2$.

Equivalently,

$$h_{\parallel}^{\text{drift}} = 0, \quad h_{\parallel}^{\text{trans}} = 0.$$

By the local invariant decomposition,

$$h_{\parallel} = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}].$$

Thus:

1. $n^{\mu}\mathbf{1}_2$ gives zero because $\text{Tr}(\sigma_3) = 0$;
2. $E^{\mu}{}_a\sigma_a$ and $J^{\mu}{}_a\sigma_a$ are worldvolume-transverse, so they vanish under contraction with n_{μ} ;
3. the remaining non-invariant structures project out of the shell-compatible $S2$ longitudinal branch.

Therefore only the $n^{\mu}\sigma_3$ structure can contribute to $h_{\parallel}$.

CXVI. PRIMITIVE COMMUTATOR-SQUARE SOURCE

Let

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B].$$

Then the direct Q_B -restriction vanishes:

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0.$$

However, the induced quadratic channel is

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \sigma_3).$$

Hence its traceless part is

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

[Primitive longitudinal source theorem] The primitive commutator-square source

$$\mathcal{M}_{\text{comm}^2}^\mu(\xi) = g_c (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu$$

contributes directly and purely to the longitudinal coefficient:

$$h_{\parallel}^{\text{comm}^2}(\xi) = \frac{1}{2} g_c \theta'(\xi)^2, \quad h_E^{\text{comm}^2} = h_O^{\text{comm}^2} = 0.$$

After multiplication by the required odd prefactor $\theta'(\xi)$, it produces the explicit longitudinal seed

$$V_{(\parallel)}^\mu(\xi) = \frac{1}{2} g_c \theta'(\xi)^3 n^\mu \sigma_3.$$

The formula for $(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}$ is explicit. Since the resulting operator is proportional to $n^\mu \sigma_3$, it contributes only to the longitudinal singlet coefficient. The extra odd factor $\theta'(\xi)$ is required so that the overlap with the even/odd low-mode pair does not vanish by parity.

CXVII. PROJECTOR-STIFFNESS SOURCE

Consider the stiffness term

$$F_{\text{align}} = \frac{\chi}{4} \int d^3x \rho^2 (\partial_i P \partial_i P).$$

[Projector-stiffness source classification] Around a locked constant-projector background $P_0 = \text{const}$, the projector stiffness term does not generically generate a new tree-level shell-enriched tensor. Therefore

$$h_{\parallel}^{\text{stiff, tree}} = 0.$$

Its contribution to $h_{\parallel}$ can arise only indirectly, after integrating out the heavy projector mode, provided an admissible shell-projector mixing source exists.

Since $\partial_i P_0 = 0$, the stiffness term vanishes on the background and begins only at quadratic order in projector fluctuations. Thus it contains no tree-level term linear in the shell gradient and linear in a test derivative, i.e. no direct source of the shell-enriched tensor. However, it gives the projector tangent mode a large positive quadratic operator; after Gaussian elimination, it can induce effective corrections to the witness data, but only indirectly.

a. Interpretation. So the stiffness sector is not a direct source of $h_{\parallel}$; it is a stabilizing/renormalizing sector.

CXVIII. ADMISSIBLE MEDIATED MIXED SOURCES

A direct mixed source must carry both a shell/worldvolume derivative index and an internal tangent label. This yields the minimal admissible mixed classes:

1. Class I: direct shell–projector derivative coupling;
2. Class II: auxiliary internal vector mediator;
3. Class III: connection/gauge-like mediator.

[Admissible mixed-source theorem] Each of Class I, II, III is symmetry-admissible as a potential source for $h_{\parallel}$, because each introduces the required (i, a) -carrier. Among them, Class II is currently the strongest explicit class.

A direct source for the shell-enriched tensor must couple:

1. shell/background derivative structure,
2. internal tangent/projector sector,
3. test-mode derivative sector,
4. shell Pauli label.

Classes I, II, III all satisfy this structural requirement. Class II is explicit because its canonical seed directions are already known:

$$c_{\parallel} \neq 0 \Rightarrow g_c \neq 0, \quad c_I \neq 0 \Rightarrow g_E \neq 0, \quad c_J \neq 0 \Rightarrow g_O \neq 0.$$

So Class II explicitly realizes all three shell witness directions, including the longitudinal one.

CXIX. UNCLOSED FULL-TECT KINETIC / QUARTIC SOURCES

There may exist additional direct contributions to $h_{\parallel}$ coming from the actual complete microscopic TECT kinetic/quartic linearization. However, at the present stage these are not theorem-level extracted in the accepted proof chain.

We therefore denote them collectively by

$$h_{\parallel}^{\text{unclosed}}.$$

[Honest unclosed-source clause] The existence of possible kinetic and quartic contributions to $h_{\parallel}$ in the full complete TECT linearization is not ruled out. But their explicit extraction is still open, because the final full microscopic coefficient functions have not yet been derived from the complete TECT projector/condensate linearization.

This is exactly the current honest status of the project: the actual local overlap formulas are known, but the explicit microscopic functions $h_{\parallel}, h_{ij}$ have not yet been extracted from the final full operator.

CXX. FINAL SOURCE FORMULA FOR $h_{\parallel}$

Collecting the classified contributions, the actual longitudinal coefficient has the source decomposition

$$h_{\parallel} = h_{\parallel}^{\text{comm}^2} + h_{\parallel}^{\text{med}} + h_{\parallel}^{\text{stiff,ind}} + h_{\parallel}^{\text{unclosed}},$$

with

$$h_{\parallel}^{0A} = h_{\parallel}^{0B} = h_{\parallel}^{0C} = 0, \quad h_{\parallel}^{\text{drift}} = 0, \quad h_{\parallel}^{\text{trans}} = 0, \quad h_{\parallel}^{\text{stiff,tree}} = 0.$$

Thus the only currently closed direct longitudinal prototype is the primitive commutator-square channel, while Class II provides the cleanest explicit admissible mediated realization.

CXXI. SHARP HONEST CONCLUSION

The strongest honest theorem of this step is:

[Microscopic source classification for $h_{\parallel}$] The actual longitudinal coefficient $h_{\parallel}$ receives no direct contribution from:

pure scalar terms, pure projector derivative terms, amplitude-only terms,

scalar drift terms, pure transverse sources, or tree-level projector stiffness on a locked constant-projector background.

Its currently closed direct prototype source is the primitive commutator-square channel

$$\theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu},$$

and its currently strongest explicit admissible mixed realization is Class II.

What remains open is the explicit extraction of any additional kinetic/quartic contributions from the full complete TECT linearization.

CXXII. FINAL SUMMARY OF THE MICROSCOPIC SOURCE CLASSIFICATION FOR $h_{\parallel}$

a. Closed classification. The actual longitudinal coefficient decomposes as

$$h_{\parallel} = h_{\parallel}^{\text{comm}^2} + h_{\parallel}^{\text{med}} + h_{\parallel}^{\text{stiff,ind}} + h_{\parallel}^{\text{unclosed}},$$

while the following direct contributions vanish:

$$h_{\parallel}^{0A} = h_{\parallel}^{0B} = h_{\parallel}^{0C} = 0, \quad h_{\parallel}^{\text{drift}} = 0, \quad h_{\parallel}^{\text{trans}} = 0, \quad h_{\parallel}^{\text{stiff,tree}} = 0.$$

b. Direct longitudinal prototype. The currently closed direct prototype for the longitudinal channel is the primitive commutator-square source:

$$\mathcal{T}_{\xi} = i[P_B, \partial_{\xi} P_B], \quad (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

Hence the parity-correct primitive transport kernel is

$$V_{(\parallel)}^{\mu}(\xi) = \frac{1}{2} g_c \theta'(\xi)^3 n^{\mu} \sigma_3,$$

which contributes purely to $h_{\parallel}$.

c. Indirect projector-stiffness role. The projector-stiffness term

$$F_{\text{align}} \propto \int \rho^2 (\partial P \partial P)$$

does not generically generate a new tree-level N_{ij}^a on a locked constant-projector background. Its effect on $h_{\parallel}$ can only be indirect, after integrating out the heavy projector mode.

d. Admissible mixed classes. Any direct mixed source for $h_{\parallel}$ must carry the required (i, a) -structure. This makes Classes I, II, III admissible in principle. Among them, Class II is currently the strongest explicit realization, because it provides canonical seed directions for

$$g_c, \quad g_E, \quad g_O.$$

e. Honest frontier. What remains open is the explicit extraction of any additional kinetic/quartic contribution from the full complete TECT linearization. So the current theorem-level status is:

the direct longitudinal prototype source is known,

but the full complete TECT source decomposition is not yet fully extra

CXXIII. ACTUAL LONGITUDINAL EXTRACTION PROTOCOL FROM THE FULL MICROSCOPIC LINEARIZATION

a. Goal. We now formulate the exact extraction protocol for the actual longitudinal coefficient. The purpose is to close the statement that, once the full microscopic TECT linearized operator is available, the longitudinal coefficient is obtained by a finite projector-and-trace computation.

CXXIV. MICROSCOPIC LINEARIZATION HYPOTHESIS

Let the full microscopic TECT condensate equation linearized around the condensed background be written in momentum space as

$$i\partial_t \Phi = L(-i\nabla)\Phi. \quad (74)$$

Fix a selected BCC shell vector

$$G_* \in S_{\text{BCC}}, \quad \hat{n} = \frac{G_*}{|G_*|},$$

and expand near that shell:

$$k = G_* + p, \quad p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2. \quad (75)$$

Assume the microscopic operator admits a first-order Taylor expansion

$$L(G_* + p) = L_* + p_i K_i + O(|p|^2), \quad L_* := L(G_*), \quad K_i := \partial_{k_i} L(k) \Big|_{k=G_*}. \quad (76)$$

Assume further that L_* has a two-dimensional low-energy eigenspace

$$H_* \subset H_{\text{full}},$$

with orthogonal projector

$$P_* : H_{\text{full}} \rightarrow H_*.$$

Fix a Pauli basis on $H_* \cong \mathbb{C}^2$:

$$\text{End}(H_*) = \text{span}\{\mathbf{1}_2, \sigma_1, \sigma_2, \sigma_3\}.$$

CXXV. PROJECTED FIRST-ORDER HAMILTONIAN

The reduced first-order Hamiltonian at the selected shell is

$$h_+^{\text{lin}}(p) = P_*(p_i K_i) P_* \Big|_{H_*}. \quad (77)$$

Since this is a Hermitian 2×2 matrix, it decomposes uniquely as

$$h_+^{\text{lin}}(p) = a_0(p) \mathbf{1}_2 + a_{\mu}(p) \sigma_{\mu}, \quad \mu = 1, 2, 3. \quad (78)$$

Define the Pauli-resolved coefficient matrix

$$M_{\mu i} := \frac{1}{2} \text{Tr}_{H_*}(\sigma_{\mu} P_* K_i P_*). \quad (79)$$

Then

$$a_{\mu}(p) = M_{\mu i} p_i, \quad h_+^{\text{lin}}(p) = M_{\mu i} p_i \sigma_{\mu}. \quad (80)$$

Thus all local Weyl data are determined directly by the projected first derivatives $P_* K_i P_*$.

CXXVI. EXTRACTION OF THE LONGITUDINAL COEFFICIENT

In the shell-adapted matched $S2$ branch, the most general symmetry-compatible reduced symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (81)$$

Comparing (80) with (81), the longitudinal coefficient is exactly

$$\lambda_{\parallel} = \hat{n}_i M_{3i} = \frac{1}{2} \hat{n}_i \text{Tr}_{H_*}(\sigma_3 P_* K_i P_*). \quad (82)$$

Likewise, the transverse coefficients are

$$\alpha = \frac{1}{2} (e_{1i} M_{1i} + e_{2i} M_{2i}), \quad \beta = \frac{1}{2} (e_{2i} M_{1i} - e_{1i} M_{2i}). \quad (83)$$

Therefore the actual local Weyl/Dirac witness problem is reduced to the finite projected derivative data $P_* K_i P_*$.

CXXVII. EQUIVALENT SHELL-TENSOR FORM

At the shell-tensor level, the same data are encoded by

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}.$$

Then the shell-matching formulas give

$$\lambda_{\parallel} = \kappa_* |G_*| A = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3, \quad (84)$$

$$\alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C. \quad (85)$$

Hence the local Weyl criterion is

$$A \neq 0, \quad (B, C) \neq (0, 0). \quad (86)$$

So the projector-trace formulation and the shell-tensor formulation are equivalent descriptions of the same extraction problem.

CXXVIII. ACTUAL LONGITUDINAL EXTRACTION PROTOCOL

We may now state the protocol explicitly.

[Actual longitudinal extraction protocol] Assume the full microscopic TECT linearization satisfies the hypotheses above. Then the actual longitudinal coefficient is extracted by the following finite procedure:

1. Compute the shell derivative operators

$$K_i = \partial_{k_i} L(k) \big|_{k=G_*}.$$

2. Identify the low shell eigenspace H_* of $L_* = L(G_*)$ and its projector P_* .
3. Form the projected derivative matrices

$$P_* K_i P_* \big|_{H_*}.$$

4. Compute the Pauli-resolved coefficients

$$M_{\mu i} = \frac{1}{2} \text{Tr}_{H_*}(\sigma_{\mu} P_* K_i P_*).$$

5. Contract with the shell axis:

$$\lambda_{\parallel} = \hat{n}_i M_{3i}.$$

6. Equivalently, read the shell tensor coefficient

$$A = \hat{n}_i \hat{n}_j N_{ij}^3,$$

and use

$$\lambda_{\parallel} = \kappa_* |G_*| A.$$

Thus the actual longitudinal coefficient-identification problem is completely reduced to a finite projector-and-contraction computation.

Steps (1)–(4) follow from the first-order Taylor expansion and Pauli decomposition of the reduced 2×2 shell Hamiltonian. Step (5) follows by comparing the coefficient of the unique shell-compatible singlet term $p_{\parallel} \sigma_3$. Step (6) is the equivalent shell-tensor form already established by the shell-matching formulas.

CXXIX. RELATION TO THE LOCAL $h_{\parallel}(\theta)$ -FORMULATION

In the local defect-frame formulation, the projected transport kernel is written as

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)),$$

with

$$h_{\parallel}(\theta) = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(\theta)].$$

Then

$$\lambda_{\parallel} = \int d\xi f_{+}(\xi) f_{-}(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)). \quad (87)$$

Therefore the global shell extraction and the local defect-profile extraction are equivalent:

$$P_* K_i P_* \iff \mathcal{M}^{\mu}(\theta) \iff h_{\parallel}(\theta).$$

So the actual longitudinal extraction may be performed either:

1. globally, through the shell-projected momentum derivative $P_* K_i P_*$, or
2. locally, through the $n^{\mu} \sigma_3$ -coefficient of $\mathcal{M}^{\mu}(\theta)$.

CXXX. MATCHING TO THE CONCRETE g_c -NORMALIZATION

Inside the concrete Class II/seed normalization, the longitudinal witness is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3. \quad (88)$$

Combining (87) with (88) gives the exact extraction formula

$$\boxed{g_c^{\text{act}} = -\frac{512}{3\sqrt{3}\pi \theta_0^3 \kappa^3} \int d\xi f_{+}(\xi) f_{-}(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi))}. \quad (89)$$

Thus once $h_{\parallel}(\theta)$ is known, the actual longitudinal coefficient g_c^{act} is fixed uniquely.

CXXXI. WHAT IS AND IS NOT CLOSED

[Sharp honest status of the protocol] At the present stage, the following statement is fully closed:

the actual longitudinal coefficient is extracted by a finite projector-trace-contraction protocol.

However, the following is still open:

the explicit computation of K_i , P_* , M_{3i} , or equivalently $h_{\parallel}(\theta)$, from the final complete microscopic TECT operator.

The closure of the protocol follows from the already established shell-level extraction theorem and the local $h_{\parallel}$ -representation. The non-closure follows from the project's explicit honest status: the final complete microscopic TECT PDE and the actual microscopic values of g_c, g_E, g_O have not yet been fully extracted.

CXXXII. SHARP HONEST CONCLUSION

The actual longitudinal frontier is now completely localized.
It is no longer:

“solve the whole full PDE in one shot.”

It is exactly:

$$L(k) \rightarrow K_i = \partial_{k_i} L|_{G_*} \rightarrow P_* K_i P_* \rightarrow M_{3i} \rightarrow \lambda_{\parallel}.$$

Equivalently:

$$\text{extract } \hat{n}_i \hat{n}_j N_{ij}^3 \text{ or } h_{\parallel}(\theta) \text{ from the full microscopic TECT linearization.}$$

CXXXIII. FINAL SUMMARY OF THE ACTUAL LONGITUDINAL EXTRACTION PROTOCOL

a. Closed result. The actual longitudinal coefficient-identification problem is fully reduced to the finite shell-projected derivative data of the microscopic linearized operator.

Given

$$L(G_* + p) = L_* + p_i K_i + O(|p|^2), \quad K_i = \partial_{k_i} L(k)|_{k=G_*},$$

and the low shell projector P_* , define

$$M_{\mu i} = \frac{1}{2} \text{Tr}_{H_*}(\sigma_{\mu} P_* K_i P_*).$$

Then the actual longitudinal coefficient is

$$\lambda_{\parallel} = \hat{n}_i M_{3i} = \frac{1}{2} \hat{n}_i \text{Tr}_{H_*}(\sigma_3 P_* K_i P_*).$$

b. Equivalent shell-tensor form. Equivalently,

$$\lambda_{\parallel} = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3.$$

c. Equivalent local profile form. In the local defect formulation,

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)), \quad h_{\parallel}(\theta) = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(\theta)],$$

so

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)).$$

d. *Actual g_c extraction.* Matching with the concrete seed normalization

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3$$

gives

$$g_c^{\text{act}} = -\frac{512}{3\sqrt{3}\pi \theta_0^3 \kappa^3} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)).$$

e. *Honest status.* So what is now closed is the protocol:

$$L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow M_{3i} \rightarrow \lambda_{\parallel}$$

or equivalently

$$\hat{n}_i \hat{n}_j N_{ij}^3 \quad \text{or} \quad h_{\parallel}(\theta) \text{ extraction.}$$

What remains open is the actual computation of these quantities from the final complete microscopic TECT operator.

CXXXIV. UNIFIED ACTUAL EXTRACTION THEOREM FOR $\lambda_{\parallel}, \alpha, \beta$

a. *Goal.* We now unify the longitudinal and transverse coefficient-identification problems into a single extraction theorem. The point is that the full microscopic TECT linearization, once available, determines all three shell coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

through one and the same projected first-derivative dataset.

CXXXV. MICROSCOPIC LINEARIZATION AND SHELL PROJECTION

Let the full microscopic TECT linearized operator around the condensed background be

$$i\partial_t \Phi = L(-i\nabla) \Phi. \tag{90}$$

Fix a selected shell point

$$G_* \neq 0, \quad \hat{n} = \frac{G_*}{|G_*|},$$

and expand around it:

$$k = G_* + p. \tag{91}$$

Assume

$$L(G_* + p) = L_* + p_i K_i + O(|p|^2), \quad L_* := L(G_*), \quad K_i := \partial_{k_i} L(k)|_{k=G_*}. \tag{92}$$

Assume further that L_* admits a two-dimensional low shell subspace

$$H_* \subset H_{\text{full}}$$

with orthogonal projector

$$P_* : H_{\text{full}} \rightarrow H_*.$$

Fix a Pauli basis on $H_* \cong \mathbb{C}^2$:

$$\text{End}(H_*) = \text{span}\{\mathbf{1}_2, \sigma_1, \sigma_2, \sigma_3\}.$$

The projected first-order shell Hamiltonian is

$$h_+^{\text{lin}}(p) = P_*(p_i K_i) P_*|_{H_*}. \tag{93}$$

CXXXVI. PAULI-RESOLVED COEFFICIENT MATRIX

Since $h_+^{\text{lin}}(p)$ is Hermitian on H_* , write

$$h_+^{\text{lin}}(p) = a_0(p)\mathbf{1}_2 + a_\mu(p)\sigma_\mu, \quad \mu = 1, 2, 3. \quad (94)$$

Define the Pauli-resolved projected derivative matrix

$$M_{\mu i} := \frac{1}{2} \text{Tr}_{H_*}(\sigma_\mu P_* K_i P_*). \quad (95)$$

Then

$$a_\mu(p) = M_{\mu i} p_i, \quad h_+^{\text{lin}}(p) = M_{\mu i} p_i \sigma_\mu. \quad (96)$$

Therefore all shell-level Weyl data are already contained in the finite set of matrices $P_* K_i P_*$, equivalently in the numbers $M_{\mu i}$.

CXXXVII. MATCHED $S2$ BRANCH AND UNIFIED COEFFICIENT READOUT

Choose the shell-adapted worldvolume frame

$$p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2.$$

In the matched singlet $\oplus$ doublet $S2$ branch, the most general symmetry-compatible local symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (97)$$

Comparing (96) with (97), one obtains the exact unified readout formulas

$$\lambda_{\parallel} = \hat{n}_i M_{3i} = \frac{1}{2} \hat{n}_i \text{Tr}_{H_*}(\sigma_3 P_* K_i P_*), \quad (98)$$

$$\alpha = \frac{1}{2} (e_{1i} M_{1i} + e_{2i} M_{2i}), \quad (99)$$

$$\beta = \frac{1}{2} (e_{2i} M_{1i} - e_{1i} M_{2i}). \quad (100)$$

Thus the full actual shell-coefficient problem is solved once $M_{\mu i}$ is known.

CXXXVIII. EQUIVALENT SHELL-TENSOR FORM

Equivalently, write the minimal shell tensor ansatz

$$N_{ij}^3 = \frac{A}{\kappa_* |G_*|} \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = \frac{1}{\kappa_* |G_*|} (B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}), \quad N_{ij}^2 = \frac{1}{\kappa_* |G_*|} (-C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}). \quad (101)$$

Then the shell contractions are exactly

$$\lambda_{\parallel} = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3 = A, \quad (102)$$

$$\alpha = \frac{\kappa_* |G_*|}{2} (e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2) = B, \quad (103)$$

$$\beta = \frac{\kappa_* |G_*|}{2} (e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2) = C. \quad (104)$$

Hence the projector-trace formulation and the shell-tensor formulation are equivalent.

CXXXIX. EQUIVALENT LOCAL DEFECT-PROFILE FORM

In the local defect/projector formulation, the first-order transport kernel is written as

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)). \quad (105)$$

Its shell-compatible invariant decomposition is

$$\mathcal{M}_{\text{inv}}^\mu(\theta) = h_0(\theta) n^\mu \mathbf{1}_2 + h_\parallel(\theta) n^\mu \sigma_3 + h_E(\theta) E^\mu{}_a \sigma_a + h_O(\theta) J^\mu{}_a \sigma_a. \quad (106)$$

Then the three actual overlaps are

$$\lambda_\parallel = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_\parallel(\theta(\xi)), \quad (107)$$

$$\alpha = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_E(\theta(\xi)), \quad (108)$$

$$\beta = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_O(\theta(\xi)). \quad (109)$$

So the global shell extraction and the local defect-profile extraction are the same coefficient-identification problem viewed in two different coordinates:

$$P_* K_i P_* \iff N_{ij}^a \iff (h_\parallel, h_E, h_O).$$

CXL. UNIFIED WEYL CRITERION

The reduced shell crossing is Weyl exactly when

$$\lambda_\parallel \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (110)$$

Equivalently,

$$A \neq 0, \quad (B, C) \neq (0, 0), \quad (111)$$

or, in shell-tensor form,

$$\hat{n}_i \hat{n}_j N_{ij}^3 \neq 0, \quad \left(e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2, e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2 \right) \neq (0, 0). \quad (112)$$

CXLI. MATCHING TO THE CONCRETE g_c, g_E, g_O NORMALIZATION

Inside the concrete seed/Class II normalization,

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad \alpha = I_\theta g_E, \quad \beta = I_\theta g_O. \quad (113)$$

Therefore the actual coefficients are

$$g_c^{\text{act}} = -\frac{512}{3\sqrt{3}\pi \theta_0^3 \kappa^3} \int d\xi f_+ f_- \theta'(\xi) h_\parallel(\theta(\xi)), \quad (114)$$

$$g_E^{\text{act}} = \frac{1}{I_\theta} \int d\xi f_+ f_- \theta'(\xi) h_E(\theta(\xi)), \quad g_O^{\text{act}} = \frac{1}{I_\theta} \int d\xi f_+ f_- \theta'(\xi) h_O(\theta(\xi)). \quad (115)$$

Thus all three actual shell coefficients are extracted by one unified protocol.

CXLII. UNIFIED EXTRACTION THEOREM

[Unified actual extraction theorem] Assume the full microscopic TECT linearization admits:

1. a shell expansion

$$L(G_* + p) = L_* + p_i K_i + O(|p|^2),$$

2. a two-dimensional low shell subspace H_* with projector P_* ,
3. the matched $S2$ shell-compatible branch.

Then the three actual shell coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

are determined uniquely and simultaneously by the projected derivative data $P_* K_i P_*$, equivalently by the shell tensor N_{ij}^a , equivalently by the local profile functions $(h_{\parallel}, h_E, h_O)$.

In particular:

$$\lambda_{\parallel} = \hat{n}_i M_{3i}, \quad \alpha = \frac{1}{2}(e_{1i} M_{1i} + e_{2i} M_{2i}), \quad \beta = \frac{1}{2}(e_{2i} M_{1i} - e_{1i} M_{2i}),$$

with

$$M_{\mu i} = \frac{1}{2} \text{Tr}_{H_*}(\sigma_{\mu} P_* K_i P_*).$$

The proof is immediate by comparing the Pauli decomposition of the projected first-order shell Hamiltonian with the unique symmetry-compatible $S2$ reduced symbol. The shell-tensor and local-profile forms follow by the already established shell matching and local overlap identities.

CXLIII. SHARP HONEST STATUS

What is now closed is the *protocol*:

$$\boxed{L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow M_{\mu i} \rightarrow (\lambda_{\parallel}, \alpha, \beta).}$$

What is still open is the *actual execution* of this protocol for the final complete microscopic TECT operator:

$$\boxed{\text{explicit computation of } K_i, P_*, M_{\mu i} \text{ (or equivalently } h_{\parallel}, h_E, h_O) \text{ remains open.}}$$

CXLIV. FINAL SUMMARY OF THE UNIFIED ACTUAL EXTRACTION THEOREM

- a. Closed result.* Once the full microscopic TECT linearization is available, the three shell coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

are extracted simultaneously from the same finite dataset:

$$\boxed{L(k) \rightarrow K_i = \partial_{k_i} L|_{G_*} \rightarrow P_* K_i P_* \rightarrow M_{\mu i}.}$$

- b. Exact formulas.* With

$$M_{\mu i} = \frac{1}{2} \text{Tr}_{H_*}(\sigma_{\mu} P_* K_i P_*),$$

the coefficients are

$$\boxed{\lambda_{\parallel} = \hat{n}_i M_{3i},}$$

$$\boxed{\alpha = \frac{1}{2}(e_{1i} M_{1i} + e_{2i} M_{2i}), \quad \beta = \frac{1}{2}(e_{2i} M_{1i} - e_{1i} M_{2i}).}$$

c. *Equivalent forms.* The same data may be written equivalently as:

$$\begin{aligned}\lambda_{\parallel} &= \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3, \\ \alpha &= \frac{\kappa_* |G_*|}{2} (e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2), \\ \beta &= \frac{\kappa_* |G_*|}{2} (e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2),\end{aligned}$$

or, in the local defect formulation,

$$\begin{aligned}\lambda_{\parallel} &= \int d\xi f_+ f_- \theta'(\xi) h_{\parallel}(\theta(\xi)), \\ \alpha &= \int d\xi f_+ f_- \theta'(\xi) h_E(\theta(\xi)), \quad \beta = \int d\xi f_+ f_- \theta'(\xi) h_O(\theta(\xi)).\end{aligned}$$

d. *Weyl criterion.* Hence the actual shell crossing is Weyl iff

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),}$$

equivalently

$$\boxed{A \neq 0, \quad (B, C) \neq (0, 0).}$$

e. *Concrete coefficient extraction.* Matching to the seed/Class II normalization gives

$$\boxed{g_c^{\text{act}} = -\frac{512}{3\sqrt{3}\pi\theta_0^3\kappa^3} \int d\xi f_+ f_- \theta'(\xi) h_{\parallel}(\theta(\xi)),}$$

$$\boxed{g_E^{\text{act}} = \frac{1}{I_{\theta}} \int d\xi f_+ f_- \theta'(\xi) h_E(\theta(\xi)), \quad g_O^{\text{act}} = \frac{1}{I_{\theta}} \int d\xi f_+ f_- \theta'(\xi) h_O(\theta(\xi)).}$$

f. *Honest frontier.* So what is now closed is the unified protocol. What remains open is only its explicit execution for the final complete microscopic TECT operator:

$$\boxed{\text{compute } K_i, P_*, M_{\mu i} \quad \text{or equivalently} \quad h_{\parallel}, h_E, h_O.}$$

CXIV. FIRST EXPLICIT TRIAL CALCULATION FROM THE MINIMAL COMPLEX-TRIPLET BRAZOVSKII CANDIDATE

a. *Goal.* We now perform the first honest explicit trial computation of the unified extraction protocol

$$L(k) \rightarrow K_i = \partial_{k_i} L|_{G_*} \rightarrow P_* K_i P_* \rightarrow M_{\mu i}.$$

The purpose is not to overclaim full microscopic closure, but to determine what the *minimal isotropic candidate PDE* actually gives at the most naive shell-projected level.

CXLVI. MINIMAL MICROSCOPIC CANDIDATE

Within the minimal local, isotropic, $U(3)$ -invariant truncation, the microscopic TECT candidate is the complex-triplet Brazovskii wave equation

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi, \quad \Psi : \mathbb{R}^{3,1} \rightarrow \mathbb{C}^3, \quad (116)$$

with

$$Z < 0, \quad Y > 0, \quad g > 0, \quad q_*^2 = -\frac{Z}{2Y}.$$

This is the correct minimal microscopic candidate, but the actual shell-enriched witness coefficients are still stated to be open in the project documents. :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7

CXLVII. NAIVE FREE-SHELL DERIVATIVE DATA

Ignore for the moment the condensate periodicity and internal projector transport, and look only at the linear symbol of the candidate equation:

$$\omega^2(k) = \mu^2 + Z|k|^2 + Y|k|^4. \quad (117)$$

Then the first shell derivative is

$$K_i^{\text{free}} := \partial_{k_i} \omega^2(k)|_{k=G_*} = (2Z + 4Y|G_*|^2)G_{*i}. \quad (118)$$

Because $|G_*| = q_*$ and $q_*^2 = -Z/(2Y)$,

$$2Z + 4Yq_*^2 = 2Z + 4Y\left(-\frac{Z}{2Y}\right) = 0. \quad (119)$$

Hence

$$\boxed{K_i^{\text{free}} = 0.} \quad (120)$$

So at the pure free-shell level, the first derivative data vanish identically at the selected shell.

CXLVIII. PROJECTED CONSEQUENCE

Let H_* be any two-dimensional low shell subspace and P_* its projector. Then immediately,

$$P_* K_i^{\text{free}} P_* = 0. \quad (121)$$

Therefore the Pauli-resolved coefficients

$$M_{\mu i}^{\text{free}} = \frac{1}{2} \text{Tr}_{H_*} (\sigma_\mu P_* K_i^{\text{free}} P_*) \quad (122)$$

all vanish:

$$\boxed{M_{\mu i}^{\text{free}} = 0.} \quad (123)$$

Hence the shell coefficients extracted from the pure free-shell symbol are

$$\boxed{\lambda_{\parallel}^{\text{free}} = 0, \quad \alpha^{\text{free}} = 0, \quad \beta^{\text{free}} = 0.} \quad (124)$$

So the naive free isotropic shell dispersion alone does not generate the Weyl witness.

CXLIX. STRONGER CHECK: LOCKED ONE-MODE SLOW-ENVELOPE EXPANSION

We now refine the same conclusion by keeping the oscillatory shell background but still remaining in the locked one-mode isotropic truncation.

Take the background

$$\Psi_0(x) = A e^{iG_* \cdot x} z_0, \quad z_0^\dagger z_0 = 1, \quad (125)$$

and a transverse fluctuation

$$\delta\Psi_\perp(x) = A e^{iG_* \cdot x} \chi(x), \quad z_0^\dagger \chi = 0. \quad (126)$$

Introduce

$$D_{G_*} := \nabla + iG_*. \quad (127)$$

Then

$$\nabla^2(e^{iG_* \cdot x} \chi) = e^{iG_* \cdot x} D_{G_*}^2 \chi, \quad (128)$$

with

$$D_{G_*}^2 = \nabla^2 + 2iG_* \cdot \nabla - q_*^2. \quad (129)$$

Therefore the linear kinetic operator becomes

$$(Z\nabla^2 - Y\nabla^4)\delta\Psi_\perp = Ae^{iG_* \cdot x} \left[ZD_{G_*}^2 - Y(D_{G_*}^2)^2 \right] \chi. \quad (130)$$

Keeping only the terms linear in the slow envelope derivative $\nabla\chi$,

$$D_{G_*}^2 = -q_*^2 + 2iG_* \cdot \nabla + O(\nabla^2), \quad (131)$$

hence

$$(D_{G_*}^2)^2 = q_*^4 - 4iq_*^2 G_* \cdot \nabla + O(\nabla^2). \quad (132)$$

So the first-order transport coefficient is

$$2i \left(Z + 2Yq_*^2 \right) G_* \cdot \nabla. \quad (133)$$

Using again $q_*^2 = -Z/(2Y)$,

$$\boxed{Z + 2Yq_*^2 = 0.} \quad (134)$$

Hence the entire first-order kinetic transport term vanishes:

$$\boxed{\left[ZD_{G_*}^2 - Y(D_{G_*}^2)^2 \right]_{\text{1st order}} = 0.} \quad (135)$$

Thus even after keeping the oscillatory shell phase, the locked one-mode isotropic kinetic truncation still gives no nontrivial first-order witness data.

CL. INTERPRETATION OF THE VANISHING RESULT

This vanishing result is not a contradiction. It means that the actual shell witness cannot come from the free isotropic shell dispersion alone. Therefore the nonzero first-order shell transport data must arise from more structured sectors, namely:

1. the local projector-defect transport matrix

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

whose coefficient functions $a_i^\mu(\theta)$ remain open in the full microscopic theory;

2. the primitive longitudinal commutator-square channel

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu,$$

which gives in the explicit representative

$$V_{(\parallel)}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3, \quad \lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \neq 0;$$

3. the explicit transverse odd-gradient channels

$$V_{(\perp, E)}^\mu(\xi) = \theta'(\xi) E^\mu{}_a \sigma_a, \quad V_{(\perp, O)}^\mu(\xi) = \theta'(\xi) J^\mu{}_a \sigma_a,$$

which already realize $(\alpha, \beta) \neq (0, 0)$;

4. admissible mixed (i, a) -carrier sectors such as Class II, whose concrete seed ansatz gives

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_\theta, \quad \beta = c_O I_\theta.$$

All of these structures are already established at the theorem/seed level, whereas the direct extraction from the final complete microscopic TECT operator remains open. :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9 :contentReference[oaicite:10]index=10 :contentReference[oaicite:11]index=11

CLI. EXPLICIT TRIAL THEOREM

[First explicit trial result for the minimal candidate PDE] For the minimal complex-triplet Brazovskii candidate

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi,$$

the naive free-shell projected first-derivative data satisfy

$$K_i^{\text{free}} = 0, \quad P_* K_i^{\text{free}} P_* = 0, \quad M_{\mu i}^{\text{free}} = 0,$$

at the shell $|G_*| = q_*$.

Moreover, in the locked one-mode isotropic slow-envelope expansion, the linear kinetic first-order transport coefficient also vanishes identically:

$$Z + 2Yq_*^2 = 0.$$

Therefore the actual shell coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

cannot be generated by the pure free isotropic shell dispersion alone. They must come from nontrivial transport sectors beyond this naive truncation.

The free-shell vanishing follows from (118) and the shell identity (119). The slow-envelope kinetic vanishing follows from (134). The projected consequences are immediate from the definitions of $M_{\mu i}$, $\lambda_{\parallel}$, α , and β .

CLII. SHARP HONEST STATUS

This trial calculation proves something genuinely useful:

the first nontrivial microscopic Weyl data are not encoded in the bare isotropic shell dispersion alone.

So the actual microscopic coefficient-identification problem is sharply localized. The next nontrivial calculation must be performed not on the free symbol $\omega^2(k)$, but on one of the genuinely structured transport sectors:

$$\mathcal{M}^\mu(\theta), \quad (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}, \quad \text{or a mixed } (i, a)\text{-carrier realization.}$$

This matches the current honest status of the project: the minimal candidate PDE is fixed, but the actual shell-enriched witness data extracted from it are still open. :contentReference[oaicite:12]index=12 :contentReference[oaicite:13]index=13

CLIII. FINAL SUMMARY OF THE FIRST EXPLICIT TRIAL CALCULATION

a. Closed result. For the minimal complex-triplet Brazovskii candidate

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi, \quad q_*^2 = -\frac{Z}{2Y},$$

the naive free-shell derivative data vanish at the selected shell:

$$K_i^{\text{free}} = \partial_{k_i} \omega^2(k) \big|_{k=G_*} = (2Z + 4Yq_*^2) G_{*i} = 0.$$

Hence

$$P_* K_i^{\text{free}} P_* = 0, \quad M_{\mu i}^{\text{free}} = 0,$$

so the pure free-shell dispersion gives

$$\lambda_{\parallel}^{\text{free}} = \alpha^{\text{free}} = \beta^{\text{free}} = 0.$$

b. Stronger locked one-mode result. Even in the locked one-mode slow-envelope expansion,

$$Z\nabla^2 - Y\nabla^4$$

produces a first-order transport coefficient proportional to

$$Z + 2Yq_*^2,$$

which also vanishes exactly at the shell:

$$\boxed{Z + 2Yq_*^2 = 0.}$$

c. Meaning. Therefore the actual shell witness coefficients do *not* come from the bare isotropic free-shell dispersion alone. They must arise from more structured transport sectors, such as:

$$V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)),$$

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi)(\Pi_Q\mathcal{T}_\xi^2\Pi_Q)_{\text{tr}}n^\mu,$$

or an admissible mixed (i, a) -carrier sector such as Class II.

d. Honest frontier. So the first explicit trial calculation closes the following statement:

$$\boxed{\text{free isotropic shell data alone are insufficient.}}$$

The next real microscopic calculation must therefore target the structured transport sector, not the bare shell dispersion.

CLIV. FIRST NONTRIVIAL TRANSPORT TRIAL CALCULATION IN THE ACTUAL-LOCAL OPERATOR CLASS

a. Goal. We now perform the first nontrivial coefficient calculation inside the already-established actual-local transport class

$$V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)).$$

The purpose is not to claim that the final complete microscopic TECT coefficients have been extracted. Rather, we compute what the leading shell witnesses look like under the most general invariant local transport ansatz.

CLV. ACTUAL-LOCAL STRUCTURAL INPUT

The strongest honest local result already established is that the shell witness coefficients admit the overlap formulas

$$\lambda_{\parallel} = \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi)h_{\parallel}(\theta(\xi)), \quad (136)$$

$$\alpha = \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi)h_E(\theta(\xi)), \quad (137)$$

$$\beta = \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi)h_O(\theta(\xi)), \quad (138)$$

where the actual local transport matrix has the shell-compatible invariant decomposition

$$\mathcal{M}_{\text{inv}}^\mu(\theta) = h_0(\theta)n^\mu\mathbf{1}_2 + h_{\parallel}(\theta)n^\mu\sigma_3 + h_E(\theta)E_a^\mu\sigma_a + h_O(\theta)J_a^\mu\sigma_a. \quad (139)$$

Thus the coefficient-identification problem is already reduced to the three scalar profile functions

$$h_{\parallel}(\theta), \quad h_E(\theta), \quad h_O(\theta).$$

CLVI. LEADING INVARIANT TRIAL ANSATZ

Assume that the actual local transport matrix is real-analytic in the even defect profile $\theta(\xi)$, so that the invariant coefficients admit expansions of the form

$$h_{\parallel}(\theta) = \Lambda_{\parallel}^{(0)} + \Lambda_{\parallel}^{(2)}\theta^2 + O(\theta^4), \quad (140)$$

$$h_E(\theta) = \Lambda_E^{(0)} + \Lambda_E^{(2)}\theta^2 + O(\theta^4), \quad (141)$$

$$h_O(\theta) = \Lambda_O^{(0)} + \Lambda_O^{(2)}\theta^2 + O(\theta^4). \quad (142)$$

Equivalently,

$$\mathcal{M}_{\text{inv}}^{\mu}(\theta) = \Lambda_0^{(0)} n^{\mu} \mathbf{1}_2 + \Lambda_{\parallel}^{(0)} n^{\mu} \sigma_3 + \Lambda_E^{(0)} E^{\mu}{}_a \sigma_a + \Lambda_O^{(0)} J^{\mu}{}_a \sigma_a + O(\theta^2). \quad (143)$$

This is the most general leading shell-compatible invariant trial ansatz inside the actual-local operator class.

CLVII. LEADING-ORDER WITNESS EXTRACTION

Substituting (140)–(142) into (136)–(138) gives

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} \underbrace{\int d\xi f_+ f_- \theta'(\xi)}_{I_{\theta}} + \Lambda_{\parallel}^{(2)} \underbrace{\int d\xi f_+ f_- \theta'(\xi) \theta(\xi)^2}_{I_{\theta^2 \theta'}} + O(\theta_0^5 \kappa), \quad (144)$$

$$\alpha = \Lambda_E^{(0)} I_{\theta} + \Lambda_E^{(2)} I_{\theta^2 \theta'} + O(\theta_0^5 \kappa), \quad (145)$$

$$\beta = \Lambda_O^{(0)} I_{\theta} + \Lambda_O^{(2)} I_{\theta^2 \theta'} + O(\theta_0^5 \kappa). \quad (146)$$

The leading overlap integral

$$I_{\theta} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \quad (147)$$

has already been evaluated explicitly:

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \neq 0 \quad (\theta_0 \kappa \neq 0). \quad (148)$$

Therefore the leading trial coefficients are

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + O(\theta_0^3 \kappa), \quad (149)$$

$$\alpha = \Lambda_E^{(0)} I_{\theta} + O(\theta_0^3 \kappa), \quad \beta = \Lambda_O^{(0)} I_{\theta} + O(\theta_0^3 \kappa). \quad (150)$$

CLVIII. LEADING WEYL CRITERION IN THE TRIAL LOCAL ANSATZ

The shell-reduced Weyl condition is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

Using (149)–(150), one gets the leading-order criterion

$$\Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0) \implies \lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0) \quad (151)$$

for a nontrivial defect profile $\theta_0 \kappa \neq 0$.

Thus the first nontrivial transport trial shows that a nonzero invariant constant coefficient in the actual-local transport matrix is already sufficient to generate the shell Weyl witness at leading order.

CLIX. RELATION TO THE EXPLICIT SEED BRANCH

The explicit seed representative obtained earlier is

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (152)$$

This corresponds not to a constant leading $h_{\parallel}$, but to an even profile behaving schematically as

$$h_{\parallel}(\theta) \sim \theta'(\xi)^2,$$

hence its overlap is

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3}, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (153)$$

So there are two logically distinct branches:

1. the **leading constant-invariant trial branch**, where

$$h_{\parallel}(\theta) = \Lambda_{\parallel}^{(0)} + O(\theta^2),$$

and therefore

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + O(\theta_0^3 \kappa);$$

2. the **primitive commutator-square branch**, where

$$h_{\parallel}(\theta) \sim \theta'^2,$$

and therefore

$$\lambda_{\parallel} \sim I_{\theta'^3}.$$

The first branch is the most general leading local transport trial. The second branch is the explicit geometrically natural primitive seed already constructed inside the concrete ansatz space.

CLX. WHAT THIS TRIAL DOES AND DOES NOT PROVE

[First nontrivial transport trial theorem] Inside the actual-local transport class

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

the most general leading shell-compatible invariant ansatz

$$\mathcal{M}_{\text{inv}}^\mu(\theta) = \Lambda_0^{(0)} n^\mu \mathbf{1}_2 + \Lambda_{\parallel}^{(0)} n^\mu \sigma_3 + \Lambda_E^{(0)} E^\mu{}_a \sigma_a + \Lambda_O^{(0)} J^\mu{}_a \sigma_a + O(\theta^2)$$

gives the leading-order shell witnesses

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + O(\theta_0^3 \kappa), \quad \alpha = \Lambda_E^{(0)} I_{\theta} + O(\theta_0^3 \kappa), \quad \beta = \Lambda_O^{(0)} I_{\theta} + O(\theta_0^3 \kappa).$$

Hence a nonzero leading invariant coefficient already yields a nontrivial Weyl witness.

However, this does *not* prove that the actual full microscopic TECT linearization produces nonzero values of

$$\Lambda_{\parallel}^{(0)}, \quad \Lambda_E^{(0)}, \quad \Lambda_O^{(0)}.$$

That microscopic coefficient extraction remains open.

The coefficient formulas follow immediately from the actual local overlap formulas and the explicit nonvanishing of I_{θ} . The final caveat follows from the current honest project status: the actual microscopic functions $h_{\parallel}, h_E, h_O$ have not yet been extracted from the final full TECT operator.

CLXI. SHARP HONEST CONCLUSION

The first nontrivial transport trial calculation proves the following:

if the actual local transport matrix has a nonzero leading invariant coefficient,

then the shell Weyl witness already appears at order $I_\theta \sim \theta_0 \kappa$.

This is stronger than the free-shell trial, where all first-derivative shell data vanished identically. But it still does not identify the true microscopic TECT coefficient point.

So the next real microscopic task is:

determine whether the actual complete TECT linearization produces $\Lambda_{\parallel}^{(0)}$, $\Lambda_E^{(0)}$, $\Lambda_O^{(0)}$ or instead starts only at higher even order

CLXII. FINAL SUMMARY OF THE FIRST NONTRIVIAL TRANSPORT TRIAL

a. Closed result. Inside the actual-local transport class

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

take the most general leading shell-compatible invariant ansatz

$$\mathcal{M}_{\text{inv}}^\mu(\theta) = \Lambda_0^{(0)} n^\mu \mathbf{1}_2 + \Lambda_{\parallel}^{(0)} n^\mu \sigma_3 + \Lambda_E^{(0)} E^\mu{}_a \sigma_a + \Lambda_O^{(0)} J^\mu{}_a \sigma_a + O(\theta^2).$$

Then the leading shell witnesses are

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_\theta + O(\theta_0^3 \kappa),$$

$$\alpha = \Lambda_E^{(0)} I_\theta + O(\theta_0^3 \kappa), \quad \beta = \Lambda_O^{(0)} I_\theta + O(\theta_0^3 \kappa).$$

Since

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \neq 0,$$

a nonzero leading invariant coefficient already yields a nontrivial Weyl witness.

b. Important distinction. This leading transport trial is *not* the same as the explicit primitive longitudinal seed branch. In the primitive branch,

$$V_{(\parallel)}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3,$$

so

$$\lambda_{\parallel} \sim I_{\theta^3},$$

not I_θ .

Thus:

constant-leading local transport branch $\neq$ primitive commutator-square branch.

c. Honest status. What is now proved is:

if nonzero leading invariant coefficients are present, the Weyl witness appears already at order I_θ .

What is still open is:

does the actual complete TECT linearization produce $\Lambda_{\parallel}^{(0)}$, $\Lambda_E^{(0)}$, $\Lambda_O^{(0)}$, or does it begin only at higher even order?

CLXIII. LEADING-ORDER SELECTION THEOREM FOR THE ACTUAL LOCAL TRANSPORT BRANCH

a. Goal. We now determine, as sharply as the current documents allow, whether the actual shell-compatible local transport matrix begins at constant invariant order

$$\Lambda^{(0)}$$

or whether some channels are forced to begin only at higher primitive order.

The correct answer is branch-sensitive: the broad actual-local operator class and the concrete exact Class II transport branch do *not* impose the same leading-order selection.

CLXIV. BROAD ACTUAL-LOCAL OPERATOR CLASS

The strongest general local form already justified is

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad \mathcal{M}^\mu(\theta) \in \text{Herm}(Q_B). \quad (154)$$

With the minimal even- θ expansion,

$$\mathcal{M}^\mu(\theta) = \Lambda_0^\mu \mathbf{1}_2 + \Lambda_i^\mu \sigma_i + O(\theta^2), \quad (155)$$

the shell-compatible invariant decomposition reads

$$\mathcal{M}_{\text{inv}}^\mu(\theta) = h_0(\theta) n^\mu \mathbf{1}_2 + h_\parallel(\theta) n^\mu \sigma_3 + h_E(\theta) E^\mu{}_a \sigma_a + h_O(\theta) J^\mu{}_a \sigma_a. \quad (156)$$

Hence the corresponding leading coefficients are

$$h_\parallel(\theta) = \Lambda_\parallel^{(0)} + O(\theta^2), \quad h_E(\theta) = \Lambda_E^{(0)} + O(\theta^2), \quad h_O(\theta) = \Lambda_O^{(0)} + O(\theta^2). \quad (157)$$

The actual overlap formulas are

$$\lambda_\parallel = \int d\xi f_+ f_- \theta'(\xi) h_\parallel(\theta(\xi)), \quad (158)$$

$$\alpha = \int d\xi f_+ f_- \theta'(\xi) h_E(\theta(\xi)), \quad \beta = \int d\xi f_+ f_- \theta'(\xi) h_O(\theta(\xi)). \quad (159)$$

Therefore, at leading order,

$$\lambda_\parallel = \Lambda_\parallel^{(0)} I_\theta + O(\theta_0^3 \kappa), \quad \alpha = \Lambda_E^{(0)} I_\theta + O(\theta_0^3 \kappa), \quad \beta = \Lambda_O^{(0)} I_\theta + O(\theta_0^3 \kappa), \quad (160)$$

where

$$I_\theta = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \neq 0. \quad (161)$$

Thus, in the broad actual-local operator class, none of the three leading invariant coefficients is symmetry-forbidden. [Non-forbiddance of constant leading coefficients] Inside the broad shell-compatible actual-local operator class (154)–(155), the coefficients

$$\Lambda_\parallel^{(0)}, \quad \Lambda_E^{(0)}, \quad \Lambda_O^{(0)}$$

are all allowed by the residual local symmetry. In particular, the general local structure does not force any of them to vanish.

The matched local S_2 branch contains the invariant singlet and doublet structures

$$n^\mu \sigma_3, \quad E^\mu{}_a \sigma_a, \quad J^\mu{}_a \sigma_a.$$

Therefore constant coefficients multiplying these structures are symmetry-allowed. Since $I_\theta \neq 0$, any nonzero such coefficient contributes to the corresponding shell witness at leading order.

a. Explicit realizable point in the same operator class. The same actual-local operator class already contains an explicit full-rank realizable point:

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0. \quad (162)$$

This shows that the local operator class itself certainly allows a nonzero constant-leading σ_3 coefficient. :contentReference[oaicite:1]index=1

CLXV. CONCRETE EXACT CLASS II BRANCH

The exact concrete Class II witness formulas are

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}. \quad (163)$$

The transverse seeds are

$$V_{(\perp, E)}^{\mu}(\xi) = \theta'(\xi) E^{\mu}{}_a \sigma_a, \quad V_{(\perp, O)}^{\mu}(\xi) = \theta'(\xi) J^{\mu}{}_a \sigma_a, \quad (164)$$

so the transverse witness lives at I_{θ} -order.

By contrast, the longitudinal seed is

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3, \quad (165)$$

hence

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3}, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \neq 0. \quad (166)$$

Therefore the concrete Class II branch exhibits a genuine mismatch of origin:

$$\boxed{\text{transverse sector: } I_{\theta}\text{-type leading branch; } \quad \text{longitudinal sector: primitive } I_{\theta'^3}\text{-type branch.}} \quad (167)$$

[Exact leading-order selection inside the concrete Class II branch] Inside the exact concrete Class II transport realization,

1. the transverse witness sector begins at I_{θ} -order:

$$\alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta};$$

2. the longitudinal witness does *not* begin at I_{θ} -order; instead its first nonzero contribution is the primitive channel

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}.$$

Equivalently, the I_{θ} -type constant-leading longitudinal branch is absent in the exact concrete Class II witness formulas.

This is exactly the content of (163), together with the explicit primitive longitudinal seed (165). The transverse sector is generated by the odd-gradient seeds (164), while the longitudinal sector is generated by the commutator-square seed and therefore starts at θ'^3 .

CLXVI. BRANCH-SENSITIVE HONEST CONCLUSION

We may now state the sharpest honest leading-order selection statement available at present.

[Branch-sensitive leading-order selection theorem] At the current stage of the TECT proof chain:

1. In the broad actual-local operator class,

$$\Lambda_{\parallel}^{(0)}, \Lambda_E^{(0)}, \Lambda_O^{(0)}$$

are all symmetry-allowed and none is forced to vanish.

2. In the exact currently constructed concrete Class II branch, the transverse coefficients begin at I_θ -order, while the longitudinal coefficient begins only at the primitive I_{θ^3} -order.
3. Therefore the documents do *not* yet determine whether the actual complete microscopic TECT coefficient point lies in:

(A) a constant-leading local branch, or (B) the primitive longitudinal branch.

Item (1) follows from the general local invariant decomposition and the explicit realizable full-rank point inside the same operator class. Item (2) follows from the exact concrete Class II witness formulas. Item (3) follows from the project's explicit honest status: the actual microscopic functions $a_i^\mu(\theta)$, equivalently $h_\parallel, h_E, h_O$, have not yet been extracted from the final complete TECT linearization.

CLXVII. INTERPRETATION FOR THE ACTUAL FULL TECT FRONTIER

The remaining microscopic question is now much sharper than before. It is not simply:

“is $\lambda_\parallel$ nonzero?”

It is specifically:

Does the actual complete TECT transport extraction produce a nonzero constant-leading $n^\mu \sigma_3$ term,

or instead

does the longitudinal sector survive only through the primitive commutator-square channel?

Thus the true leading-order selection audit is:

$$\Lambda_\parallel^{(0)} \stackrel{?}{\neq} 0 \quad \text{or} \quad \Lambda_\parallel^{(0)} = 0 \text{ with first nonzero term in the primitive } I_{\theta^3} \text{ branch.} \tag{168}$$

CLXVIII. FINAL SUMMARY OF THE LEADING-ORDER SELECTION THEOREM

- a. What is now closed.* The leading-order selection problem is branch-sensitive.
- b. Broad actual-local operator class.* Inside the broad shell-compatible local operator class

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad \mathcal{M}^\mu(\theta) = \Lambda_0^\mu \mathbf{1}_2 + \Lambda_i^\mu \sigma_i + O(\theta^2),$$

none of the invariant leading coefficients

$$\Lambda_\parallel^{(0)}, \quad \Lambda_E^{(0)}, \quad \Lambda_O^{(0)}$$

is symmetry-forbidden. In particular, a constant-leading longitudinal branch is allowed.

- c. Concrete exact Class II branch.* In the exact concrete Class II witness formulas,

$$\alpha = g_E I_\theta, \quad \beta = g_O I_\theta, \quad \lambda_\parallel = \frac{g_c}{2} I_{\theta^3}.$$

Hence:

transverse sector starts at I_θ , longitudinal sector starts at primitive I_{θ^3} .

So the I_θ -type leading longitudinal branch is absent in this exact concrete Class II realization.

d. Sharp honest conclusion. Therefore the current documents do *not* yet determine whether the actual complete microscopic TECT coefficient point lies in

(A) a constant-leading local branch or (B) the primitive longitudinal branch.

What is now honestly reduced is the actual audit:

$$\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0 \quad \text{or} \quad \Lambda_{\parallel}^{(0)} = 0 \text{ with the first nonzero longitudinal term in the } I_{\theta^3} \text{ branch.}$$

CLXIX. TWO-BRANCH LONGITUDINAL DECISION THEOREM

a. Goal. We now convert the longitudinal leading-order ambiguity into an explicit decision protocol. The issue is not whether a longitudinal witness can exist in principle. That is already established. The actual unresolved question is only which branch the complete microscopic TECT transport extraction realizes:

Branch A: constant-leading I_{θ} -type longitudinal branch,

or

Branch B: primitive I_{θ^3} -type longitudinal branch.

CLXX. GENERAL ACTUAL-LOCAL LONGITUDINAL EXPANSION

The strongest honest local transport form already established is

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)), \quad (169)$$

with shell-compatible invariant decomposition

$$\mathcal{M}_{\text{inv}}^{\mu}(\theta) = h_0(\theta) n^{\mu} \mathbf{1}_2 + h_{\parallel}(\theta) n^{\mu} \sigma_3 + h_E(\theta) E^{\mu}{}_a \sigma_a + h_O(\theta) J^{\mu}{}_a \sigma_a. \quad (170)$$

The actual longitudinal overlap is

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)). \quad (171)$$

Assume the even- θ expansion

$$h_{\parallel}(\theta) = \Lambda_{\parallel}^{(0)} + \Lambda_{\parallel}^{(2)} \theta^2 + O(\theta^4). \quad (172)$$

Then

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + \Lambda_{\parallel}^{(2)} I_{\theta^2 \theta'} + O(\theta_0^5 \kappa), \quad (173)$$

where

$$I_{\theta} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \neq 0. \quad (174)$$

Thus, if

$$\Lambda_{\parallel}^{(0)} \neq 0, \quad (175)$$

then the longitudinal witness already survives on the constant-leading branch:

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + O(\theta_0^3 \kappa). \quad (176)$$

CLXXI. PRIMITIVE LONGITUDINAL BRANCH

The exact primitive longitudinal seed already established in the concrete seed/Class II construction is

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (177)$$

Its projected witness is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0.} \quad (178)$$

Equivalently, in the concrete Class II coefficients,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}. \quad (179)$$

This is the exact primitive branch. It does *not* begin at I_θ -order.

CLXXII. TRANSVERSE SECTOR IS ALREADY RESOLVED

The transverse witness sector is already explicitly realized in the same concrete seed/Class II framework:

$$\alpha = g_E I_\theta, \quad \beta = g_O I_\theta, \quad I_\theta \neq 0. \quad (180)$$

Hence the actual remaining branch ambiguity is purely longitudinal.

CLXXIII. DECISION FUNCTIONAL FOR THE CONSTANT-LEADING BRANCH

Define the constant-leading longitudinal projector by

$$\Lambda_{\parallel}^{(0)} := \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(0)]. \quad (181)$$

This is the unique shell-compatible constant longitudinal invariant coefficient.

If $\Lambda_{\parallel}^{(0)} \neq 0$, then the decision is finished: the actual full TECT local transport extraction lies in the constant-leading longitudinal branch.

If instead

$$\Lambda_{\parallel}^{(0)} = 0,$$

one must continue to the primitive-channel audit.

CLXXIV. DECISION FUNCTIONAL FOR THE PRIMITIVE BRANCH

If the constant-leading coefficient vanishes, define the primitive longitudinal coefficient λ_c by the factorization

$$V_{\parallel, \text{tr}}^\mu(\xi) = \lambda_c \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu. \quad (182)$$

Then the primitive overlap is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.} \quad (183)$$

Hence:

$$\lambda_c \neq 0 \implies \lambda_{\parallel} \neq 0 \quad \text{on the primitive branch.} \quad (184)$$

CLXXV. TWO-BRANCH DECISION THEOREM

[Two-branch longitudinal decision theorem] Assume the actual full microscopic TECT transport extraction has already been reduced to the shell-compatible local form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).$$

Then the longitudinal witness decision is the following finite branch test:

1. Compute the constant-leading longitudinal invariant coefficient

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(0)].$$

2. If

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

then the actual longitudinal witness lies on the constant-leading branch:

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_\theta + O(\theta_0^3 \kappa).$$

3. If

$$\Lambda_{\parallel}^{(0)} = 0,$$

then inspect the primitive longitudinal channel

$$V_{\parallel, \text{tr}}^\mu(\xi) = \lambda_c \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu.$$

4. If

$$\lambda_c \neq 0,$$

then the longitudinal witness lies on the primitive branch:

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

5. If

$$\Lambda_{\parallel}^{(0)} = 0 \quad \text{and} \quad \lambda_c = 0,$$

then the currently established two branches both fail, and one must either:

- (a) search for a higher even-order longitudinal branch, or
- (b) conclude that the longitudinal witness is absent in that microscopic realization.

The constant-leading branch follows from the general local expansion

$$h_{\parallel}(\theta) = \Lambda_{\parallel}^{(0)} + O(\theta^2)$$

and the overlap identity

$$\lambda_{\parallel} = \int d\xi f_+ f_- \theta' h_{\parallel}(\theta(\xi)).$$

Since $I_\theta \neq 0$, a nonzero $\Lambda_{\parallel}^{(0)}$ produces a nonzero longitudinal witness at leading order.

If $\Lambda_{\parallel}^{(0)} = 0$, the first currently established concrete longitudinal realization is the primitive channel

$$\theta' (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu,$$

whose projected overlap is exactly

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

Since $I_{\theta'^3} \neq 0$, nonzero λ_c implies nonzero longitudinal witness on the primitive branch.

If both coefficients vanish, the current theorem-level construction does not determine a nonzero longitudinal witness, so further higher-order structure would need to be audited explicitly.

CLXXVI. BRANCH-SENSITIVE HONEST STATUS

The strongest honest conclusion is not that one of the branches has already been microscopically selected by complete TECT. Rather, the documents imply:

1. the broad actual-local operator class allows a constant-leading longitudinal branch;
2. the exact currently constructed concrete Class II witness formulas realize the primitive longitudinal branch;
3. the actual full microscopic TECT coefficient point has not yet been extracted, so the final branch choice remains open.

Therefore the actual remaining microscopic audit is:

$$\boxed{\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0 \quad \text{or} \quad \Lambda_{\parallel}^{(0)} = 0 \text{ with } \lambda_c \stackrel{?}{\neq} 0.} \quad (185)$$

CLXXVII. PRACTICAL DECISION PROTOCOL

The concrete execution order is:

- a. *Step A.* Extract the shell-compatible local transport matrix $\mathcal{M}^\mu(\theta)$ from the full microscopic linearization.
- b. *Step B.* Compute

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(0)].$$

- c. *Step C.* If $\Lambda_{\parallel}^{(0)} \neq 0$, stop. The actual longitudinal witness is already present at I_θ -order.
- d. *Step D.* If $\Lambda_{\parallel}^{(0)} = 0$, project onto the primitive channel

$$\theta'(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu$$

and extract λ_c .

- e. *Step E.* If $\lambda_c \neq 0$, stop. The actual longitudinal witness is present on the primitive $I_{\theta'^3}$ -branch.
- f. *Step F.* If both vanish, the current theorem-level branch audit remains inconclusive and one must inspect higher-order longitudinal channels explicitly.

CLXXVIII. FINAL SUMMARY OF THE TWO-BRANCH LONGITUDINAL DECISION THEOREM

- a. *Closed result.* The actual longitudinal witness problem is reduced to a finite branch decision:

$$\boxed{\text{Branch A: constant-leading } I_\theta\text{-type longitudinal branch}}$$

versus

$$\boxed{\text{Branch B: primitive } I_{\theta'^3}\text{-type longitudinal branch.}}$$

- b. *Branch A.* Define

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(0)].$$

If

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

then

$$\boxed{\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_\theta + O(\theta_0^3 \kappa),}$$

so the longitudinal witness is already present at I_θ -order.

c. *Branch B.* If

$$\Lambda_{\parallel}^{(0)} = 0,$$

one must inspect the primitive channel

$$V_{\parallel, \text{tr}}^{\mu}(\xi) = \lambda_c \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu}.$$

If

$$\lambda_c \neq 0,$$

then

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3},$$

so the longitudinal witness survives on the primitive branch.

d. *Honest status.* At the current stage:

1. the broad actual-local operator class allows Branch A;
2. the exact currently constructed concrete Class II formulas realize Branch B;
3. the actual complete microscopic TECT coefficient point has not yet been extracted, so the final branch choice remains open.

e. *Exact remaining audit.* Therefore the actual microscopic longitudinal decision problem is exactly:

$$\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0 \quad \text{or} \quad \Lambda_{\parallel}^{(0)} = 0 \text{ with } \lambda_c \stackrel{?}{\neq} 0.$$

CLXXIX. UNIFIED LONGITUDINAL AUDIT PROTOCOL

a. *Goal.* We now turn the branch-sensitive longitudinal decision into a concrete audit protocol. The purpose is to fix the exact order of extraction from the actual full microscopic TECT transport kernel, without overclaiming that the final coefficient values have already been computed.

CLXXX. INPUT DATA FROM THE ACTUAL LOCAL TRANSPORT EXTRACTION

Assume the full microscopic TECT linearization has already been reduced to the shell-compatible local form

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)), \quad (186)$$

with invariant decomposition

$$\mathcal{M}_{\text{inv}}^{\mu}(\theta) = h_0(\theta) n^{\mu} \mathbf{1}_2 + h_{\parallel}(\theta) n^{\mu} \sigma_3 + h_E(\theta) E^{\mu}_a \sigma_a + h_O(\theta) J^{\mu}_a \sigma_a. \quad (187)$$

The actual witness formulas are

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)), \quad (188)$$

$$\alpha = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_E(\theta(\xi)), \quad \beta = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_O(\theta(\xi)). \quad (189)$$

Thus the transverse and longitudinal audits are already separated at the level of the scalar profile functions

$$h_{\parallel}, \quad h_E, \quad h_O.$$

CLXXXI. STEP 0: TRANSVERSE SECTOR IS ALREADY IN I_θ -FORM

The currently established exact concrete Class II formulas give

$$\alpha = g_E I_\theta, \quad \beta = g_O I_\theta, \quad I_\theta \neq 0, \quad (190)$$

while the longitudinal coefficient is

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}. \quad (191)$$

Therefore the unresolved branch ambiguity is purely longitudinal. The transverse audit is simply:

$$\boxed{(g_E, g_O) \neq (0, 0) \iff (\alpha, \beta) \neq (0, 0).} \quad (192)$$

So the unified longitudinal audit may be performed after the transverse sector has been checked once.

CLXXXII. BRANCH A: CONSTANT-LEADING LONGITUDINAL TEST

Expand the longitudinal profile function as

$$h_{\parallel}(\theta) = \Lambda_{\parallel}^{(0)} + \Lambda_{\parallel}^{(2)} \theta^2 + O(\theta^4). \quad (193)$$

Define the constant-leading longitudinal coefficient by

$$\boxed{\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(0)].} \quad (194)$$

Then

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_\theta + \Lambda_{\parallel}^{(2)} I_{\theta^2 \theta'} + O(\theta_0^5 \kappa), \quad (195)$$

where

$$I_\theta = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \neq 0. \quad (196)$$

Hence:

[Branch A decision criterion] If

$$\Lambda_{\parallel}^{(0)} \neq 0, \quad (197)$$

then the longitudinal witness is already present on the constant-leading branch:

$$\boxed{\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_\theta + O(\theta_0^3 \kappa).} \quad (198)$$

Immediate from (195) and $I_\theta \neq 0$.

a. Interpretation. So if the actual extracted local transport matrix contains a nonzero $n^\mu \sigma_3$ constant invariant coefficient at $\theta = 0$, the longitudinal audit is finished at once.

CLXXXIII. BRANCH B: PRIMITIVE LONGITUDINAL TEST

If

$$\Lambda_{\parallel}^{(0)} = 0,$$

one proceeds to the currently established primitive longitudinal branch.

Define the quadratic traceless Q_B -response

$$S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}, \quad \mathcal{T}_\xi := i[P_B, \partial_\xi P_B]. \quad (199)$$

The primitive longitudinal seed is

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu. \quad (200)$$

Factor the actual primitive longitudinal component as

$$V_{\parallel, \text{act}}^\mu(\xi) = \lambda_c \theta'(\xi) S_\xi n^\mu + V_{\parallel, \text{rem}}^\mu(\xi), \quad (201)$$

where $V_{\parallel, \text{rem}}^\mu$ is orthogonal to the primitive target covector in the reduced witness pairing.

Then the primitive overlap is

$$J_\parallel := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) S_\xi. \quad (202)$$

Inside the explicit seed representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3, \quad J_\parallel = \frac{1}{2} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0. \quad (203)$$

Therefore

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}} \quad (204)$$

on the primitive branch.

[Branch B decision criterion] If

$$\Lambda_\parallel^{(0)} = 0 \quad \text{and} \quad \lambda_c \neq 0, \quad (205)$$

then the longitudinal witness survives on the primitive branch:

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3} \neq 0.} \quad (206)$$

This is exactly the primitive commutator-square overlap formula already established for the concrete seed/Class II branch. Since $I_{\theta'^3} \neq 0$, nonzero λ_c implies nonzero longitudinal witness.

CLXXXIV. UNIFIED DECISION TREE

The two branches combine into one finite decision tree.

[Unified longitudinal audit protocol] Assume the actual full microscopic TECT transport kernel has been reduced to the local shell-compatible form (186)–(187). Then the longitudinal audit is performed in the following order:

1. Compute the constant-leading coefficient

$$\Lambda_\parallel^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(0)].$$

2. If

$$\Lambda_\parallel^{(0)} \neq 0,$$

stop: the actual theory is on Branch A, and

$$\lambda_\parallel = \Lambda_\parallel^{(0)} I_\theta + O(\theta_0^3 \kappa).$$

3. If

$$\Lambda_{\parallel}^{(0)} = 0,$$

compute the primitive quadratic Q_B -response S_{ξ} and the primitive coefficient λ_c .

4. If

$$\lambda_c \neq 0,$$

stop: the actual theory is on Branch B, and

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

5. If both

$$\Lambda_{\parallel}^{(0)} = 0 \quad \text{and} \quad \lambda_c = 0,$$

then the current theorem-level audit is inconclusive at these two established branches, and one must inspect higher even-order longitudinal channels explicitly.

Branch A follows from the even expansion of $h_{\parallel}(\theta)$ and the nonvanishing of I_{θ} . Branch B follows from the primitive commutator-square seed and the nonvanishing of $I_{\theta'^3}$. If both fail, the currently established branch library contains no nonzero longitudinal seed, so additional higher-order structures must be audited separately.

CLXXXV. WHAT IS ACTUALLY CLOSED AND WHAT REMAINS OPEN

[Sharp honest status] At the present stage, the following is closed:

$$\boxed{\text{the longitudinal witness problem is reduced to a two-branch finite decision protocol.}} \quad (207)$$

What remains open is the actual full microscopic input:

$$\boxed{\text{the complete microscopic TECT transport extraction has not yet determined whether } \Lambda_{\parallel}^{(0)} \neq 0 \text{ or instead } \Lambda_{\parallel}^{(0)} = 0 \text{ with } \lambda_c \neq 0} \quad (208)$$

The broad actual-local operator class allows a constant-leading longitudinal coefficient. The exact concrete Class II formulas realize the primitive longitudinal branch. But the actual full microscopic coefficient functions $h_{\parallel}, h_E, h_O$ have not yet been extracted from the final complete TECT operator, so the actual branch selection remains open.

CLXXXVI. PRACTICAL EXECUTION ORDER

The practical longitudinal audit now has the exact order:

- a. (U1) Extract the shell-compatible local transport matrix $\mathcal{M}^{\mu}(\theta)$.
- b. (U2) Compute

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(0)].$$

- c. (U3) If $\Lambda_{\parallel}^{(0)} \neq 0$, conclude Branch A and stop.
- d. (U4) If $\Lambda_{\parallel}^{(0)} = 0$, compute

$$S_{\xi} = (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}$$

and project the actual longitudinal transport onto

$$\theta'(\xi) S_{\xi} n^{\mu}$$

to extract λ_c .

- e. (U5) If $\lambda_c \neq 0$, conclude Branch B and stop.
- f. (U6) If both vanish, inspect higher-order longitudinal channels.

CLXXXVII. FINAL SUMMARY OF THE UNIFIED LONGITUDINAL AUDIT PROTOCOL

a. *Closed result.* The actual longitudinal witness problem is no longer vague. It is reduced to the following exact finite decision protocol.

b. *Branch A: constant-leading longitudinal branch.* Define

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(0)].$$

If

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

then

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + O(\theta_0^3 \kappa),$$

so the longitudinal witness is present already at I_{θ} -order.

c. *Branch B: primitive longitudinal branch.* If

$$\Lambda_{\parallel}^{(0)} = 0,$$

inspect the primitive channel

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu}.$$

If its coefficient $\lambda_c \neq 0$, then

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3},$$

so the longitudinal witness survives on the primitive branch.

d. *Decision tree.* Therefore the exact remaining longitudinal audit is:

$$\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0 \quad \text{or} \quad \Lambda_{\parallel}^{(0)} = 0 \quad \text{with} \quad \lambda_c \stackrel{?}{\neq} 0.$$

e. *Honest status.* At the current stage:

1. the broad actual-local operator class allows Branch A;
2. the exact concrete Class II formulas realize Branch B;
3. the actual complete microscopic TECT extraction has not yet selected which branch is realized physically.

f. *Meaning.* So the longitudinal frontier is now completely localized:

$$\text{extract } \Lambda_{\parallel}^{(0)} \text{ first; if it vanishes, extract the primitive coefficient } \lambda_c.$$

CLXXXVIII. FINAL MICROSCOPIC WITNESS AUDIT PROTOCOL

a. *Goal.* We now combine the transverse audit and the branch-sensitive longitudinal audit into a single final microscopic witness protocol. The purpose is to state the exact finite test that the complete microscopic TECT linearization must satisfy in order for the selected shell crossing to be Weyl.

CLXXXIX. WITNESS CRITERION AT THE SHELL-REDUCED LEVEL

The local shell-reduced Weyl criterion is

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (209)$$

Equivalently, in the shell-tensor language,

$$\boxed{W_{\parallel} \neq 0, \quad (W_I, W_J) \neq (0, 0),} \quad (210)$$

and the corresponding local Weyl determinant is

$$\det M = (\kappa_* |G_*|)^3 W_{\parallel} (W_I^2 + W_J^2). \quad (211)$$

Thus the microscopic witness audit is complete exactly when one longitudinal witness and one transverse witness survive.

CXC. ACTUAL MICROSCOPIC EXTRACTION CHAIN

The full microscopic extraction chain is:

$$\boxed{P_B(\xi) \implies V^\mu(\xi) \implies u_a^\pm \implies (M^\mu)_{ab} \implies P_* K_i P_* \implies (\lambda_{\parallel}, \alpha, \beta).} \quad (212)$$

Equivalently, one may formulate the same extraction at the tensor level:

$$\boxed{V_{\text{micro}} \subset W = \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}},} \quad (213)$$

and for each required channel $\alpha \in \{\parallel, I, J\}$, define the shell target covector

$$T^{(\alpha)}.$$

Then the channel survives iff

$$\boxed{P_V T^{(\alpha)} \neq 0.} \quad (214)$$

So the witness problem is no longer an infinite-dimensional PDE mystery. It is a finite projection / seed-embedding audit.

CXCI. TRANSVERSE AUDIT

The exact currently established concrete Class II formulas give

$$\boxed{\alpha = g_E I_\theta, \quad \beta = g_O I_\theta, \quad I_\theta \neq 0.} \quad (215)$$

Therefore the transverse sector survives iff

$$\boxed{(g_E, g_O) \neq (0, 0).} \quad (216)$$

Equivalently, it is enough to produce one admissible transverse seed tensor $N_{\text{seed}}^{(\perp)} \in V_{\text{micro}}$ such that

$$\boxed{\det T_{\perp}(N_{\text{seed}}^{(\perp)}) \neq 0,} \quad (217)$$

or in explicit witness language,

$$(\alpha, \beta) \neq (0, 0).$$

Inside the concrete seed space, two canonical transverse seeds are already known:

$$N_{\text{seed}}^{(\perp, E)}, \quad N_{\text{seed}}^{(\perp, O)},$$

giving

$$\alpha = I_\theta, \beta = 0 \quad \text{or} \quad \alpha = 0, \beta = I_\theta.$$

Hence the transverse witness sector is already explicitly closed inside the concrete ansatz space.

CXCII. LONGITUDINAL AUDIT: TWO-BRANCH DECISION

The longitudinal sector is branch-sensitive.

Branch A: constant-leading branch

From the actual local transport extraction

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

define the constant-leading longitudinal coefficient

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(0)]. \quad (218)$$

If

$$\Lambda_{\parallel}^{(0)} \neq 0, \quad (219)$$

then the longitudinal witness is already present at I_θ -order:

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_\theta + O(\theta_0^3 \kappa). \quad (220)$$

Branch B: primitive branch

If

$$\Lambda_{\parallel}^{(0)} = 0,$$

then inspect the primitive commutator-square channel

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu, \quad \mathcal{T}_\xi = i[P_B, \partial_\xi P_B]. \quad (221)$$

Factor its actual coefficient as

$$\lambda_c.$$

Then the primitive overlap formula is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0. \quad (222)$$

Hence if

$$\lambda_c \neq 0, \quad (223)$$

the longitudinal witness survives on the primitive branch.

Longitudinal decision rule

Therefore the longitudinal sector survives iff

$$\Lambda_{\parallel}^{(0)} \neq 0 \quad \text{or} \quad [\Lambda_{\parallel}^{(0)} = 0 \text{ and } \lambda_c \neq 0]. \quad (224)$$

CXCIII. UNIFIED FINAL WITNESS PROTOCOL

Combining the transverse and longitudinal conditions, the full microscopic witness audit succeeds iff

$$\boxed{(\alpha, \beta) \neq (0, 0) \quad \text{and} \quad [\Lambda_{\parallel}^{(0)} \neq 0 \text{ or } (\Lambda_{\parallel}^{(0)} = 0 \text{ and } \lambda_c \neq 0)]}. \quad (225)$$

Equivalently, in seed language, it is sufficient to find:

1. one admissible transverse seed direction with

$$(\alpha, \beta) \neq (0, 0),$$

2. one admissible longitudinal seed direction, either:

- (a) a constant-leading $n^\mu \sigma_3$ seed with $\Lambda_{\parallel}^{(0)} \neq 0$, or
- (b) a primitive commutator-square seed with $\lambda_c \neq 0$.

[Final microscopic witness audit theorem] Let V_{micro} be the actual symmetry-allowed transport tensor space extracted from the full microscopic TECT linearization. Assume:

1. the transverse sector contains one admissible seed with $(\alpha, \beta) \neq (0, 0)$;
2. the longitudinal sector satisfies either:
 - (a) Branch A: $\Lambda_{\parallel}^{(0)} \neq 0$, or
 - (b) Branch B: $\Lambda_{\parallel}^{(0)} = 0$ and $\lambda_c \neq 0$.

Then the shell-reduced Weyl witness condition

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)$$

holds, and the selected shell crossing is Weyl.

The transverse part is immediate from (215). The longitudinal part follows from the two-branch decision theorem: either a constant-leading I_θ -type longitudinal coefficient survives, or else the primitive I_{θ^3} -type coefficient survives. In either case $\lambda_{\parallel} \neq 0$. Combining the two gives the Weyl condition (209).

CXCIV. PRACTICAL OUTPUT OF THE PROTOCOL

The full microscopic witness protocol produces four logically distinct outputs:

- a. (W1) *Carrier survival*. Do we obtain at least one longitudinal and one transverse seed direction?
- b. (W2) *Valley witness*. Do the extracted coefficients satisfy

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)?$$

- c. (W3) *Opposite-valley matching*. Repeat the same extraction at $-G_*$ and compare the witness triples:

$$(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+).$$

- d. (W4) *Data for later audits*. The same extraction supplies the low-energy block data needed later for:

$$V(0) = 0, \quad \Delta > 0, \quad \text{mass protection}, \quad M_Z.$$

CXCV. SHARP HONEST STATUS

What is now fully closed is the witness protocol:

$$\boxed{\text{full microscopic witness audit} = (\alpha, \beta)\text{-audit} + \lambda_{\parallel}\text{-audit}} \quad (226)$$

with the exact longitudinal decision rule (224).

What remains open is not the logic of the protocol, but its actual execution in the final complete microscopic TECT linearization:

$$\boxed{\text{compute the actual full transport extraction and determine } (\alpha, \beta), \Lambda_{\parallel}^{(0)}, \lambda_c.} \quad (227)$$

CXCVI. FINAL SUMMARY OF THE FINAL MICROSCOPIC WITNESS AUDIT PROTOCOL

a. Closed result. The complete microscopic witness problem for the selected shell crossing is reduced to the exact finite condition

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).}$$

b. Transverse audit. The transverse sector is already in explicit I_{θ} -form:

$$\alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}, \quad I_{\theta} \neq 0.$$

So the transverse witness survives iff

$$\boxed{(g_E, g_O) \neq (0, 0).}$$

c. Longitudinal audit. The longitudinal sector obeys the two-branch decision rule:

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0 \quad \text{or} \quad [\Lambda_{\parallel}^{(0)} = 0 \text{ and } \lambda_c \neq 0].}$$

If Branch A holds,

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + O(\theta_0^3 \kappa).$$

If Branch B holds,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

d. Unified final protocol. Therefore the final microscopic witness audit succeeds iff

$$\boxed{(\alpha, \beta) \neq (0, 0) \quad \text{and} \quad [\Lambda_{\parallel}^{(0)} \neq 0 \text{ or } (\Lambda_{\parallel}^{(0)} = 0 \text{ and } \lambda_c \neq 0)].}$$

e. Operational meaning. Equivalently, it is enough to find:

1. one admissible transverse seed direction,
2. one admissible longitudinal seed direction, either

constant-leading $n^{\mu} \sigma_3$ or primitive commutator-square.

f. Honest frontier. So the witness protocol itself is now completely explicit. What remains open is only the actual full microscopic extraction of

$$(\alpha, \beta), \quad \Lambda_{\parallel}^{(0)}, \quad \lambda_c$$

from the final complete TECT linearization.

CXCVII. OPPOSITE-VALLEY MATCHING THEOREM

a. Goal. We now close the exact relation between the two shell valleys $+G_*$ and $-G_*$. The point is to show that opposite-valley pairing is not an additional vague dynamical requirement, but a finite matched-frame equality of the witness triples.

CXCVIII. LOCAL WEYL BLOCKS AT THE TWO VALLEYS

Let $h_+(p)$ and $h_-(p)$ denote the shell-projected 2×2 Weyl blocks at the two opposite valleys $+G_*$ and $-G_*$, respectively.

Choose an adapted frame at $+G_*$:

$$\hat{n}_+ = \frac{G_*}{|G_*|}, \quad \{e_{1+}, e_{2+}, \hat{n}_+\}.$$

At the opposite valley we choose the matched frame

$$\hat{n}_- = -\hat{n}_+, \quad e_{1-} = e_{1+}, \quad e_{2-} = e_{2+},$$

or any equivalent consistently matched choice.

Then the two local Weyl blocks have the form

$$h_+(p) = W_{\parallel}^+ p_{\parallel} \sigma_3 + W_I^+ (p_1 \sigma_1 + p_2 \sigma_2) + W_J^+ (-p_2 \sigma_1 + p_1 \sigma_2), \quad (228)$$

$$h_-(p) = W_{\parallel}^- p_{\parallel} \sigma_3 + W_I^- (p_1 \sigma_1 + p_2 \sigma_2) + W_J^- (-p_2 \sigma_1 + p_1 \sigma_2), \quad (229)$$

where the coefficients are the witness triples extracted in the matched frames.

CXCIX. MATCHED WITNESS TRIPLES

Define the positive-valley witness functionals by

$$W_{\parallel}^+[N] := \hat{n}_{+i} \hat{n}_{+j} N_{ij}^3(+), \quad (230)$$

$$W_I^+[N] := \frac{1}{2} \left(e_{1+j} \hat{n}_{+i} N_{ij}^1(+) + e_{2+j} \hat{n}_{+i} N_{ij}^2(+) \right), \quad (231)$$

$$W_J^+[N] := \frac{1}{2} \left(e_{1+j} \hat{n}_{+i} N_{ij}^2(+) - e_{2+j} \hat{n}_{+i} N_{ij}^1(+) \right), \quad (232)$$

and analogously the negative-valley matched-frame witness functionals

$$W_{\parallel}^-[N] := \hat{n}_{-i} \hat{n}_{-j} N_{ij}^3(-), \quad (233)$$

$$W_I^-[N] := \frac{1}{2} \left(e_{1-j} \hat{n}_{-i} N_{ij}^1(-) + e_{2-j} \hat{n}_{-i} N_{ij}^2(-) \right), \quad (234)$$

$$W_J^-[N] := \frac{1}{2} \left(e_{1-j} \hat{n}_{-i} N_{ij}^2(-) - e_{2-j} \hat{n}_{-i} N_{ij}^1(-) \right). \quad (235)$$

The opposite-valley matching condition is

$$\boxed{(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+).} \quad (236)$$

CC. WHY MATCHED EQUALITY IMPLIES OPPOSITE-VALLEY PAIRING

[Matched witness equality implies opposite-valley pairing] Assume the two shell-projected Weyl blocks are written in matched opposite-valley frames and satisfy

$$(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+).$$

Then

$$\boxed{h_-(p) = -h_+(-p).} \quad (237)$$

Using (228),

$$-h_+(-p) = -\left[W_{\parallel}^+(-p_{\parallel})\sigma_3 + W_I^+((-p_1)\sigma_1 + (-p_2)\sigma_2) + W_J^+(-(-p_2)\sigma_1 + (-p_1)\sigma_2)\right] \quad (238)$$

$$= W_{\parallel}^+p_{\parallel}\sigma_3 + W_I^+(p_1\sigma_1 + p_2\sigma_2) + W_J^+(-p_2\sigma_1 + p_1\sigma_2). \quad (239)$$

If the matched witness triples are equal,

$$W_{\parallel}^- = W_{\parallel}^+, \quad W_I^- = W_I^+, \quad W_J^- = W_J^+,$$

then the right-hand side is exactly $h_-(p)$ as written in (229). Hence

$$h_-(p) = -h_+(-p).$$

CCI. DIRAC DOUBLING

Define the doubled valley spinor

$$\Phi = \begin{pmatrix} \phi_+ \\ \phi_- \end{pmatrix} \in \mathbb{C}^4.$$

Then the doubled local Hamiltonian is

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}. \quad (240)$$

Under the pairing relation (237), after a suitable basis choice this becomes

$$H_D(q) = \begin{pmatrix} q_i\sigma_i & 0 \\ 0 & -q_i\sigma_i \end{pmatrix} = \sum_{i=1}^3 q_i (\tau_z \otimes \sigma_i), \quad (241)$$

which is the massless Dirac Hamiltonian.

[Opposite-valley matching yields local Dirac doubling] If each valley is Weyl and the matched witness triples are equal, then the doubled local shell block is Dirac.

Weyl at each valley gives the two 2×2 blocks. The matching relation gives the standard opposite-valley doubling

$$h_-(p) = -h_+(-p),$$

and then (240) is exactly a doubled Dirac block.

CCII. CONCRETE CLASS II SEED FAMILY

Inside the concrete Class II seed family, the induced shell tensors at $+G_*$ and $-G_*$ take the matched-frame forms

$$\delta N_{ij}^3(+) = -\mu c_{\parallel} \hat{n}_{+i} \hat{n}_{+j}, \quad \delta N_{ij}^1(+) = -\mu (c_I \hat{n}_{+i} e_{1+j} + c_J \hat{n}_{+i} e_{2+j}), \quad \delta N_{ij}^2(+) = -\mu (-c_J \hat{n}_{+i} e_{1+j} + c_I \hat{n}_{+i} e_{2+j}), \quad (242)$$

and

$$\delta N_{ij}^3(-) = -\mu c_{\parallel} \hat{n}_{-i} \hat{n}_{-j}, \quad \delta N_{ij}^1(-) = -\mu (c_I \hat{n}_{-i} e_{1-j} + c_J \hat{n}_{-i} e_{2-j}), \quad \delta N_{ij}^2(-) = -\mu (-c_J \hat{n}_{-i} e_{1-j} + c_I \hat{n}_{-i} e_{2-j}). \quad (243)$$

Therefore the witness triples agree identically in the matched frames:

$$\boxed{(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+).} \quad (244)$$

So opposite-valley matching is already closed inside the concrete seed/Class II realization.

CCIII. SHARP HONEST CONCLUSION

The opposite-valley matching problem has now been reduced to a finite equality test.

[Opposite-valley matching theorem] At the shell-reduced level, opposite-valley pairing is equivalent to the matched-frame equality of the witness triples:

$$(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+).$$

When this holds, one has

$$h_-(p) = -h_+(-p),$$

and therefore the doubled local shell block is Dirac.

Inside the concrete Class II seed family, this matching is already satisfied identically.

a. What remains open. What is still not yet fully extracted microscopically is whether the *actual full microscopic TECT* realization produces the same matched witness triple at both valleys. So the remaining audit is not conceptual; it is the explicit finite comparison:

$$(W_{\parallel}^-, W_I^-, W_J^-) \stackrel{?}{=} (W_{\parallel}^+, W_I^+, W_J^+)$$

in the final complete microscopic extraction.

CCIV. FINAL SUMMARY OF THE OPPOSITE-VALLEY MATCHING THEOREM

a. Closed result. Opposite-valley pairing is reduced to one finite matched-frame condition:

$$(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+).$$

b. Exact consequence. If the two valleys are already Weyl and the witness triples agree in the matched opposite-valley frames, then

$$h_-(p) = -h_+(-p).$$

Therefore the doubled local shell Hamiltonian

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}$$

is unitarily equivalent to a massless Dirac Hamiltonian.

c. Concrete seed-level closure. Inside the concrete Class II seed family, the induced tensors at $+G_*$ and $-G_*$ have the same axial-family coefficients in their own matched local frames. Hence

$$(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+)$$

already holds identically at the witness level.

d. Honest frontier. So what remains open is not the logic of opposite-valley doubling. It is only the actual microscopic extraction check:

$$(W_{\parallel}^-, W_I^-, W_J^-) \stackrel{?}{=} (W_{\parallel}^+, W_I^+, W_J^+)$$

for the final complete TECT realization.

CCV. ZERO CONSTANT MIXING THEOREM AND THE ACTUAL REMOTE-SECTOR AUDIT

a. Goal. We now close the first remote-sector condition in the local Dirac closure chain:

$$V(0) = 0.$$

The objective is to state the exact theorem-level criterion, prove its finite sectorwise reduction, and isolate the remaining actual microscopic verification task.

CCVI. FULL MICROSCOPIC BLOCK FORM

Near the selected shell crossing $p = 0$, write the full microscopic Hamiltonian in block form

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \quad (245)$$

where:

- H_D acts on the local shell-relevant doubled Dirac sector $\mathcal{H}_D$,
- H_R acts on the remote complement $\mathcal{H}_R$,
- $V(p)$ is the inter-sector mixing block.

At $p = 0$, let H_* denote the shell-relevant residual symmetry group (little group / residual symmetry at the crossing), and assume the decomposition

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$$

is H_* -invariant.

CCVII. WHY $V(0)$ IS AN INTERTWINER

Assume the microscopic Hamiltonian is H_* -equivariant at $p = 0$:

$$U(h) H_{\text{full}}(0) U(h)^{-1} = H_{\text{full}}(0), \quad h \in H_*. \quad (246)$$

Then the off-diagonal block must satisfy

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D). \quad (247)$$

Because $H_{\text{full}}(0)$ is H_* -equivariant and the splitting

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$$

is H_* -invariant, each block in (245) must transform covariantly. In particular, the off-diagonal block intertwines the H_* -action on $\mathcal{H}_R$ and $\mathcal{H}_D$, hence belongs to

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

CCVIII. ZERO CONSTANT MIXING THEOREM

[Little-group mismatch implies zero constant mixing] Assume:

1. the microscopic Hamiltonian is H_* -equivariant at $p = 0$;
2. the decomposition

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$$

is H_* -invariant;

- 3.

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0. \quad (248)$$

Then

$$\boxed{V(0) = 0.} \quad (249)$$

Moreover, the first nontrivial mixing begins at least linearly:

$$\|V(p)\| \leq c|p| \quad \text{for small } p. \quad (250)$$

By (247),

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

If the latter space vanishes, then the only possible intertwiner is the zero map. Hence

$$V(0) = 0.$$

The linear estimate (250) is then the first allowed order in the momentum expansion.

a. Meaning. So $V(0) = 0$ is not heuristic. It is the exact representation-theoretic consequence of the absence of common shell-relevant irreducible content between the local Dirac block and the remote sector. :contentReference[oaicite:10]index=10 :contentReference[oaicite:11]index=11

CCIX. SECTORWISE REDUCTION

Decompose the remote sector into H_* -invariant pieces:

$$\mathcal{H}_R = \bigoplus_a K_a. \quad (251)$$

Then

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_a \text{Hom}_{H_*}(K_a, \mathcal{H}_D). \quad (252)$$

Therefore it is enough to prove

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \quad \forall a. \quad (253)$$

[Sectorwise zero-mixing theorem] Assume:

1. $\mathcal{H}_R = \bigoplus_a K_a$ is an H_* -invariant decomposition;
2. for every sector K_a , at least one shell-relevant label mismatches that of $\mathcal{H}_D$.

Then

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \quad \forall a, \quad (254)$$

hence

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0, \quad (255)$$

and therefore

$$\boxed{V(0) = 0.} \quad (256)$$

If K_a differs from $\mathcal{H}_D$ in at least one shell-relevant conserved label, then no nonzero H_* -intertwiner can map K_a into $\mathcal{H}_D$. Thus

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0$$

for every a . Using (252), the full Hom-space also vanishes, so by the previous theorem $V(0) = 0$.

CCX. PRACTICAL SHELL-RELEVANT LABEL CHECKLIST

The finite microscopic audit is therefore:

$$\boxed{\text{For every } K_a \subset \mathcal{H}_R, \text{ find at least one mismatch label relative to } \mathcal{H}_D.} \quad (257)$$

The established checklist of shell-relevant mismatch labels is:

1. selected shell-pair label $\pm G_*$,
2. transverse versus longitudinal/volumetric character,
3. presence or absence of the enriched shell-relevant internal doublet,
4. residual parity / grading / chirality,
5. any extra conserved microscopic label.

Thus a remote sector is excluded from constant mixing as soon as it fails to match $\mathcal{H}_D$ in at least one of these labels. :contentReference[oaicite:12]index=12 :contentReference[oaicite:13]index=13 :contentReference[oaicite:14]index=14

CCXI. MINIMAL DANGEROUS TRUNCATION

In the minimal dangerous truncation, the remote sector may be organized schematically as

$$\mathcal{H}_R = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}} \oplus \mathcal{H}_{\text{extra}}. \quad (258)$$

The default eliminations are:

$$\mathcal{H}_{\text{vol}} : \text{volumetric mismatch}, \quad (259)$$

$$\mathcal{H}_{\text{high}} : \text{different shell-orbit mismatch}, \quad (260)$$

$$\mathcal{H}_{\text{non-enr}} : \text{no enriched internal doublet}, \quad (261)$$

$$\mathcal{H}_{\text{wrong-int}} : \text{wrong internal representation}. \quad (262)$$

Only $\mathcal{H}_{\text{extra}}$ must be checked case by case.

So the actual remote-sector audit is a finite classification task, not an uncontrolled infinite-dimensional problem. :contentReference[oaicite:15]index=15

CCXII. WHY THIS IS ENOUGH FOR SPECTRAL ISOLATION

Once

$$V(0) = 0, \quad \|V(p)\| \leq c|p|,$$

the Schur-complement self-energy satisfies

$$\Sigma(E, p) = V(p) (H_R(p) - E)^{-1} V(p)^\dagger = O(|p|^2/\Delta) \quad (263)$$

provided the remote sector has a positive gap $\Delta > 0$. Thus zero constant mixing is exactly the representation-theoretic reason the local linear Dirac cone is not destroyed at leading order by remote hybridization. :contentReference[oaicite:16]index=16 :contentReference[oaicite:17]index=17 :contentReference[oaicite:18]index=18

CCXIII. FINAL ZERO-MIXING AUDIT PROTOCOL

The actual microscopic verification protocol is now completely explicit:

- a. (Z1) Determine the shell-relevant residual symmetry group H_* at the selected crossing.
- b. (Z2) Decompose the local Dirac block $\mathcal{H}_D$ into its shell-relevant H_* -representation content.
- c. (Z3) Decompose the remote sector $\mathcal{H}_R$ into H_* -invariant pieces

$$\mathcal{H}_R = \bigoplus_a K_a.$$

- d. (Z4) For each K_a , check whether it differs from $\mathcal{H}_D$ in at least one shell-relevant label.
- e. (Z5) If every K_a mismatches $\mathcal{H}_D$, conclude

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0$$

and therefore

$$\boxed{V(0) = 0.}$$

CCXIV. SHARP HONEST CONCLUSION

We can now state the strongest honest result of this step.

[Zero constant mixing theorem in audit form] The zero constant mixing condition is fully reduced to the finite representation-theoretic check

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.}$$

Equivalently, it is enough to verify sectorwise that every remote sector K_a differs from the local Dirac block $\mathcal{H}_D$ in at least one shell-relevant label.

If this holds, then

$$\boxed{V(0) = 0, \quad \|V(p)\| \leq c|p|.}$$

a. *What remains open.* The theorem itself is closed. What remains open is the actual microscopic verification that the final complete TECT remote-sector decomposition indeed satisfies this irrep/label mismatch condition.

CCXV. FINAL SUMMARY OF THE ZERO CONSTANT MIXING STEP

a. *Closed theorem.* At the selected shell crossing $p = 0$, the constant inter-sector mixing block satisfies

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Therefore, if

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,}$$

then automatically

$$\boxed{V(0) = 0.}$$

b. *Sectorwise reduction.* If the remote sector decomposes as

$$\mathcal{H}_R = \bigoplus_a K_a,$$

then it is enough to prove

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \quad \forall a.$$

This follows once every remote sector K_a differs from $\mathcal{H}_D$ in at least one shell-relevant label.

c. Practical mismatch checklist. A remote sector is excluded from constant mixing if it mismatches the local Dirac block in at least one of:

$\pm G_*$, transverse vs volumetric character, enriched internal doublet, parity / grading / chirality, extra conserved

d. Consequence. Once

$$V(0) = 0 \quad \text{and} \quad \|V(p)\| \leq c|p|,$$

the Schur-complement correction is only

$$\Sigma(E, p) = O(|p|^2/\Delta),$$

so remote states do not destroy the leading linear Dirac cone.

e. Honest status. So the zero-mixing theorem itself is closed. What remains open is only the actual microscopic verification that the final complete TECT remote-sector decomposition satisfies

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.}$$

CCXVI. REMOTE GAP THEOREM AS A FINITE BLOCH-SPECTRAL INEQUALITY

a. Goal. We now close Step 4 of the local Dirac closure chain:

$$\Delta > 0.$$

The correct theorem-level statement is that the remote gap is an *independent spectral input*, and that in the present TECT framework it is reduced to either:

1. a finite Bloch-spectral lower-bound inequality, or
2. a bounded-perturbation stability estimate from an already-gapped realization.

CCXVII. WHY $\det M \neq 0$ IS NOT THE REMOTE GAP

Let the local shell-reduced Weyl block be

$$H_D(p) = \Gamma_i(M_{\text{eff}}p)_i, \quad \det M_{\text{eff}} \neq 0.$$

Equivalently, in the A, B, C notation,

$$M_{\text{eff}} = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}, \quad \det M_{\text{eff}} = A(B^2 + C^2) \neq 0.$$

This controls only the nondegeneracy of the *local Dirac block*. It does *not* determine the remote spectrum of $H_R(p)$. Therefore:

$$\boxed{\det M_{\text{eff}} \neq 0 \not\Rightarrow \text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset.}$$

So the remote-gap theorem must be spectral, not determinant-based.

CCXVIII. REMOTE BLOCK AND THE GAP CONDITION

Write the full microscopic Hamiltonian in block form

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \tag{264}$$

where:

- $H_D(p)$ is the isolated local doubled Dirac block,
- $H_R(p)$ is the remote sector,
- $V(p)$ is the inter-sector mixing block.

The remote-gap condition is

$$\boxed{\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset \quad (|p| \leq \rho)} \quad (265)$$

for some $\Delta > 0$ and sufficiently small ρ .

CCXIX. DIRECT FINITE BLOCH-SPECTRAL ROUTE

Choose a first nontrivial truncated remote basis

$$\mathcal{B}_R^{(N)} = \{\text{remote shell/Bloch basis states up to order } N\},$$

and construct the truncated remote matrix

$$H_R^{(N)}(p).$$

Define the truncated spectral distance

$$\delta_{0,\rho}^{(N)} := \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(H_R^{(N)}(p))). \quad (266)$$

Let the total truncation/tail/mixing error budget be

$$\eta_{R,\rho}^{(N)} := \|H_R - H_R^{(N)}\|_\rho, \quad (267)$$

or more generally any explicit upper bound for the total remote correction outside the benchmark block.

Then the finite gap criterion is:

[Finite Bloch-spectral inequality theorem] If

$$\boxed{\delta_{0,\rho}^{(N)} > \eta_{R,\rho}^{(N)},} \quad (268)$$

then the full remote block has a positive gap

$$\boxed{\Delta_\rho \geq \delta_{0,\rho}^{(N)} - \eta_{R,\rho}^{(N)} > 0.} \quad (269)$$

By assumption, every spectral point of the truncated remote block lies at distance at least $\delta_{0,\rho}^{(N)}$ from zero. The full remote block differs from the truncated one by at most $\eta_{R,\rho}^{(N)}$ in operator norm. Therefore standard self-adjoint spectral stability implies that every spectral point of $H_R(p)$ lies at distance at least

$$\delta_{0,\rho}^{(N)} - \eta_{R,\rho}^{(N)}$$

from zero. If this number is positive, the claimed uniform remote gap follows.

a. Interpretation. So the remote-gap problem is reduced to one explicit finite inequality:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho}.}$$

CCXX. PERTURBATIVE STABILITY ROUTE

Suppose instead that one already has a gapped unperturbed realization

$$\text{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \emptyset \quad (|p| \leq p_*), \quad (270)$$

with $\Delta_0 > 0$, and that the actual remote block is

$$H_R(p) = H_R^{(0)}(p) + \delta H_R(p), \quad \sup_{|p| \leq p_*} \|\delta H_R(p)\| \leq \varepsilon. \quad (271)$$

Then:

[Remote-gap stability under bounded perturbation] If

$$\boxed{\varepsilon < \Delta_0}, \quad (272)$$

then the perturbed remote block still has a positive uniform gap:

$$\boxed{\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset, \quad \Delta := \Delta_0 - \varepsilon > 0.} \quad (273)$$

For each p , the spectrum of a self-adjoint operator moves by at most the operator norm of the perturbation. Hence every spectral point of $H_R(p)$ remains at distance at least

$$\Delta_0 - \|\delta H_R(p)\| \geq \Delta_0 - \varepsilon$$

from zero. If $\varepsilon < \Delta_0$, this lower bound is positive.

a. Stronger convenient form. If one has the stronger estimate

$$\varepsilon < \frac{\Delta_0}{2},$$

then automatically

$$\Delta > \frac{\Delta_0}{2} > 0.$$

CCXXI. MASTER EXPLICIT SPECTRAL INEQUALITY

The project documents also provide a more structured benchmark formulation. Let

$$\Delta_{\text{bench},\rho} > 0$$

be an explicit diagonal benchmark lower bound on the remote block, built from sectorwise lower bounds such as:

1. higher-shell radial separation,
2. heavy volumetric mass gap,
3. wrong-internal spectral separation,
4. extra-sector coercivity.

Let

$$\eta_{\text{rem},\rho}$$

be the total remote mixing/tail/off-diagonal budget.

Then the master inequality is

$$\boxed{\Delta_{\text{bench},\rho} > \eta_{\text{rem},\rho}.} \quad (274)$$

[Master explicit spectral inequality theorem] Assume:

1. the local shell Dirac block exists and is nondegenerate;
2. $V(0) = 0$ and $\|V(p)\| \leq c|p|$;
3. a residual symmetry forbids constant Dirac masses;
4. the diagonal benchmark is positive:

$$\Delta_{\text{bench},\rho} > 0;$$

5. the master inequality (274) holds.

Then the full remote block has a uniform positive gap

$$\Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho} - \eta_{\text{rem},\rho} > 0, \quad (275)$$

and therefore, for $|E| < \Delta_{\text{full},\rho}/2$,

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta_{\text{full},\rho}}. \quad (276)$$

The diagonal benchmark bounds the absolute value of every diagonal remote eigenvalue from below by $\Delta_{\text{bench},\rho}$. The total remote off-diagonal/tail correction is bounded by $\eta_{\text{rem},\rho}$. Hence the full remote spectrum remains at distance at least

$$\Delta_{\text{bench},\rho} - \eta_{\text{rem},\rho}$$

from zero. The resolvent bound is then the standard consequence for $|E| < \Delta_{\text{full},\rho}/2$.

CCXXII. SCHUR-COMPLEMENT CONSEQUENCE

Once Step 3 has already given

$$V(0) = 0, \quad \|V(p)\| \leq c|p|,$$

and Step 4 gives $\Delta > 0$, the remote resolvent is bounded by

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta} \quad (|E| < \Delta/2).$$

Therefore the self-energy satisfies

$$\|\Sigma(E, p)\| = \|V(p)(H_R(p) - E)^{-1}V(p)^\dagger\| \leq \frac{2c^2}{\Delta}|p|^2, \quad (277)$$

and so the effective low-energy Hamiltonian becomes

$$H_{\text{eff}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right). \quad (278)$$

Thus the remote sector affects only the subleading curvature and does not destroy the leading linear Dirac cone.

CCXXIII. STEP-4 CLOSURE IN THE CONCRETE CLASS II SETTING

Inside the concrete Class II setting, the theorem-level Step-4 statement is:

[Step-4 closure in the concrete Class II setting] Assume:

1. the final microscopic TECT truncation admits a remote block decomposition $H_R(p)$;

2. either:

(a) the direct Bloch-spectral inequality holds:

$$\delta_{0,\rho} > \eta_{R,\rho},$$

or

(b) there exists an already-gapped realization with gap $\Delta_0 > 0$, and the Class II correction obeys

$$\sup_{|p| \leq p_*} \|\delta H_R(p)\| < \Delta_0;$$

3. Step 3 already holds:

$$V(0) = 0, \quad \|V(p)\| \leq c|p|.$$

Then the remote sector has a positive gap

$$\Delta > 0,$$

and the Schur-complement correction to the local Dirac block is only

$$O(|p|^2/\Delta).$$

In case (2a), use the direct finite spectral inequality. In case (2b), use bounded perturbation stability. Then combine the resolvent bound with $\|V(p)\| \leq c|p|$.

CCXXIV. WHAT IS AND IS NOT CLOSED

[Sharp honest status of Step 4] At the present stage, the following is theorem-level closed:

the remote-gap problem is fully reduced to a finite Bloch-spectral inequality,

or equivalently to a bounded-perturbation gap-stability audit.

What is still open is the actual microscopic verification of either:

$$\delta_{0,\rho} > \eta_{R,\rho},$$

or

$$\text{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \emptyset \text{ with } \sup_{|p| \leq p_*} \|\delta H_R(p)\| < \Delta_0.$$

a. Meaning. So Step 4 is closed at the reduction-theorem level, but not yet unconditionally closed for the final complete microscopic TECT realization.

CCXXV. FINAL SUMMARY OF THE REMOTE GAP STEP

a. Closed theorem-level statement. The remote-gap condition

$$\Delta > 0$$

is an independent spectral input. It is not implied by

$$\det M \neq 0.$$

b. Direct finite spectral route. Choose a finite remote Bloch/shell truncation and define

$$\delta_{0,\rho} = \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(H_R^{(N)}(p))), \quad \eta_{R,\rho} = \|H_R - H_R^{(N)}\|_\rho.$$

Then

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \implies \Delta_\rho \geq \delta_{0,\rho} - \eta_{R,\rho} > 0.}$$

c. Perturbative route. If an already-gapped realization satisfies

$$\text{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \emptyset$$

and

$$\sup_{|p| \leq p_*} \|\delta H_R(p)\| < \Delta_0,$$

then

$$\boxed{\Delta \geq \Delta_0 - \|\delta H_R\| > 0.}$$

d. Master explicit inequality. Equivalently, if an explicit diagonal benchmark $\Delta_{\text{bench},\rho}$ and total remote budget $\eta_{\text{rem},\rho}$ satisfy

$$\boxed{\Delta_{\text{bench},\rho} > \eta_{\text{rem},\rho},}$$

then

$$\boxed{\Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho} - \eta_{\text{rem},\rho} > 0.}$$

e. Schur-complement consequence. Combining Step 3 and Step 4,

$$V(0) = 0, \quad \|V(p)\| \leq c|p|, \quad \Delta > 0,$$

gives

$$\boxed{\Sigma(E, p) = O\left(\frac{|p|^2}{\Delta}\right), \quad H_{\text{eff}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right).}$$

So the remote sector changes only subleading curvature and does not destroy the leading linear Dirac cone.

f. Honest frontier. Therefore Step 4 is fully closed at the reduction-theorem level. What remains open is only the actual microscopic verification of one of:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho}}$$

or

$$\boxed{\text{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \emptyset \quad \text{and} \quad \|\delta H_R\| < \Delta_0.}$$

CCXXVI. MASS PROTECTION THEOREM AS A FINITE RESIDUAL-SYMMETRY EXCLUSION TEST

a. Goal. We now close the last local audit in the local Dirac stability chain: the absence of a symmetry-allowed constant Dirac mass. The correct theorem-level statement is that mass protection is not automatic from the doubled Dirac algebra itself. It requires an additional residual low-energy symmetry, and the entire question reduces to a finite classification problem on constant 4×4 Hermitian operators.

CCXXVII. DOUBLED LOCAL DIRAC BLOCK AND THE MASS SPACE

Assume the two opposite valleys have already been matched, so that the local doubled shell block is Dirac:

$$H_D(q) = \Gamma_i q_i. \quad (279)$$

In the standard valley $\otimes$ internal basis, one may write

$$H_D(q) = \tau_z \otimes (q_1 \sigma_1 + q_2 \sigma_2 + q_3 \sigma_3). \quad (280)$$

A constant Hermitian operator K is a Dirac mass iff

$$\{K, \Gamma_i\} = 0 \quad (i = 1, 2, 3). \quad (281)$$

For the doubled local Dirac block, the full constant mass space is exactly

$$\mathcal{M} = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}. \quad (282)$$

Therefore the doubled Dirac algebra by itself does not protect the node: without extra symmetry, a constant mass is allowed.

CCXXVIII. ALLOWED CONSTANT ALGEBRA

Let G_{res} be the residual low-energy symmetry group acting on the doubled local block, with representation $U(g)$. Define the allowed constant Hermitian algebra

$$A_{\text{res}} := \{K \in \text{Herm}(4) : U(g)KU(g)^{-1} = K \ \forall g \in G_{\text{res}}\}. \quad (283)$$

Then the mass-protection question is exactly:

$$A_{\text{res}} \cap \mathcal{M} \stackrel{?}{=} \{0\}. \quad (284)$$

[Finite mass-protection criterion] If

$$A_{\text{res}} \cap \mathcal{M} = \{0\}, \quad (285)$$

then no symmetry-allowed constant Dirac mass exists, and the local Dirac node is mass-protected.

Any constant gap-opening Dirac perturbation must lie in the mass space $\mathcal{M}$. A symmetry-allowed constant perturbation must lie in A_{res} . Therefore a symmetry-allowed constant Dirac mass exists iff

$$A_{\text{res}} \cap \mathcal{M} \neq \{0\}.$$

If the intersection is trivial, then no such mass exists.

CCXXIX. RESIDUAL VALLEY $U(1)_V$ AS A SUFFICIENT PROTECTION MECHANISM

Define the valley-charge generator

$$Q_V := \tau_z \otimes \mathbf{1}_2, \quad U_\theta := e^{i\theta Q_V/2}. \quad (286)$$

Since

$$[Q_V, H_D(q)] = 0, \quad (287)$$

the doubled kinetic block is compatible with a residual valley symmetry.

Now the two mass generators transform as

$$U_\theta(\tau_x \otimes \mathbf{1}_2)U_\theta^{-1} = \cos \theta (\tau_x \otimes \mathbf{1}_2) + \sin \theta (\tau_y \otimes \mathbf{1}_2), \quad (288)$$

$$U_\theta(\tau_y \otimes \mathbf{1}_2)U_\theta^{-1} = -\sin \theta (\tau_x \otimes \mathbf{1}_2) + \cos \theta (\tau_y \otimes \mathbf{1}_2). \quad (289)$$

Equivalently,

$$U_\theta((\tau_x \pm i\tau_y) \otimes \mathbf{1}_2)U_\theta^{-1} = e^{\pm i\theta}((\tau_x \pm i\tau_y) \otimes \mathbf{1}_2). \quad (290)$$

So the two mass generators carry nonzero valley charge.

[Mass protection from exact residual valley $U(1)_V$] Assume:

1. the doubled low-energy block has the kinetic form (280);
2. the constant operator under consideration preserves the little-group action at $p = 0$;
3. the residual valley symmetry generated by $Q_V = \tau_z \otimes \mathbf{1}_2$ is exact on the isolated doubled block.

Then every symmetry-allowed constant Hermitian operator is of the form

$$K = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2, \quad (291)$$

and in particular the genuine mass coefficients vanish:

$$m_1 = m_2 = 0. \quad (292)$$

Hence no constant Dirac mass is allowed.

By little-group scalarity, every constant Hermitian operator has the form

$$K = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2 + m_1 \tau_x \otimes \mathbf{1}_2 + m_2 \tau_y \otimes \mathbf{1}_2.$$

Exact $U(1)_V$ invariance requires

$$U_\theta K U_\theta^{-1} = K \quad \forall \theta.$$

The terms $\mathbf{1}_4$ and $\tau_z \otimes \mathbf{1}_2$ are invariant, but $\tau_x \otimes \mathbf{1}_2$ and $\tau_y \otimes \mathbf{1}_2$ rotate as a charged doublet by (288)–(290). Therefore they are forbidden unless

$$m_1 = m_2 = 0.$$

These are precisely the genuine constant Dirac mass coefficients.

CCXXX. WEAKER SUFFICIENT MECHANISM: RESIDUAL $\mathbb{Z}_2^V$

The full continuous $U(1)_V$ is stronger than necessary. The discrete remnant

$$S_V := \tau_z \otimes \mathbf{1}_2, \quad S_V^2 = \mathbf{1}_4 \quad (293)$$

already suffices, because

$$S_V(\tau_x \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_x \otimes \mathbf{1}_2), \quad S_V(\tau_y \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_y \otimes \mathbf{1}_2). \quad (294)$$

Hence

$$\boxed{A_{\mathbb{Z}_2^V} \cap \mathcal{M} = \{0\}}. \quad (295)$$

[Mass protection from exact residual $\mathbb{Z}_2^V$] An exact residual valley $\mathbb{Z}_2^V$ also forbids all genuine constant Dirac masses.

CCXXXI. WHY OPPOSITE-VALLEY MATCHING ALONE IS NOT ENOUGH

Opposite-valley matching gives

$$h_-(p) = -h_+(-p),$$

and therefore produces the doubled Dirac structure. However, by itself it does not remove the two constant mass generators

$$\tau_x \otimes \mathbf{1}_2, \quad \tau_y \otimes \mathbf{1}_2.$$

So opposite-valley matching is necessary for doubling, but not sufficient for mass protection.

CCXXXII. RADIATIVE STABILITY UNDER REMOTE-SECTOR ELIMINATION

Suppose the residual valley symmetry is exact in the full microscopic theory and the remote-sector elimination preserves it. Then the zero-momentum self-energy satisfies

$$[Q_V, \Sigma(0, 0)] = 0. \quad (296)$$

Under little-group scalarity this implies

$$\Sigma(0, 0) = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2, \quad (297)$$

so no radiatively generated term proportional to

$$\tau_x \otimes \mathbf{1}_2 \quad \text{or} \quad \tau_y \otimes \mathbf{1}_2$$

can appear.

[Radiative stability of mass protection] If the residual valley symmetry is exact in the full microscopic theory and remote-sector elimination preserves that symmetry, then no symmetry-preserving self-energy correction can generate a constant Dirac mass.

CCXXXIII. CONCRETE CLASS II COMPATIBILITY

The canonical minimal Class II representative

$$m^a(P_0) = m\delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j} \quad (298)$$

renormalizes the derivative witness coefficients, but does not by itself insert an independent constant 4×4 mass operator.

Thus Class II is mass-protection compatible, but

Class II compatibility is not itself a proof of mass protection.

The independent residual-symmetry audit is still required.

CCXXXIV. FINAL FINITE AUDIT PROTOCOL

The actual microscopic mass-protection check is therefore the following finite classification problem:

- a. (M1) Identify the actual residual low-energy symmetry group G_{res} acting on the local doubled Dirac block.
- b. (M2) Construct its 4×4 representation $U(g)$.
- c. (M3) Solve

$$U(g)KU(g)^{-1} = K \quad \forall g \in G_{\text{res}}$$

to obtain the allowed constant algebra $A_{\text{res}} \subset \text{Herm}(4)$.

- d. (M4) Inside A_{res} , solve the anticommutation equations

$$\{K, \Gamma_i\} = 0 \quad (i = 1, 2, 3).$$

- e. (M5) If the only solution is $K = 0$, conclude mass protection.

CCXXXV. SHARP HONEST CONCLUSION

The strongest honest statement is:

Mass protection is completely reduced to the finite residual-symmetry exclusion test (299)

$$A_{\text{res}} \cap \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\} = \{0\}. \quad (300)$$

A sufficient mechanism is exact residual valley $U(1)_V$, and a weaker but still sufficient mechanism is exact residual valley $\mathbb{Z}_2^V$.

What remains open is not the theorem. It is only the actual microscopic identification of the residual symmetry preserved by the final complete TECT realization.

CCXXXVI. FINAL SUMMARY OF THE MASS PROTECTION STEP

a. Closed theorem-level result. For the doubled local Dirac block,

$$\mathcal{M} = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}$$

is the full constant Dirac-mass space. Therefore mass protection never follows from the Dirac algebra alone.

b. Finite classification criterion. Let

$$A_{\text{res}} = \{K \in \text{Herm}(4) : U(g)KU(g)^{-1} = K \ \forall g \in G_{\text{res}}\}$$

be the residual symmetry-allowed constant algebra. Then mass protection is exactly the finite test

$$A_{\text{res}} \cap \mathcal{M} = \{0\}.$$

c. Sufficient symmetry mechanisms. An exact residual valley

$$Q_V = \tau_z \otimes \mathbf{1}_2$$

generates a $U(1)_V$ symmetry under which the two mass generators rotate as a charged doublet, so

$$A_{U(1)_V} \cap \mathcal{M} = \{0\}.$$

A weaker but still sufficient condition is the residual

$$\mathbb{Z}_2^V : \quad S_V = \tau_z \otimes \mathbf{1}_2,$$

under which both mass generators are odd, so again

$$A_{\mathbb{Z}_2^V} \cap \mathcal{M} = \{0\}.$$

d. Important limitation. Opposite-valley matching by itself does *not* imply mass protection. It produces the doubled Dirac structure, but does not by itself eliminate

$$\tau_x \otimes \mathbf{1}_2, \quad \tau_y \otimes \mathbf{1}_2.$$

e. Radiative stability. If the same residual valley symmetry is exact in the full microscopic theory and preserved under remote-sector elimination, then the zero-momentum self-energy satisfies

$$\Sigma(0,0) = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2,$$

so no radiatively generated constant Dirac mass can appear.

f. Honest frontier. So the mass-protection theorem itself is closed. What remains open is only the actual microscopic identification of the residual symmetry preserved by the final complete TECT realization:

$$\boxed{\text{determine } G_{\text{res}} \text{ and verify } A_{\text{res}} \cap \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\} = \{0\}.$$

CCXXXVII. LOCAL DIRAC CLOSURE THEOREM IN THE CONCRETE CLASS II SETTING

a. Goal. We now collect the five local closure steps into a single theorem. The point is to separate clearly:

1. what is already theorem-level closed inside the concrete Class II framework,
2. what still remains as actual microscopic verification for the full complete TECT realization.

CCXXXVIII. INPUT DATA

Let the shell-relevant local doubled block be built from an opposite shell pair

$$\pm G_*,$$

with matched shell-adapted frames and local Weyl witnesses

$$(W_{\parallel}^{\pm}, W_I^{\pm}, W_J^{\pm}).$$

Assume the concrete Class II shell-adapted witness coefficients

$$g_c, \quad g_E, \quad g_O$$

are extracted from the induced microscopic tensor via the established shell/defect dictionary.

The exact shell witness formulas in the concrete Class II setting are

$$\boxed{\lambda_{\parallel} = \frac{1}{2} g_c I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta},} \quad (301)$$

with

$$I_{\theta'^3} \neq 0, \quad I_{\theta} \neq 0.$$

Therefore local Weyl nondegeneracy is exactly

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).} \quad (302)$$

CCXXXIX. FIVE-STEP LOCAL CLOSURE HYPOTHESES

Assume the following five inputs:

a. (1) Weyl witness nonvanishing at each valley. At the two valleys $\pm G_*$,

$$W_{\parallel}^{\pm} \neq 0, \quad (W_I^{\pm}, W_J^{\pm}) \neq (0, 0). \quad (303)$$

b. (2) Opposite-valley witness matching. In the matched opposite-valley frame,

$$\boxed{(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+).} \quad (304)$$

c. (3) Zero constant mixing. For the block decomposition

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^{\dagger} & H_R(p) \end{pmatrix}, \quad (305)$$

assume

$$\boxed{V(0) = 0, \quad \|V(p)\| \leq c|p|.} \quad (306)$$

d. (4) Remote spectral gap. Assume the remote block is spectrally separated:

$$\boxed{\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset \quad \text{for small } p,} \quad (307)$$

for some $\Delta > 0$.

e. (5) Mass protection. Assume no symmetry-allowed constant Dirac mass exists on the isolated doubled block. Equivalently,

$$\boxed{A_{\text{res}} \cap \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\} = \{0\}.} \quad (308)$$

CCXL. MAIN THEOREM

[Local Dirac closure theorem in the concrete Class II setting] Assume the five inputs (303)–(308). Then the full microscopic TECT truncation contains a spectrally isolated four-dimensional low-energy sector whose effective Hamiltonian is

$$H_{\text{eff}}(E, p) = \Gamma_i(M_{\text{eff}}p)_i + O\left(\frac{|p|^2}{\Delta}\right), \quad \det M_{\text{eff}} \neq 0. \quad (309)$$

Consequently, the low-energy theory contains a stable massless local Dirac cone.

By input (1), each valley is Weyl. Indeed, the local shell witness determinant is nonzero exactly when

$$W_{\parallel}^{\pm} \neq 0, \quad (W_I^{\pm}, W_J^{\pm}) \neq (0, 0).$$

By input (2), the two opposite valleys carry identical witness triples in the matched frame. Hence

$$h_{-}(p) = -h_{+}(-p),$$

so the doubled shell block is local Dirac.

By input (3), the constant inter-sector mixing vanishes at the crossing, and the first allowed mixing is at least linear in momentum:

$$V(0) = 0, \quad \|V(p)\| \leq c|p|.$$

By input (4), the remote sector is uniformly gapped by $\Delta > 0$, so the Schur-complement self-energy obeys

$$\Sigma(E, p) = V(p)(H_R(p) - E)^{-1}V(p)^{\dagger} = O\left(\frac{|p|^2}{\Delta}\right).$$

By input (5), no symmetry-allowed constant Dirac mass can be generated on the isolated doubled block. Therefore the effective low-energy Hamiltonian is

$$H_{\text{eff}}(E, p) = \Gamma_i(M_{\text{eff}}p)_i + O\left(\frac{|p|^2}{\Delta}\right),$$

with nondegenerate linear coefficient matrix M_{eff} , and the resulting local Dirac cone is both spectrally isolated and massless.

CCXLI. CONCRETE EXPLICIT SUFFICIENT FORM

A concrete sufficient realization is the canonical minimal Class II representative

$$m^a(P_0) = m\delta^{a3}, \quad C_{ij}^{33} = c_{\parallel}\hat{n}_i\hat{n}_j, \quad C_{ij}^{31} = c_I\hat{n}_ie_{1j} + c_J\hat{n}_ie_{2j}, \quad C_{ij}^{32} = -c_J\hat{n}_ie_{1j} + c_I\hat{n}_ie_{2j}. \quad (310)$$

Then

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J. \quad (311)$$

Hence, if

$$\mu \neq 0, \quad c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0), \quad (312)$$

and inputs (2)–(5) above also hold, then the resulting shell block is Weyl at each valley and the local low-energy theory contains a stable massless Dirac fermion.

In the purely induced case, the effective shell coefficients are

$$A_{\text{eff}} = -\mu c_{\parallel}, \quad B_{\text{eff}} = -\mu c_I, \quad C_{\text{eff}} = -\mu c_J, \quad (313)$$

so the shell-level Weyl condition becomes simply

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0). \quad (314)$$

CCXLII. WHAT IS ACTUALLY CLOSED

The theorem above closes the following chain:

$$\boxed{\text{explicit Class II witness} \implies \text{Weyl at each valley} \implies \text{opposite-valley Dirac doubling} \implies \text{spectrally isolated stable massless Dirac}} \quad (315)$$

In particular, inside the concrete Class II truncation, the finite witness problem is already closed:

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0)$$

hold generically on an open dense full-measure subset of the allowed Class II tensor space once one admissible longitudinal seed and one admissible transverse seed exist.

CCXLIII. WHAT IS STILL OPEN

The theorem is *conditional* with respect to the final complete microscopic TECT realization. The remaining genuinely microscopic verification tasks are:

1. that the actual complete TECT action realizes a nontrivial Class II witness carrier sector or equivalent admissible carrier;
2. that the actual remote sector satisfies the little-group / shell-label mismatch audit for

$$V(0) = 0;$$

3. that the actual remote block satisfies the positive-gap condition

$$\Delta > 0;$$

4. that the actual residual symmetry algebra forbids all constant Dirac masses.

Thus the exact final status is

$$\boxed{\text{the local Dirac problem is fully reduced to a finite conditional closure theorem in the concrete Class II setting.}} \quad (316)$$

It is not yet a fully unconditional microscopic theorem for the complete TECT action.

CCXLIV. FINAL SUMMARY OF THE LOCAL DIRAC CLOSURE THEOREM IN THE CONCRETE CLASS II SETTING

a. Closed theorem-level content. Inside the concrete Class II framework, the local Dirac problem is reduced to five finite inputs:

- (1) Weyl witness nonvanishing at each valley,
- (2) opposite-valley witness matching,
- (3) zero constant mixing $V(0) = 0$, $\|V(p)\| \leq c|p|$,
- (4) remote spectral gap $\Delta > 0$,
- (5) mass protection.

If these hold, then the full microscopic low-energy theory contains a spectrally isolated four-dimensional sector with

$$\boxed{H_{\text{eff}}(E, p) = \Gamma_i (M_{\text{eff}} p)_i + O\left(\frac{|p|^2}{\Delta}\right), \quad \det M_{\text{eff}} \neq 0,}$$

hence a stable massless local Dirac cone.

b. Concrete explicit sufficient form. For the canonical minimal Class II representative

$$m^a(P_0) = m\delta^{a3}, \quad C_{ij}^{33} = c_{\parallel}\hat{n}_i\hat{n}_j, \quad C_{ij}^{31} = c_I\hat{n}_ie_{1j} + c_J\hat{n}_ie_{2j}, \quad C_{ij}^{32} = -c_J\hat{n}_ie_{1j} + c_I\hat{n}_ie_{2j},$$

one has

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

So if

$$\mu \neq 0, \quad c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0),$$

and the remaining three finite audits also hold, the local Dirac cone follows.

c. Exact honest status. What is now closed is the theorem chain

$$\boxed{\text{explicit Class II witness} \implies \text{Weyl} \implies \text{Dirac doubling} \implies \text{stable isolated massless Dirac cone.}}$$

What is still not unconditionally closed for the final complete TECT action is:

$$\boxed{\text{actual realization of the Class II carrier,}}$$

$$\boxed{\text{actual little-group mismatch for } V(0) = 0,}$$

$$\boxed{\text{actual remote-gap positivity } \Delta > 0,}$$

$$\boxed{\text{actual residual symmetry forbidding all constant masses.}}$$

Therefore the exact final status is:

$$\boxed{\text{the local Dirac problem is fully reduced to a finite conditional closure theorem in the concrete Class II setting,}}$$

but

$$\boxed{\text{it is not yet a fully unconditional microscopic theorem for the complete TECT action.}}$$

CCXLV. WHY THE CONVERSATION FELT REPETITIVE AND THE FIRST GENUINE CALCULATIONAL FORK

a. Honest status. The repetition occurred because the conservative proof path has essentially exhausted all theorem-level reductions that can be obtained without an explicit final microscopic operator $L(k)$. The uploaded documents themselves state that the following are still open on the full-PDE side:

$$L(k) \text{ in final form,} \quad K_i = \partial_{k_i} L(k) \Big|_{k=G_*}, \quad g_c, \ g_E, \ g_O.$$

So if one insists on staying fully conservative, there is no longer a new numerical coefficient to extract.

b. Computational fork. At this point there are only two honest options:

1. remain on the conservative path and stop at reduction theorems;
2. choose an explicit working microscopic branch and actually compute inside that branch, while clearly labeling it as a working branch rather than the final unconditional theorem.

The uploaded microscopic derivation draft provides exactly such a working branch:

$$N_{ij}^a = \frac{4Z}{A} \delta_{ij} \delta^{a0} + \frac{gA^2}{q_*^2} \hat{G}_{*i} \hat{G}_{*j} \sigma^a.$$

The same draft also explicitly says that the multi-mode BCC completion and polarization locking are still open, so this branch is computationally concrete but not yet the final full theorem.

CCXLVI. FIRST REAL CALCULATION IN THE WORKING MICROSCOPIC BRANCH

a. Input. Adopt the explicit enriched shell tensor

$$\boxed{\mathbf{N}_{ij}^a = \kappa_Z \delta_{ij} \delta^{a0} + \kappa_g \hat{G}_{*i} \hat{G}_{*j} \sigma^a,} \quad (317)$$

with

$$\kappa_Z = \frac{4Z}{A}, \quad \kappa_g = \frac{gA^2}{q_*^2}, \quad q_*^2 = -\frac{Z}{2Y}. \quad (318)$$

b. Shell-adapted witness contractions. In the shell-adapted frame $(\hat{n}, e_1, e_2)$, the uploaded derivation computes the three witness contractions as

$$\boxed{W_{\parallel} = \hat{n}_i \hat{n}_j \mathbf{N}_{ij}^3 = \kappa_g,} \quad (319)$$

$$\boxed{W_I = \frac{1}{2} \left(e_{1j} \hat{n}_i \mathbf{N}_{ij}^1 + e_{2j} \hat{n}_i \mathbf{N}_{ij}^2 \right) = \kappa_Z,} \quad (320)$$

$$\boxed{W_J = \frac{1}{2} \left(e_{1j} \hat{n}_i \mathbf{N}_{ij}^2 - e_{2j} \hat{n}_i \mathbf{N}_{ij}^1 \right) = 0.} \quad (321)$$

c. Immediate Weyl test. The shell-level Weyl determinant is

$$\det M = (\kappa_* |G_*|)^3 W_{\parallel} (W_I^2 + W_J^2). \quad (322)$$

Substituting (319)–(321),

$$\det M = (\kappa_* q_*)^3 \kappa_g \kappa_Z^2. \quad (323)$$

Using (318),

$$\det M = (\kappa_* q_*)^3 \left(\frac{gA^2}{q_*^2} \right) \left(\frac{4Z}{A} \right)^2 \quad (324)$$

$$= (\kappa_* q_*)^3 \frac{gA^2}{q_*^2} \frac{16Z^2}{A^2} \quad (325)$$

$$= 16 \kappa_* g Z^2 q_*. \quad (326)$$

Therefore

$$\boxed{\det M = 16 \kappa_* g Z^2 q_* > 0} \quad (327)$$

whenever

$$g > 0, \quad Z \neq 0, \quad q_* > 0.$$

d. Velocity data in this branch. The same derivation gives the explicit shell velocity matrix

$$M_{ai} = \begin{pmatrix} \kappa_Z & 0 & 0 \\ 0 & \kappa_Z & 0 \\ \kappa_g & 0 & 0 \end{pmatrix}, \quad (328)$$

so the branch predicts

$$v_{\parallel} = \kappa_g = \frac{gA^2}{q_*^2}, \quad v_1 = v_2 = |\kappa_Z| = \frac{4|Z|}{A}. \quad (329)$$

CCXLVII. WHAT THIS CALCULATION ACTUALLY ACHIEVES

This is a *real calculation*, not a reduction theorem. Inside this working branch we have explicitly obtained:

$$N_{ij}^a, \quad W_{\parallel}, W_I, W_J, \quad \det M.$$

So within this branch, the shell-level Weyl witness is already nonzero.

a. But it is not yet the final full theorem. The same uploaded draft explicitly says the following are still open:

multi-mode BCC completion, polarization locking from dynamics.

Therefore this calculation is best understood as:

$$\boxed{\text{a concrete microscopic working branch, not yet the final unconditional full-TECT theorem.}} \quad (330)$$

CCXLVIII. WHY THIS IS THE CORRECT NEXT MOVE

So the honest situation is now sharply split:

a. Conservative route. No further actual coefficient can be extracted without a final explicit $L(k)$.

b. Working-branch route. The uploaded microscopic derivation draft gives an explicit branch in which one can really calculate:

$$N_{ij}^a \rightarrow (W_{\parallel}, W_I, W_J) \rightarrow \det M.$$

Thus the conversation felt repetitive because we stayed too long on the conservative route. The correct non-repetitive next step is to continue the working branch calculation and attack its first remaining open problem:

$$\boxed{\text{extend the single-mode locked BCC derivation to the full 12-mode BCC shell sum.}}$$

CCXLIX. FINAL SUMMARY: WHY IT FELT REPETITIVE AND WHERE THE FIRST REAL CALCULATION BEGINS

a. Why it felt repetitive. The conversation repeated because the conservative route had already exhausted its theorem-level reductions. The uploaded documents themselves still say that the final full microscopic operator $L(k)$, its derivatives K_i , and the actual coefficients g_c, g_E, g_O are not yet extracted in final form.

b. What the first real calculation is. To move beyond repetition, one must choose a concrete working microscopic branch. The uploaded derivation draft provides exactly such a branch:

$$N_{ij}^a = \frac{4Z}{A} \delta_{ij} \delta^{a0} + \frac{gA^2}{q_*^2} \hat{G}_{*i} \hat{G}_{*j} \sigma^a.$$

Inside this branch one can really compute:

$$W_{\parallel} = \kappa_g, \quad W_I = \kappa_Z, \quad W_J = 0,$$

and therefore

$$\boxed{\det M = 16 \kappa_* g Z^2 q_* > 0.}$$

c. Honest status. So this branch already gives a genuine microscopic calculation of the shell witness. However, it is still not the final full theorem, because the same draft leaves open:

multi-mode BCC completion, polarization locking from dynamics.

d. What this means. The repetition came from staying on the conservative route too long. A real next step *does* exist, but it requires committing to the working microscopic branch and then extending it from the single-mode locked representative to the full 12-mode BCC shell sum.

CCL. A FIRST GENUINE 12-MODE BCC SHELL-SUM CALCULATION

a. Working branch. We now stop the theorem-only reductions and perform an actual shell-sum calculation in the first explicit calculational branch that is already available from the uploaded documents:

$$\Psi_{\text{BCC}}(x) = z \phi_0(x), \quad z^\dagger z = 1,$$

with the polarization-locked equal-amplitude BCC background

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x) = A \sum_{Q \in S_{\text{BCC}}} e^{iQ \cdot x}. \quad (331)$$

This locked sector is rigorously equivalent to the scalar Brazovskii shell theory, but the unrestricted triplet polarization-locking problem is still open. Thus the present calculation is a genuine microscopic working branch, not yet the final unrestricted full-TECT theorem.

CCLI. STEP 1: SHELL-PROJECTED KINETIC CONTRIBUTION REMAINS SINGLE-MODE LOCAL

From (331), the Fourier coefficient of the background at each shell vector is

$$\widehat{\phi}_0(Q) = A \quad (Q \in S_{\text{BCC}}).$$

Therefore, when we project onto a selected shell valley $G_* \in S_{\text{BCC}}$, the phase-matched kinetic cross-term sees exactly the same local shell amplitude A as in the single-mode representative.

Hence the identity-channel enriched tensor contribution stays

$$\boxed{\kappa_Z^{(12)} = \frac{4Z}{A}}. \quad (332)$$

So the transverse witness contribution inherited from the kinetic sector is unchanged at the shell-projected level:

$$\boxed{W_I^{(12)} = \kappa_Z^{(12)} = \frac{4Z}{A}, \quad W_J^{(12)} = 0}. \quad (333)$$

CCLII. STEP 2: FULL 12-MODE BCC COMBINATORIAL FACTOR AT THE SELECTED SHELL

The new effect of the full 12-mode BCC background appears in the shell-projected quartic combinatorics. For the selected shell vector G_* , define

$$N(G_*) := \#\{(Q_1, Q_2) \in S_{\text{BCC}}^2 : Q_1 + Q_2 = G_*\}. \quad (334)$$

The uploaded BCC shell computation gives

$$\boxed{N(G_*) = 4 \quad \text{for every } G_* \in S_{\text{BCC}}}. \quad (335)$$

To see this explicitly, choose for instance

$$G_* = \frac{1}{\sqrt{2}}(1, 1, 0).$$

Then the ordered shell pairs satisfying $Q_1 + Q_2 = G_*$ are exactly:

$$\frac{1}{\sqrt{2}}(1, 0, 1) + \frac{1}{\sqrt{2}}(0, 1, -1) = G_*, \quad (336)$$

$$\frac{1}{\sqrt{2}}(1, 0, -1) + \frac{1}{\sqrt{2}}(0, 1, 1) = G_*, \quad (337)$$

together with the two order-reversed pairs, giving four ordered pairs in total.

Therefore the shell-projected quartic contribution picks up the positive BCC multiplicity factor

$$\mathcal{C}_{\text{BCC}} = N(G_*) = 4.$$

CCLIII. STEP 3: 12-MODE LOCKED CONTINUATION OF THE LONGITUDINAL COEFFICIENT

In the single-mode working branch, the quartic longitudinal coefficient was

$$\kappa_g^{(1)} = \frac{gA^2}{q_*^2}.$$

Under the phase-matched 12-mode locked shell projection, the natural continuation is therefore

$$\boxed{\kappa_g^{(12)} = \mathcal{C}_{\text{BCC}} \frac{gA^2}{q_*^2} = 4 \frac{gA^2}{q_*^2}.} \quad (338)$$

Thus the longitudinal witness becomes

$$\boxed{W_{\parallel}^{(12)} = \kappa_g^{(12)} = 4 \frac{gA^2}{q_*^2}.} \quad (339)$$

So the full 12-mode locked shell sum does not wash out the longitudinal witness. It enhances it by a positive combinatorial factor.

CCLIV. STEP 4: WEYL DETERMINANT IN THE 12-MODE LOCKED BRANCH

Using the shell-level determinant formula

$$\det M = \kappa_* |G_*|^3 W_{\parallel} (W_I^2 + W_J^2), \quad (340)$$

and substituting

$$|G_*| = q_*, \quad W_{\parallel}^{(12)} = 4 \frac{gA^2}{q_*^2}, \quad W_I^{(12)} = \frac{4Z}{A}, \quad W_J^{(12)} = 0,$$

we obtain

$$\det M^{(12)} = \kappa_* q_*^3 \left(4 \frac{gA^2}{q_*^2} \right) \left(\frac{4Z}{A} \right)^2 \quad (341)$$

$$= \kappa_* q_*^3 \cdot 4 \frac{gA^2}{q_*^2} \cdot \frac{16Z^2}{A^2} \quad (342)$$

$$= 64 \kappa_* g Z^2 q_*. \quad (343)$$

Therefore

$$\boxed{\det M^{(12)} = 64 \kappa_* g Z^2 q_* > 0} \quad (344)$$

for

$$g > 0, \quad Z < 0, \quad q_* > 0.$$

So the shell-projected Weyl witness remains strictly nondegenerate in the locked 12-mode BCC continuation.

CCLV. WHAT THIS CALCULATION ACTUALLY PROVES

This is now a real calculational result.

Inside the polarization-locked equal-amplitude 12-mode BCC branch, the shell-projected enriched witness data are

$$\boxed{W_{\parallel}^{(12)} = 4 \frac{gA^2}{q_*^2}, \quad W_I^{(12)} = \frac{4Z}{A}, \quad W_J^{(12)} = 0,} \quad (345)$$

and hence

$$\boxed{\det M^{(12)} = 64 \kappa_* g Z^2 q_* > 0.} \quad (346)$$

So the single-mode witness does survive the first genuine 12-mode locked BCC shell sum.

CCLVI. WHAT THIS CALCULATION DOES NOT YET PROVE

It does *not* yet prove the final unrestricted full-TECT theorem, for two reasons:

1. the unrestricted triplet polarization-locking problem is still open;
2. the exact full multi-mode projector/BdG linearization has not yet been carried out in the complete microscopic theory.

Therefore the present result should be read as:

$$\boxed{\text{12-mode locked equal-amplitude BCC shell sum: closed and nondegenerate.}} \quad (347)$$

but not yet:

$$\boxed{\text{full unrestricted triplet multi-mode microscopic theorem: closed.}} \quad (348)$$

CCLVII. FINAL SUMMARY OF THE FIRST 12-MODE BCC SHELL-SUM CALCULATION

a. What was actually calculated. Inside the polarization-locked equal-amplitude 12-mode BCC working branch

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x) = A \sum_{Q \in S_{\text{BCC}}} e^{iQ \cdot x},$$

the selected-shell phase-matched continuation gives

$$\kappa_Z^{(12)} = \frac{4Z}{A}.$$

The BCC shell combinatorics at the selected shell vector satisfies

$$N(G_*) = \#\{(Q_1, Q_2) \in S_{\text{BCC}}^2 : Q_1 + Q_2 = G_*\} = 4.$$

Therefore the quartic longitudinal coefficient is enhanced to

$$\kappa_g^{(12)} = 4 \frac{gA^2}{q_*^2}.$$

Hence the shell-projected witness data are

$$\boxed{W_{\parallel}^{(12)} = 4 \frac{gA^2}{q_*^2}, \quad W_I^{(12)} = \frac{4Z}{A}, \quad W_J^{(12)} = 0.}$$

b. Weyl determinant. The determinant becomes

$$\boxed{\det M^{(12)} = 64 \kappa_* g Z^2 q_* > 0.}$$

So the Weyl witness survives the first genuine 12-mode locked BCC shell sum.

c. Honest status. This is a real calculational advance beyond the single-mode representative. But it is still only a result in the

locked equal-amplitude 12-mode BCC branch,

not yet the final unrestricted triplet multi-mode theorem, because

polarization locking is still open,

and

the full multi-mode microscopic projector/BdG linearization is still open.

d. Meaning. So we now know something genuinely new:

the shell witness does not disappear when the single-mode representative is lifted to the locked 12-mode BCC shell sum.

CCLVIII. FIRST EXPLICIT BLOCH-PROJECTED CALCULATION IN THE LOCKED 12-MODE BCC BRANCH

a. Goal. We now perform the first genuinely informative Bloch-projected calculation around the locked 12-mode BCC background. The purpose is to compute the projected 4×4 shell Hamiltonian at the Bragg point and identify the first actual microscopic obstruction.

CCLIX. LOCKED 12-MODE BCC BACKGROUND AND BRAGG SHELL BASIS

Work in the polarization-locked equal-amplitude BCC branch:

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x) = A \sum_{Q \in S_{\text{BCC}}} e^{iQ \cdot x}, \quad A_{G_*} = A. \quad (349)$$

Fix a selected shell vector $G_* \in S_{\text{BCC}}$, and the Bragg quasimomentum

$$k_* = \frac{G_*}{2}. \quad (350)$$

The genuine minimal shell-Dirac carrier is

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2. \quad (351)$$

Choose the constant-envelope shell basis

$$u_a(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_a, \quad \xi_a^\dagger \xi_b = \delta_{ab}, \quad a, b = 1, 2, \quad (352)$$

and define

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x). \quad (353)$$

Projecting the Bloch Hamiltonian gives

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}, \quad p = k - k_*. \quad (354)$$

CCLX. ACTUAL CALCULATION OF THE CONSTANT BRAGG MASS B_0

For the constant-envelope ansatz, the shell-pair mixing at $p = 0$ is

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2. \quad (355)$$

Now take the literal quadratic scalar periodic potential

$$U_{\text{per}}(x) = c \phi_0(x)^2. \quad (356)$$

Using

$$\phi_0(x)^2 = A^2 \sum_{Q_1, Q_2 \in S_{\text{BCC}}} e^{i(Q_1 + Q_2) \cdot x}, \quad (357)$$

the first-shell Fourier coefficient is

$$\widehat{\phi_0^2}(G_*) = A^2 N(G_*), \quad N(G_*) := \#\{(Q_1, Q_2) \in S_{\text{BCC}}^2 : Q_1 + Q_2 = G_*\}. \quad (358)$$

For every $G_* \in S_{\text{BCC}}$,

$$N(G_*) = 4. \quad (359)$$

Hence

$$\boxed{\widehat{U}(G_*) = 4cA^2, \quad B_0 = 4cA^2 \mathbf{1}_2 \neq 0.} \quad (360)$$

Therefore the literal quadratic scalar periodic potential produces a nonzero constant Bragg mass.

a. Standard scalar ϕ^4 Hessian. For the standard scalar Hessian,

$$U_{\text{per}}^{\text{tot}}(x) = u|\phi_0(x)|^2,$$

and in the usual notation $u = 3g$, so

$$\boxed{B_0 = 12gA^2 \mathbf{1}_2.} \quad (361)$$

Thus the plain scalar periodic sector by itself is not a shell-Dirac realization.

CCLXI. ENRICHED DERIVATIVE COUPLINGS SURVIVE AT FIRST SHELL

Now add the minimal enriched periodic coefficients

$$T^\alpha(x) = C_{\alpha j} \partial_j \phi_0(x) \otimes s_\alpha, \quad \alpha = 1, 2. \quad (362)$$

Since

$$\partial_j \phi_0(x) = \sum_{Q \in S_{\text{BCC}}} iQ_j A_Q e^{iQ \cdot x}, \quad (363)$$

their first-shell Fourier coefficients are

$$\widehat{T}^\alpha(G_*) = i(C_\alpha \cdot G_*) A_{G_*}. \quad (364)$$

Therefore

$$W^\alpha = \widehat{T}^\alpha(G_*) s_\alpha = i(C_\alpha \cdot G_*) A_{G_*} s_\alpha. \quad (365)$$

Hence the exact first-shell survival criterion is

$$\boxed{A_{G_*} \neq 0, \quad C_1 \cdot G_* \neq 0, \quad C_2 \cdot G_* \neq 0.} \quad (366)$$

Under this condition,

$$W^1 \neq 0, \quad W^2 \neq 0,$$

and because s_1, s_2 anticommute, the transverse shell-Dirac structure is already present.

CCLXII. PROJECTED 4×4 SHELL HAMILTONIAN

Combining the nonzero scalar Bragg mass with the enriched derivative couplings, the projected shell Hamiltonian takes the form

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + B_0 \tau_1 \otimes \mathbf{1}_2 + p_1 \tau_1 \otimes W^1 + p_2 \tau_1 \otimes W^2 + O(|p|^2). \quad (367)$$

Using (360) and (365),

$$\boxed{H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + 4cA^2 \tau_1 \otimes \mathbf{1}_2 + i(C_1 \cdot G_*) A_{p_1} \tau_1 \otimes s_1 + i(C_2 \cdot G_*) A_{p_2} \tau_1 \otimes s_2 + O(|p|^2).} \quad (368)$$

CCLXIII. IMMEDIATE SPECTRAL CONSEQUENCE

At $p = 0$, the shell-pair sector is already gapped by the constant Bragg mass:

$$H_{\text{proj}}(0) = E_0(0)\mathbf{1}_4 + 4cA^2 \tau_1 \otimes \mathbf{1}_2. \quad (369)$$

So the eigenvalues are

$$E(0) = E_0(0) \pm 4cA^2, \quad (370)$$

each doubly degenerate. Therefore there is no exact massless shell crossing at the Bragg point unless an additional mechanism cancels B_0 .

$B_0 \neq 0 \implies \text{no exact shell Dirac point at } p = 0 \text{ in the plain scalar periodic branch.}$

(371)

CCLXIV. WHAT THIS CALCULATION ACTUALLY SHOWS

This is the first real Bloch-projected calculation in the locked 12-mode BCC branch, and it shows something nontrivial:

1. The enriched derivative couplings W^1, W^2 can survive explicitly at the selected shell.
2. The literal scalar periodic background also produces a nonzero constant Bragg mass B_0 .
3. Therefore the first actual microscopic obstruction is not the absence of transverse Dirac structure, but the presence of a nonzero scalar pair-mixing mass.

So the next honest microscopic task is no longer vague. It is:

find the actual cancellation or symmetry mechanism that kills B_0 while retaining $W^1, W^2, v_{\parallel} \neq 0$.

(372)

CCLXV. SHARP HONEST CONCLUSION

Inside the locked 12-mode BCC constant-envelope Bloch reduction, the plain scalar periodic Hessian does not yet produce a shell Dirac point, because

$$B_0 = \widehat{U}(G_*)\mathbf{1}_2$$

is generically nonzero.

At the same time, the enriched derivative sector already provides the correct two transverse traceless matrices. So the true remaining microscopic obstruction is isolated exactly:

the shell-Dirac problem has been reduced to the elimination of the constant Bragg mass B_0 .

(373)

CCLXVI. FINAL SUMMARY OF THE FIRST BLOCH-PROJECTED 12-MODE CALCULATION

a. What was actually computed. In the locked equal-amplitude 12-mode BCC branch, using the constant-envelope shell basis at

$$k_* = \frac{G_*}{2},$$

the projected shell Hamiltonian takes the form

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}.$$

For the literal quadratic scalar periodic potential

$$U_{\text{per}}(x) = c \phi_0(x)^2, \quad \phi_0(x) = A \sum_{Q \in S_{\text{BCC}}} e^{iQ \cdot x},$$

one finds

$$\widehat{\phi_0^2}(G_*) = A^2 N(G_*), \quad N(G_*) = 4,$$

hence

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2 = 4cA^2 \mathbf{1}_2 \neq 0.$$

b. Meaning. Therefore the plain scalar periodic branch does *not* yet give an exact shell Dirac point:

$$B_0 \neq 0 \implies \text{there is a constant Bragg mass obstruction at } p = 0.$$

c. At the same time, the enriched transverse sector survives. For

$$T^\alpha(x) = C_{\alpha j} \partial_j \phi_0(x) \otimes s_\alpha,$$

one gets

$$W^\alpha = i(C_\alpha \cdot G_*) A_{G_*} s_\alpha.$$

So if

$$A_{G_*} \neq 0, \quad C_1 \cdot G_* \neq 0, \quad C_2 \cdot G_* \neq 0,$$

then the two transverse traceless shell matrices are already present.

d. True obstruction isolated. So the real microscopic obstruction is now sharply identified:

$$\text{not the absence of transverse Dirac structure, but the presence of } B_0 \neq 0.$$

e. Next actual task. The next real calculation is therefore:

$$\text{find the actual cancellation or symmetry mechanism that kills } B_0 \text{ while keeping } W^1, W^2, v_{\parallel} \neq 0.$$

CCLXVII. EXACT CANCELLATION CONDITION FOR THE BRAGG MASS IN THE LOCKED 12-MODE BCC BRANCH

a. Goal. We now perform an actual new calculation. Inside the locked equal-amplitude 12-mode BCC branch, we compute the first-shell Fourier coefficients of the most natural scalar periodic operators and derive the exact linear cancellation condition for

$$B_0^{\text{tot}} = 0.$$

CCLXVIII. LOCKED 12-MODE BCC BACKGROUND

Work in the equal-amplitude locked BCC branch

$$\phi_0(x) = A \sum_{Q \in S_{\text{BCC}}} e^{iQ \cdot x}, \quad |Q| = q_* \quad (Q \in S_{\text{BCC}}), \quad (374)$$

with

$$S_{\text{BCC}} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}.$$

Fix a selected shell vector

$$G_* = \frac{q_*}{\sqrt{2}} (1, 1, 0).$$

We need the first-shell Fourier coefficient at G_* .

CCLXIX. ORDERED-PAIR COMBINATORICS

Define

$$N(G_*) := \#\{(Q_1, Q_2) \in S_{\text{BCC}}^2 : Q_1 + Q_2 = G_*\}. \quad (375)$$

For the BCC first shell, the ordered solutions are exactly:

$$Q_1 = \frac{q_*}{\sqrt{2}}(1, 0, 1), \quad Q_2 = \frac{q_*}{\sqrt{2}}(0, 1, -1), \quad (376)$$

$$Q_1 = \frac{q_*}{\sqrt{2}}(1, 0, -1), \quad Q_2 = \frac{q_*}{\sqrt{2}}(0, 1, 1), \quad (377)$$

together with the two order-reversed pairs. Hence

$$\boxed{N(G_*) = 4.} \quad (378)$$

For every such contributing pair one also has

$$Q_1 \cdot Q_2 = -\frac{q_*^2}{2}. \quad (379)$$

CCLXX. FIRST-SHELL FOURIER COEFFICIENTS OF NATURAL SCALAR OPERATORS

We now compute three natural scalar periodic operators.

A. A. The basic quadratic scalar block ϕ_0^2

Using

$$\phi_0(x)^2 = A^2 \sum_{Q_1, Q_2 \in S_{\text{BCC}}} e^{i(Q_1 + Q_2) \cdot x},$$

the G_* -Fourier coefficient is

$$\widehat{\phi_0^2}(G_*) = A^2 N(G_*) = 4A^2. \quad (380)$$

B. B. The gradient scalar block $|\nabla \phi_0|^2$

Since

$$\nabla \phi_0 = iA \sum_Q Q e^{iQ \cdot x},$$

we have

$$|\nabla \phi_0|^2 = -A^2 \sum_{Q_1, Q_2} (Q_1 \cdot Q_2) e^{i(Q_1 + Q_2) \cdot x}.$$

Therefore

$$\widehat{|\nabla \phi_0|^2}(G_*) = -A^2 \sum_{Q_1 + Q_2 = G_*} Q_1 \cdot Q_2. \quad (381)$$

Using (379) and $N(G_*) = 4$,

$$\boxed{\widehat{|\nabla \phi_0|^2}(G_*) = -4A^2 \left(-\frac{q_*^2}{2}\right) = 2q_*^2 A^2.} \quad (382)$$

C. C. The mixed Laplacian scalar block $\phi_0(-\Delta)\phi_0$

Because every shell vector has length q_* ,

$$(-\Delta)e^{iQ \cdot x} = q_*^2 e^{iQ \cdot x}.$$

Hence

$$\phi_0(-\Delta)\phi_0 = q_*^2 \phi_0^2,$$

pointwise in this locked first-shell branch. Therefore

$$\boxed{\phi_0(\widehat{-\Delta})\phi_0(G_*) = q_*^2 \widehat{\phi_0^2}(G_*) = 4q_*^2 A^2.} \quad (383)$$

CCLXXI. GENERAL SCALAR BRAGG BLOCK AND THE EXACT CANCELLATION CONDITION

Take the most general scalar periodic contribution built from these three channels:

$$U_{\text{tot}}(x) = c_0 \phi_0(x)^2 + c_1 |\nabla \phi_0(x)|^2 + c_2 \phi_0(x)(-\Delta)\phi_0(x). \quad (384)$$

Then the constant Bragg mass in the shell-pair sector is

$$B_0^{\text{tot}} = \widehat{U}_{\text{tot}}(G_*) \mathbf{1}_2. \quad (385)$$

Using (380), (382), and (383),

$$\widehat{U}_{\text{tot}}(G_*) = c_0 (4A^2) + c_1 (2q_*^2 A^2) + c_2 (4q_*^2 A^2) \quad (386)$$

$$= 2A^2 (2c_0 + q_*^2 c_1 + 2q_*^2 c_2). \quad (387)$$

Therefore

$$\boxed{B_0^{\text{tot}} = 2A^2 (2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2.} \quad (388)$$

Hence the exact cancellation condition is

$$\boxed{2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0.} \quad (389)$$

This is the first exact linear cancellation law for the Bragg mass in the locked 12-mode BCC branch.

CCLXXII. STANDARD ϕ^4 HESSIAN AS A SPECIAL CASE

For the standard scalar quartic Hessian, the periodic scalar block is

$$U_{\text{per}}^{(\phi^4)}(x) = 3g \phi_0(x)^2,$$

so

$$c_0 = 3g, \quad c_1 = 0, \quad c_2 = 0.$$

Then

$$\boxed{B_0^{(\phi^4)} = 12g A^2 \mathbf{1}_2,} \quad (390)$$

which is nonzero for $g > 0$.

Thus the plain scalar quartic branch is indeed obstructed.

CCLXXIII. HOW TO KILL B_0 WITHOUT TOUCHING THE DERIVATIVE WITNESS SECTOR

Suppose we add extra scalar periodic operators while keeping the derivative-enriched sector unchanged. Then $W^1, W^2, v_{\parallel}$ are controlled by the enriched derivative couplings, whereas B_0 is controlled by the scalar periodic block. So at this shell-projected level one can kill B_0 independently by imposing (389).

For example:

a. Case 1. If only ϕ_0^2 and $|\nabla\phi_0|^2$ are present, i.e. $c_2 = 0$, then

$$\boxed{c_1 = -\frac{2c_0}{q_*^2}.} \quad (391)$$

b. Case 2. If only ϕ_0^2 and $\phi_0(-\Delta)\phi_0$ are present, i.e. $c_1 = 0$, then

$$\boxed{c_2 = -\frac{c_0}{q_*^2}.} \quad (392)$$

c. Case 3. For the standard scalar quartic obstruction $c_0 = 3g$, the general exact cancellation condition is

$$\boxed{c_1 + 2c_2 = -\frac{6g}{q_*^2}.} \quad (393)$$

So the obstruction is not mysterious anymore: it is a one-equation linear tuning problem in the scalar periodic sector.

CCLXXIV. WHAT THIS CALCULATION ACTUALLY ACHIEVES

This is a genuinely new calculational step.

It proves that, in the locked 12-mode BCC branch:

1. the plain scalar quartic contribution gives

$$B_0^{(\phi^4)} = 12gA^2 \mathbf{1}_2 \neq 0;$$

2. the first-shell Fourier coefficients of the natural scalar operators are

$$\widehat{\phi_0^2}(G_*) = 4A^2, \quad \widehat{|\nabla\phi_0|^2}(G_*) = 2q_*^2 A^2, \quad \phi_0(-\Delta)\phi_0(G_*) = 4q_*^2 A^2;$$

3. the exact cancellation condition for the constant Bragg mass is

$$2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0.$$

So the B_0 -obstruction is now reduced to a concrete scalar-coefficient cancellation equation.

CCLXXV. HONEST STATUS

This still does not prove that complete unrestricted TECT realizes such a cancellation dynamically. What is now established is only:

$$\boxed{\text{if the scalar periodic sector of the actual locked 12-mode branch satisfies } 2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0, \text{ then } B_0^{\text{tot}} = 0.} \quad (394)$$

The remaining microscopic question is whether complete TECT naturally generates such a relation, or whether it must arise from an additional symmetry or tuning mechanism.

CCLXXVI. FINAL SUMMARY OF THE EXACT B_0 -CANCELLATION CALCULATION

a. What was newly computed. In the locked equal-amplitude 12-mode BCC branch, for a selected shell vector G_* , the exact first-shell Fourier coefficients are

$$\widehat{\phi_0^2}(G_*) = 4A^2,$$

$$\widehat{|\nabla\phi_0|^2}(G_*) = 2q_*^2 A^2,$$

$$\phi_0(\widehat{-\Delta})\phi_0(G_*) = 4q_*^2 A^2.$$

b. General scalar Bragg block. For

$$U_{\text{tot}}(x) = c_0 \phi_0^2 + c_1 |\nabla\phi_0|^2 + c_2 \phi_0(-\Delta)\phi_0,$$

the constant Bragg mass is

$$B_0^{\text{tot}} = 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2.$$

c. Exact cancellation condition. Therefore

$$B_0^{\text{tot}} = 0 \iff 2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0.$$

d. Standard scalar quartic obstruction. For the plain scalar ϕ^4 Hessian,

$$c_0 = 3g, \quad c_1 = c_2 = 0,$$

so

$$B_0^{(\phi^4)} = 12gA^2 \mathbf{1}_2 \neq 0.$$

e. Meaning. So the B_0 -obstruction is no longer vague. In the locked 12-mode BCC branch it is reduced to one explicit linear scalar-coefficient equation.

f. Honest frontier. What remains open is whether complete unrestricted TECT produces this cancellation dynamically, or whether an additional symmetry / operator / tuning mechanism is required.

CCLXXVII. NO-GO CALCULATION FOR B_0 -CANCELLATION INSIDE THE CURRENTLY KNOWN SAFE OPERATOR LIBRARY

a. Goal. We now perform the next genuinely decisive calculation. The question is:

Does the currently known microscopic operator library already contain a sector that can cancel B_0 while leaving $W^1, W^2, v_{\parallel}$ int

The answer, with the currently uploaded operator set, is no.

CCLXXVIII. EXACT CANCELLATION LAW FROM THE PREVIOUS STEP

In the locked equal-amplitude 12-mode BCC branch, we already derived the exact scalar Bragg-mass formula

$$B_0^{\text{tot}} = 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2. \quad (395)$$

Hence the exact cancellation condition is

$$B_0^{\text{tot}} = 0 \iff 2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0. \quad (396)$$

So the problem is now completely explicit: we only need to identify the actual coefficients c_0, c_1, c_2 contributed by the currently known microscopic sectors.

CCLXXIX. SECTOR 1: ACTUAL SCALAR HESSIAN

Inside the actual Q -sector scalar Hessian, the uploaded derivation states that the only periodic scalar contribution is

$$U_{\text{per}}^{\text{tot}}(x) = u |\phi_0(x)|^2, \quad (397)$$

and that the other rank-one Hessian terms vanish identically on Q . Therefore, in the notation of (395),

$$\boxed{c_0^{(\text{scalar})} = u, \quad c_1^{(\text{scalar})} = 0, \quad c_2^{(\text{scalar})} = 0.} \quad (398)$$

Substituting into (395),

$$\boxed{B_0^{(\text{scalar})} = 4uA^2 \mathbf{1}_2.} \quad (399)$$

Thus the actual scalar Hessian channel by itself is obstructed.

CCLXXX. SECTOR 2: ALIGNMENT TERM

The refined alignment term is

$$F_{\text{align}} = \frac{\chi}{4} \int d^3x \rho^2 \text{Tr}(\partial_i P \partial_i P). \quad (400)$$

But on the locked sector

$$\Psi = z \phi, \quad P = zz^\dagger = \text{const},$$

one has

$$\partial_i P = 0 \quad \implies \quad F_{\text{align}} = 0. \quad (401)$$

Therefore the alignment term contributes no periodic scalar block at all on the locked branch:

$$\boxed{c_0^{(\text{align})} = 0, \quad c_1^{(\text{align})} = 0, \quad c_2^{(\text{align})} = 0.} \quad (402)$$

Equivalently,

$$\boxed{B_0^{(\text{align})} = 0.} \quad (403)$$

So the alignment term is structurally safe, but it does not help with the Bragg-mass cancellation.

CCLXXXI. SECTOR 3: CLASS II

The exact Class II witness formulas modify only the shell-relevant derivative witness coefficients:

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J. \quad (404)$$

This means Class II acts in the derivative witness sector

$$(W_{\parallel}, W_I, W_J)$$

rather than in the scalar identity-channel Bragg block.

Therefore, at the level of the scalar Bragg-mass coefficients,

$$\boxed{c_0^{(\text{Class II})} = 0, \quad c_1^{(\text{Class II})} = 0, \quad c_2^{(\text{Class II})} = 0,} \quad (405)$$

so

$$\boxed{B_0^{(\text{Class II})} = 0 \quad \text{as a direct scalar cancellation channel.}} \quad (406)$$

Hence Class II renormalizes the desired derivative witness sector, but does not cancel the dangerous scalar Bragg mass.

CCLXXXII. TOTAL KNOWN MICROSCOPIC OPERATOR LIBRARY

Collecting (398), (402), and (405), the total currently known safe operator library gives

$$\boxed{c_0^{(\text{known})} = u, \quad c_1^{(\text{known})} = 0, \quad c_2^{(\text{known})} = 0.} \quad (407)$$

Therefore

$$2c_0^{(\text{known})} + q_*^2 c_1^{(\text{known})} + 2q_*^2 c_2^{(\text{known})} = 2u. \quad (408)$$

If $u \neq 0$, then the exact cancellation condition (396) fails. Hence

$$\boxed{B_0^{(\text{known})} = 4uA^2 \mathbf{1}_2 \neq 0.} \quad (409)$$

CCLXXXIII. NO-GO THEOREM

[No-go for B_0 -cancellation inside the currently known safe operators] Inside the presently known microscopic operator library consisting of:

1. the actual Q -sector scalar Hessian,
2. the projector-alignment term F_{align} ,
3. and the Class II derivative witness sector,

the scalar Bragg-mass coefficients are

$$(c_0, c_1, c_2) = (u, 0, 0).$$

Therefore the exact cancellation condition

$$2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0$$

cannot hold unless $u = 0$. For the condensed phase with nontrivial scalar Hessian $u \neq 0$,

$$\boxed{B_0^{\text{tot}} = 4uA^2 \mathbf{1}_2 \neq 0.}$$

The scalar-Hessian contribution is (398). The alignment term vanishes on the locked sector by (401), hence gives (402). Class II modifies only the derivative witness sector, giving (405) as far as the scalar Bragg block is concerned. Summing the three contributions yields (407), and then (409) follows immediately from (395).

CCLXXXIV. MEANING OF THE NO-GO RESULT

This is not just another reduction theorem. It is an actual negative calculation:

$$\boxed{\text{there is no currently known minimal safe operator that both preserves the witness sector and cancels } B_0.} \quad (410)$$

So the next microscopic escape route must come from something outside the current safe operator library.

CCLXXXV. WHAT SURVIVES AFTER THE NO-GO RESULT

Importantly, the no-go result is only about the scalar Bragg block. It does *not* kill the derivative witness sector. Indeed, the enriched shell witness mechanism remains intact:

$$W^1, \quad W^2, \quad v_{\parallel}$$

are still carried by the derivative-enriched structure, while the obstruction sits entirely in

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

Thus the problem has become even sharper:

$$\boxed{\text{keep the derivative witness sector, but remove the scalar identity-channel Bragg block by a new microscopic mechanism.}} \quad (411)$$

CCLXXXVI. WHAT THE ACTUAL REMAINING ROUTES ARE

The uploaded analysis already isolates the only remaining possibilities:

1. an exact cancellation involving additional *non-scalar* sectors not present in the actual scalar Hessian;
2. a graded/projection mechanism in which the dangerous scalar pair block is absent from the relevant reduced operator;
3. a genuinely different microscopic completion beyond the plain scalar-Hessian channel.

So the next real calculation is no longer ambiguous. It must target one of these three routes.

CCLXXXVII. FINAL SUMMARY OF THE B_0 -CANCELLATION NO-GO CALCULATION

a. What was actually computed. Using the exact cancellation law

$$B_0^{\text{tot}} = 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2,$$

we evaluated the contributions of the currently known safe microscopic sectors.

b. Results. For the actual scalar Hessian:

$$(c_0, c_1, c_2) = (u, 0, 0).$$

For the alignment term on the locked sector:

$$(c_0, c_1, c_2) = (0, 0, 0).$$

For Class II as a direct scalar cancellation channel:

$$(c_0, c_1, c_2) = (0, 0, 0).$$

Therefore the total currently known operator library gives

$$(c_0, c_1, c_2)_{\text{known}} = (u, 0, 0).$$

c. Exact consequence. Hence

$$B_0^{\text{known}} = 4uA^2 \mathbf{1}_2 \neq 0 \quad (u \neq 0).$$

d. Meaning. So there is no currently known minimal safe operator that both preserves the derivative witness sector and cancels the Bragg mass B_0 .

e. What remains. The only remaining microscopic escape routes are:

- (i) an additional non-scalar sector,
- (ii) a graded/projection mechanism removing the scalar pair block,
- (iii) a genuinely different microscopic completion.

This is a real no-go calculation, not just another reformulation.

CCLXXXVIII. FIRST ACTUAL CALCULATION IN THE GRADED $2+2$ BYPASS ROUTE

a. Goal. We now stop discussing cancellation inside the plain scalar Bragg-pair block and move entirely to the graded $2+2$ bypass mechanism.

The objective is to compute explicitly:

1. why the dangerous constant scalar Bragg block does not appear in the relevant graded principal operator;
2. the determinant and spectrum of the graded first-order reduced operator;
3. the exact finite nondegeneracy condition that replaces the old B_0 -cancellation problem.

CCLXXXIX. GRADED 2 + 2 LOW-ENERGY SECTOR

Let

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2,$$

and let J be the grading involution

$$J^{\dagger} = J, \quad J^2 = 1, \quad J|_{K_{\pm}} = \pm 1. \quad (412)$$

Let the reduced effective operator be

$$D_{\text{eff}}(p) = \Pi_K L(p) \Pi_K = D_1(p) + O(|p|^2), \quad (413)$$

and assume the leading term is:

1. first-order in momentum,
2. J -odd:

$$J D_1(p) J^{-1} = -D_1(p). \quad (414)$$

Then in the J -graded basis,

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^{\dagger} & 0 \end{pmatrix}, \quad (415)$$

with

$$A_1(p) = p_i M_i. \quad (416)$$

This is exactly the theorem-level graded bypass setup already established in the manuscripts.

CCXC. WHY THE SCALAR BRAGG MASS IS ABSENT HERE

The dangerous scalar obstruction in the plain shell-pair route was

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2,$$

which is momentum-independent.

But in the graded bypass route, the relevant principal infrared operator is by definition first-order:

$$A_1(p) = p_i M_i.$$

Hence

$$A_1(0) = 0, \quad D_1(0) = 0. \quad (417)$$

So the scalar Bragg block is not cancelled inside D_1 ; it is absent because the graded principal operator starts at first order in momentum. This is the precise meaning of the bypass mechanism.

CCXCI. GENERAL 2 × 2 MIXED SYMBOL

Write the mixed symbol in Pauli form:

$$A_1(p) = \ell_0(p) \mathbf{1}_2 + \ell_1(p) \sigma_1 + \ell_2(p) \sigma_2 + \ell_3(p) \sigma_3, \quad (418)$$

where each coefficient is linear in momentum:

$$\ell_{\alpha}(p) = C_{\alpha}^i p_i, \quad \alpha = 0, 1, 2, 3. \quad (419)$$

Equivalently,

$$A_1(p) = p_i \left(C_0^i \mathbf{1}_2 + C_a^i \sigma_a \right). \quad (420)$$

a. Important point. The identity part

$$\ell_0(p)\mathbf{1}_2$$

is not a mass obstruction, because it is still linear in p . The manuscripts already state this explicitly: the identity component is principal-level irrelevant once the full symbol is nondegenerate.

CCXCII. EXACT DETERMINANT CALCULATION

For any 2×2 matrix of the form

$$A = \ell_0\mathbf{1}_2 + \ell_a\sigma_a,$$

the standard Pauli identity gives

$$\det A = \ell_0^2 - \ell_1^2 - \ell_2^2 - \ell_3^2. \quad (421)$$

Therefore

$$\det A_1(p) = \ell_0(p)^2 - \ell_1(p)^2 - \ell_2(p)^2 - \ell_3(p)^2. \quad (422)$$

Now define the symmetric matrix

$$G_{ij} := \sum_{a=1}^3 C_a^i C_a^j - C_0^i C_0^j. \quad (423)$$

Then

$$\det A_1(p) = -p_i G_{ij} p_j. \quad (424)$$

So the entire nondegeneracy problem is reduced to a finite quadratic form.

CCXCIII. DETERMINANT OF THE FULL GRADED PRINCIPAL OPERATOR

Since

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix},$$

one has

$$\det D_1(p) = \det(A_1(p)A_1(p)^\dagger) = |\det A_1(p)|^2. \quad (425)$$

Hence

$$\det D_1(p) = (p_i G_{ij} p_j)^2. \quad (426)$$

Therefore the graded principal operator is nondegenerate for generic $p \neq 0$ iff

$$p_i G_{ij} p_j \neq 0 \quad \text{for generic } p \neq 0. \quad (427)$$

A particularly clean sufficient condition is:

$$G_{ij} > 0 \quad (\text{positive definite}). \quad (428)$$

Then

$$p_i G_{ij} p_j > 0 \quad (\forall p \neq 0),$$

so

$$\det A_1(p) < 0, \quad \det D_1(p) > 0 \quad (\forall p \neq 0).$$

CCXCIV. SPECTRUM OF THE GRADED PRINCIPAL OPERATOR

Assume for simplicity that the coefficients $\ell_\alpha(p)$ are real. Then $A_1(p)$ is Hermitian and its eigenvalues are

$$\lambda_\pm(A_1) = \ell_0(p) \pm |\vec{\ell}(p)|, \quad |\vec{\ell}| := \sqrt{\ell_1^2 + \ell_2^2 + \ell_3^2}. \quad (429)$$

Therefore the eigenvalues of $D_1(p)$ are

$$\boxed{E(p) = \pm(\ell_0(p) \pm |\vec{\ell}(p)|)}. \quad (430)$$

So the zero-energy cone is determined by

$$\ell_0(p)^2 = |\vec{\ell}(p)|^2,$$

equivalently

$$p_i G_{ij} p_j = 0.$$

a. Interpretation. The identity coefficient C_0^i produces a tilt, not a constant mass. The true load-bearing data are the traceless coefficient vectors C_a^i , together with the effective quadratic form G_{ij} .

CCXCV. CANONICAL DIRAC REDUCTION

If G_{ij} is nondegenerate and has Euclidean signature in the spatial momenta, then by an invertible linear change of coordinates in p and a unitary change of basis in $K_\pm$, the mixed symbol can be brought to the canonical form

$$\boxed{A_1(p) \sim p_i \sigma_i}. \quad (431)$$

Consequently,

$$\boxed{D_1(p) \sim \begin{pmatrix} 0 & p_i \sigma_i \\ p_i \sigma_i & 0 \end{pmatrix} = p_i \alpha_i}, \quad (432)$$

with

$$\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}, \quad \{\alpha_i, \alpha_j\} = 2\delta_{ij} \mathbf{1}_4.$$

Thus the graded bypass produces the massless Dirac principal operator once the finite nondegeneracy test on G_{ij} passes.

CCXCVI. WHAT THIS CALCULATION ACTUALLY ACHIEVES

This is the first actual computation in the graded bypass route. It proves explicitly that:

1. the dangerous scalar Bragg block is absent from the relevant principal operator because

$$D_1(0) = 0;$$

2. the identity part of the mixed symbol is not a constant mass, but only a linear tilt contribution;
3. the entire Dirac nondegeneracy problem is reduced to the finite quadratic-form test

$$G_{ij} = \sum_{a=1}^3 C_a^i C_a^j - C_0^i C_0^j;$$

4. if G_{ij} is positive definite, then the graded principal operator is nondegenerate and canonically Dirac.

CCXCVII. SHARP HONEST CONCLUSION

The graded/projection mechanism has now been pushed one step further than the theorem statement alone. The real remaining microscopic question is no longer

“how do we cancel B_0 ?”

but rather

does the actual projected TECT low-energy $2 + 2$ sector produce C_α^i such that $G_{ij} > 0$?

So the graded bypass route is now reduced to an explicit finite coefficient-extraction problem:

extract $C_0^i, C_1^i, C_2^i, C_3^i$ from the projected microscopic transport operator.

CCXCVIII. FINAL SUMMARY OF THE FIRST ACTUAL CALCULATION IN THE GRADED BYPASS ROUTE

a. What was actually computed. In the graded $2 + 2$ bypass, the relevant principal infrared operator is

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_i M_i.$$

Therefore

$$D_1(0) = 0,$$

so the dangerous constant scalar Bragg block is absent from the relevant graded principal operator.

b. General mixed symbol. Writing

$$A_1(p) = \ell_0(p) \mathbf{1}_2 + \ell_a(p) \sigma_a, \quad \ell_\alpha(p) = C_\alpha^i p_i,$$

one gets

$$\det A_1(p) = \ell_0^2 - \ell_1^2 - \ell_2^2 - \ell_3^2.$$

c. Finite nondegeneracy matrix. Defining

$$G_{ij} = \sum_{a=1}^3 C_a^i C_a^j - C_0^i C_0^j,$$

we obtain

$$\det A_1(p) = -p_i G_{ij} p_j, \quad \det D_1(p) = (p_i G_{ij} p_j)^2.$$

d. Meaning. So the graded bypass route is no longer about cancelling B_0 . It is about extracting the coefficient vectors C_α^i and checking whether

$$G_{ij} > 0$$

(or more generally whether $p_i G_{ij} p_j \neq 0$ for generic $p \neq 0$).

e. If this holds. Then the graded principal operator is nondegenerate and can be reduced to

$$D_1(p) \sim p_i \alpha_i,$$

i.e. the massless Dirac principal operator.

f. Honest frontier. Therefore the real remaining microscopic question in the graded route is:

does the actual projected TECT 2 + 2 sector produce coefficients C_α^i with $G_{ij} > 0$?

CCXCIX. ACTUAL EXTRACTION OF C_α^i IN THE GRADED 2 + 2 ROUTE

A. Goal

We continue from the previous graded 2 + 2 bypass calculation. The abstract principal operator was

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = \ell_0(p)\mathbf{1}_2 + \ell_a(p)\sigma_a, \quad \ell_\alpha(p) = C_\alpha^i p_i.$$

The next real task is to extract the coefficient vectors

$$C_0^i, \quad C_1^i, \quad C_2^i, \quad C_3^i$$

from the actual projected transport operator.

B. Projected overlap matrices

Let the graded low modes be

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \quad u_a^-(\xi) = f_-(\xi)e_a(\xi), \quad a = 1, 2.$$

The projected mixed overlap matrices are

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle = \int d\xi (u_a^+)^\dagger V^\mu(\xi) u_b^-. \quad (433)$$

This is the actual local projector-defect extraction formula.

The reduced mixed symbol is then

$$A_1(p) = p_\mu M^\mu. \quad (434)$$

C. Pauli decomposition and exact coefficient extraction

Every Hermitian 2×2 matrix M^μ decomposes uniquely as

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i, \quad i = 1, 2, 3. \quad (435)$$

The coefficients are extracted by traces:

$$C_0^\mu = \frac{1}{2} \text{Tr}(M^\mu), \quad C_i^\mu = \frac{1}{2} \text{Tr}(\sigma_i M^\mu). \quad (436)$$

Substituting into (434),

$$A_1(p) = p_\mu \left[C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i \right]. \quad (437)$$

Therefore

$$\ell_0(p) = p_\mu C_0^\mu, \quad \ell_i(p) = p_\mu C_i^\mu.$$

This closes the extraction map

$$V^\mu(\xi) \implies M^\mu \implies C_\alpha^\mu.$$

D. Actual local minimal transport class

For the actual local projector-defect operator class already identified in the TECT chain, the projected matrices take the form

$$M^\mu = I_\theta \left[\Lambda_\parallel n^\mu \sigma_3 + \Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right]. \quad (438)$$

Equivalently, the reduced S2 symbol is

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (439)$$

with

$$\lambda_\parallel = I_\theta \Lambda_\parallel, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (440)$$

E. Adapted-basis coefficient vectors

Choose the shell/worldvolume momentum basis

$$p = (p_1, p_2, p_\parallel).$$

Comparing (437) with (439), the Pauli coefficients are read off explicitly.

For the p_1 -direction:

$$A_1(p) \supset p_1(\alpha \sigma_1 + \beta \sigma_2),$$

hence

$$C_0^1 = 0, \quad C_1^1 = \alpha = I_\theta \Lambda_E, \quad C_2^1 = \beta = I_\theta \Lambda_O, \quad C_3^1 = 0. \quad (441)$$

For the p_2 -direction:

$$A_1(p) \supset p_2(-\beta \sigma_1 + \alpha \sigma_2),$$

so

$$C_0^2 = 0, \quad C_1^2 = -\beta = -I_\theta \Lambda_O, \quad C_2^2 = \alpha = I_\theta \Lambda_E, \quad C_3^2 = 0. \quad (442)$$

For the longitudinal direction:

$$A_1(p) \supset p_\parallel \lambda_\parallel \sigma_3,$$

thus

$$C_0^\parallel = 0, \quad C_1^\parallel = 0, \quad C_2^\parallel = 0, \quad C_3^\parallel = \lambda_\parallel = I_\theta \Lambda_\parallel. \quad (443)$$

Therefore, in the minimal actual local transport class,

$$C_0^i = 0 \quad \text{for all } i = 1, 2, \parallel. \quad (444)$$

F. The finite nondegeneracy matrix G_{ij}

Recall

$$G_{ij} = \sum_{a=1}^3 C_a^i C_a^j - C_0^i C_0^j. \quad (445)$$

Since $C_0^i = 0$, this reduces to

$$G_{ij} = \sum_{a=1}^3 C_a^i C_a^j. \quad (446)$$

Using (441), (442), and (443), we get:

a. Diagonal entries

$$\begin{aligned} G_{11} &= (C_1^1)^2 + (C_2^1)^2 + (C_3^1)^2 \\ &= \alpha^2 + \beta^2 = I_\theta^2(\Lambda_E^2 + \Lambda_O^2), \end{aligned} \quad (447)$$

$$\begin{aligned} G_{22} &= (C_1^2)^2 + (C_2^2)^2 + (C_3^2)^2 \\ &= \beta^2 + \alpha^2 = I_\theta^2(\Lambda_E^2 + \Lambda_O^2), \end{aligned} \quad (448)$$

$$\begin{aligned} G_{\parallel\parallel} &= (C_1^\parallel)^2 + (C_2^\parallel)^2 + (C_3^\parallel)^2 \\ &= \lambda_\parallel^2 = I_\theta^2 \Lambda_\parallel^2. \end{aligned} \quad (449)$$

b. Off-diagonal entries

$$\begin{aligned} G_{12} &= C_1^1 C_1^2 + C_2^1 C_2^2 + C_3^1 C_3^2 \\ &= \alpha(-\beta) + \beta(\alpha) + 0 = 0, \end{aligned} \quad (450)$$

$$G_{1\parallel} = C_1^1 C_1^\parallel + C_2^1 C_2^\parallel + C_3^1 C_3^\parallel = 0, \quad (451)$$

$$G_{2\parallel} = C_1^2 C_1^\parallel + C_2^2 C_2^\parallel + C_3^2 C_3^\parallel = 0. \quad (452)$$

Hence

$$G_{ij} = I_\theta^2 \begin{pmatrix} \Lambda_E^2 + \Lambda_O^2 & 0 & 0 \\ 0 & \Lambda_E^2 + \Lambda_O^2 & 0 \\ 0 & 0 & \Lambda_\parallel^2 \end{pmatrix}. \quad (453)$$

G. Equivalence with the witness condition

Therefore

$$G_{ij} > 0 \iff I_\theta \neq 0, \quad \Lambda_\parallel \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (454)$$

Since the explicit defect profile already gives

$$I_\theta \neq 0,$$

we conclude

$$G_{ij} > 0 \iff \Lambda_\parallel \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (455)$$

But by (440), this is equivalently

$$G_{ij} > 0 \iff \lambda_\parallel \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (456)$$

So the graded nondegeneracy criterion is exactly the actual witness criterion in the local projector-defect operator class.

H. Determinant and Dirac reduction

From the previous graded calculation,

$$\det A_1(p) = -p_i G_{ij} p_j, \quad \det D_1(p) = (p_i G_{ij} p_j)^2.$$

Using (453),

$$\det A_1(p) = -I_\theta^2 \left[(\Lambda_E^2 + \Lambda_O^2)(p_1^2 + p_2^2) + \Lambda_\parallel^2 p_\parallel^2 \right]. \quad (457)$$

Hence for generic $p \neq 0$,

$$\boxed{\det A_1(p) \neq 0 \iff \Lambda_{\parallel} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (458)$$

Therefore the graded principal operator is nondegenerate exactly when the actual witness coefficients are nondegenerate.

After unitary basis changes in $K_{\pm}$ and an invertible linear redefinition of $(p_1, p_2, p_{\parallel})$, this yields the canonical Dirac symbol

$$D_1(p) \sim p_i \alpha_i.$$

I. Sharp honest conclusion

The next step beyond the abstract graded theorem is now fully explicit:

$$\boxed{V^{\mu}(\xi) \implies M^{\mu} \implies C_{\alpha}^i \implies G_{ij}} \quad (459)$$

and in the actual local projector-defect operator class this reduces to

$$\boxed{G_{ij} > 0 \iff \Lambda_{\parallel} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (460)$$

Thus the real remaining microscopic problem in the graded bypass route is simply: extract the actual transport coefficients

$$\Lambda_{\parallel}, \quad \Lambda_E, \quad \Lambda_O$$

from the full TECT defect-projector transport kernel.

CCC. FINAL SUMMARY OF THE NEXT STEP IN THE GRADED BYPASS ROUTE

a. What was actually computed. The abstract coefficient vectors

$$C_{\alpha}^i$$

of the graded $2 + 2$ principal symbol can be extracted directly from the projected overlap matrices

$$(M^{\mu})_{ab} = \langle u_a^+, V^{\mu} u_b^- \rangle, \quad M^{\mu} = C_0^{\mu} \mathbf{1}_2 + C_i^{\mu} \sigma_i.$$

b. In the actual local projector-defect transport class. Using

$$M^{\mu} = I_{\theta} \left[\Lambda_{\parallel} n^{\mu} \sigma_3 + \Lambda_E E^{\mu}_a \sigma_a + \Lambda_O J^{\mu}_a \sigma_a \right],$$

the coefficient vectors are explicitly

$$C_0^i = 0,$$

$$(C_1^1, C_2^1, C_3^1) = (I_{\theta} \Lambda_E, I_{\theta} \Lambda_O, 0),$$

$$(C_1^2, C_2^2, C_3^2) = (-I_{\theta} \Lambda_O, I_{\theta} \Lambda_E, 0),$$

$$(C_1^{\parallel}, C_2^{\parallel}, C_3^{\parallel}) = (0, 0, I_{\theta} \Lambda_{\parallel}).$$

c. Finite nondegeneracy matrix. Therefore

$$\boxed{G_{ij} = I_{\theta}^2 \begin{pmatrix} \Lambda_E^2 + \Lambda_O^2 & 0 & 0 \\ 0 & \Lambda_E^2 + \Lambda_O^2 & 0 \\ 0 & 0 & \Lambda_{\parallel}^2 \end{pmatrix}.}$$

d. Key equivalence. Hence the graded bypass nondegeneracy condition is exactly

$$G_{ij} > 0 \iff \Lambda_{\parallel} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

Equivalently,

$$G_{ij} > 0 \iff \lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

e. Meaning. So the graded bypass route is no longer an independent mystery. Inside the actual local projector-defect operator class, it reduces exactly to the same three transport coefficients

$$\Lambda_{\parallel}, \quad \Lambda_E, \quad \Lambda_O$$

that already control the witness problem.

f. Honest frontier. Therefore the true remaining microscopic task is now sharply localized:

$$\boxed{\text{derive the actual transport kernel } V^{\mu}(\xi) \text{ and extract } \Lambda_{\parallel}, \Lambda_E, \Lambda_O \text{ from the full TECT dynamics.}}$$

CCCI. QUADRATIC CLASS II REDUCTION AND EXPLICIT PROJECTION TO (g_c, g_E, g_O)

A. Scope and honest input

We start from the Class II UV action

$$L_{UV}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a[\Psi] + \beta X_i^a K_i^a[P, \psi]. \quad (461)$$

At the current document level, the full unreduced closed forms of $J_i^a[\Psi]$ and $K_i^a[P, \psi]$ are not yet available in the complete TECT action. What *is* available is the shell-relevant quadratic reduction template and its induced tensor summary:

$$\boxed{\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.} \quad (462)$$

The task is therefore to derive (462) from the reduced quadratic Class II structure and then project it to the shell-adapted coefficients (A, B, C) , equivalently (g_c, g_E, g_O) .

B. Background and fluctuations

Fix the refined locked background

$$\Psi_0 = \phi_0 z_0, \quad P_0 = z_0 z_0^{\dagger}, \quad z_0^{\dagger} z_0 = 1, \quad (463)$$

and write the shell-relevant fluctuations as

$$\Psi = \Psi_0 + \psi, \quad P = P_0 + \delta P.$$

For the projector fluctuation, parameterize

$$\delta P = \xi^b T^b + \dots, \quad (464)$$

where $b = 1, 2, 3$ labels the shell-relevant internal tangent directions.

C. Reduced quadratic Class II ansatz

At shell-relevant quadratic order, the only pieces needed for the induced first-order enriched tensor are:

1. a background source linear in X_i^a and $\partial_i \phi_0$,
2. a shell-fluctuation source linear in X_i^a , $D_j \psi$, and ξ^b .

Thus the reduced quadratic Class II Lagrangian has the form

$$L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda m^a X_i^a (\partial_i \phi_0) + g_X X_i^a C_{ij}^{ab} (D_j \psi) \xi^b + \text{h.c.} \quad (465)$$

Here:

- m^a is the internal background orientation source,
- C_{ij}^{ab} is the shell-relevant carrier tensor,
- λ and g_X are reduced couplings inherited from α, β and the reduction of $J_i^a[\Psi]$, $K_i^a[P, \psi]$.

D. Integrating out the heavy mediator

Because X_i^a is heavy, its leading equation of motion is algebraic:

$$M_X^2 X_i^a + \lambda m^a \partial_i \phi_0 + g_X C_{ij}^{ab} (D_j \psi) \xi^b = 0. \quad (466)$$

Hence

$$X_i^a = -\frac{1}{M_X^2} \left[\lambda m^a \partial_i \phi_0 + g_X C_{ij}^{ab} (D_j \psi) \xi^b \right] + O(M_X^{-4}). \quad (467)$$

Substituting (467) back into (465), the mixed bilinear term is

$$\delta L_{\text{mix}}^{(2)} = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab} (\partial_i \phi_0) (D_j \psi) \xi^b + \text{h.c.} \quad (468)$$

Therefore the induced shell-enriched tensor is exactly

$$\boxed{\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.} \quad (469)$$

E. Shell-adapted minimal Class II seed family

Choose the shell frame

$$\hat{n} = \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\} \text{ orthonormal,}$$

and the minimal shell-adapted Class II seed family

$$m^a(P_0) = m \delta^{a3}, \quad (470)$$

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad (471)$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}. \quad (472)$$

All other shell-irrelevant components are set to zero in the minimal seed truncation.

Define

$$\Gamma := \frac{\lambda g_X m}{M_X^2}. \quad (473)$$

Then (469) gives

$$\delta N_{ij}^3 = -\Gamma c_{\parallel} \hat{n}_i \hat{n}_j, \quad (474)$$

$$\delta N_{ij}^1 = -\Gamma (c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}), \quad (475)$$

$$\delta N_{ij}^2 = -\Gamma (-c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}). \quad (476)$$

F. Projection to the axial shell basis

Compare (474)–(476) with the canonical axial shell basis

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (477)$$

Reading off coefficients directly:

$$\boxed{\delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J.} \quad (478)$$

Thus the induced Class II correction preserves the exact shell-axial structure and only renormalizes the three coefficients.

G. Dictionary to (g_c, g_E, g_O)

Using the already established shell/defect dictionary

$$A_{\text{eff}} \leftrightarrow g_c, \quad B_{\text{eff}} \leftrightarrow g_E, \quad C_{\text{eff}} \leftrightarrow g_O,$$

equation (478) implies

$$\boxed{g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.} \quad (479)$$

The overall proportionality is controlled by Γ and by the defect-overlap normalization.

a. Important honest point. At the current document level, (479) is the strongest fully honest result. The nonzero witness directions are extracted explicitly, but the final absolute numerical normalization of (g_c, g_E, g_O) still depends on the detailed completion of the shell/defect overlap dictionary in the full complete TECT action.

H. Immediate seed consequences

From (479) one gets the three canonical witness directions:

a. Longitudinal seed. If

$$c_{\parallel} \neq 0, \quad c_I = c_J = 0,$$

then

$$g_c \neq 0.$$

b. Even transverse seed. If

$$c_{\parallel} = 0, \quad c_I \neq 0, \quad c_J = 0,$$

then

$$g_E \neq 0.$$

c. Odd transverse seed. If

$$c_{\parallel} = 0, \quad c_I = 0, \quad c_J \neq 0,$$

then

$$g_O \neq 0.$$

Thus, provided the full complete TECT action actually realizes a nontrivial Class II sector in the relevant defect background, the witness channels are present on an open dense full-measure subset of the allowed Class II parameter space.

I. What is now really closed

The actual quadratic Class II computation closes the chain

$$L_{\text{UV}}^{(II)} \implies \delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab} \implies (\delta A, \delta B, \delta C) = -\Gamma(c_{\parallel}, c_I, c_J). \quad (480)$$

Equivalently,

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J. \quad (481)$$

J. What still remains open

What remains open is not the shell-adapted projection itself, but:

1. the exact microscopic realization of the Class II carrier sector in the true complete TECT action;
2. the final absolute normalization of (g_c, g_E, g_O) in the full dynamics.

CCCII. FINAL SUMMARY OF THE CLASS II QUADRATIC PROJECTION CALCULATION

a. What was actually done. Starting from the Class II UV action

$$L_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a[\Psi] + \beta X_i^a K_i^a[P, \psi],$$

and reducing it to the shell-relevant quadratic sector around the locked background (Ψ_0, P_0) , one obtains

$$L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda m^a X_i^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (D_j \psi) \xi^b + \text{h.c.}$$

Integrating out X_i^a yields the induced enriched tensor

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.$$

b. Shell-adapted minimal Class II seed. For

$$m^a(P_0) = m \delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j,$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j},$$

this gives

$$\delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J, \quad \Gamma = \frac{\lambda g_X m}{M_X^2}.$$

c. *Witness coefficients.* Using the shell/defect dictionary

$$A_{\text{eff}} \leftrightarrow g_c, \quad B_{\text{eff}} \leftrightarrow g_E, \quad C_{\text{eff}} \leftrightarrow g_O,$$

one gets

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

d. *Honest status.* So the corrected Path C is no longer merely schematic: its shell-adapted induced tensor and witness directions are explicitly derived at the quadratic template level.

What remains open is:

$$\text{whether the full complete TECT action really realizes the Class II carrier sector,}$$

and

$$\text{the final absolute normalization of } (g_c, g_E, g_O) \text{ in the complete dynamics.}$$

CCCI. NORMALIZATION EXTRACTION AND EFFECTIVE WITNESS AUDIT FOR THE CLASS II ROUTE

CCCIV. PART I. STRONGEST HONEST NORMALIZATION EXTRACTION AVAILABLE NOW

a. *Scope.* At the present document level, the complete unreduced microscopic forms of

$$J_i^a[\Psi], \quad K_i^a[P, \psi]$$

are not yet available in the final full TECT action. Therefore the absolute microscopic values of

$$g_c, \quad g_E, \quad g_O$$

cannot yet be extracted to the very end from first principles.

What *is* already closed is the shell-relevant quadratic Class II reduction:

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}. \quad (482)$$

A. Quadratic reduction and raw normalization scale

Starting from the shell-relevant quadratic Class II template

$$L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda m^a X_i^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (D_j \psi) \xi^b + \text{h.c.}, \quad (483)$$

the heavy mediator equation is

$$M_X^2 X_i^a + \lambda m^a \partial_i \phi_0 + g_X C_{ij}^{ab} (D_j \psi) \xi^b = 0. \quad (484)$$

Thus

$$X_i^a = -\frac{1}{M_X^2} \left[\lambda m^a \partial_i \phi_0 + g_X C_{ij}^{ab} (D_j \psi) \xi^b \right] + O(M_X^{-4}), \quad (485)$$

and substituting back gives the induced mixed bilinear

$$\delta L_{\text{mix}}^{(2)} = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab} (\partial_i \phi_0) (D_j \psi) \xi^b + \text{h.c.} \quad (486)$$

Hence (482) follows.

Define the raw Class II normalization scale

$$\Gamma := \frac{\lambda g_X m}{M_X^2}. \quad (487)$$

B. Shell-adapted minimal seed family

Take the axial shell frame

$$\hat{n} = \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\} \text{ orthonormal},$$

and the minimal Class II seed parameterization

$$m^a(P_0) = m \delta^{a3}, \quad (488)$$

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad (489)$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}. \quad (490)$$

Then

$$\delta N_{ij}^3 = -\Gamma c_{\parallel} \hat{n}_i \hat{n}_j, \quad (491)$$

$$\delta N_{ij}^1 = -\Gamma (c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}), \quad (492)$$

$$\delta N_{ij}^2 = -\Gamma (-c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}). \quad (493)$$

Comparing with the canonical axial shell basis

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}, \quad (494)$$

we read off

$$\boxed{\delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J.} \quad (495)$$

C. Reduced witness normalization

Using the already established dictionary

$$A_{\text{eff}} \leftrightarrow g_c, \quad B_{\text{eff}} \leftrightarrow g_E, \quad C_{\text{eff}} \leftrightarrow g_O,$$

the strongest honest current extraction is

$$\boxed{g_c = \nu_{\parallel} \Gamma c_{\parallel}, \quad g_E = \nu_{\perp} \Gamma c_I, \quad g_O = \nu_{\perp} \Gamma c_J,} \quad (496)$$

where $\nu_{\parallel}, \nu_{\perp}$ are the remaining shell/defect-overlap normalization constants of the complete microscopic reduction.

Equivalently, at the transport-symbol level,

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad (497)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa. \quad (498)$$

Substituting (496) into (497)–(498),

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \nu_{\parallel} \Gamma c_{\parallel} \theta_0^3 \kappa^3,} \quad (499)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16}\nu_{\perp}\Gamma c_I\theta_0\kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16}\nu_{\perp}\Gamma c_J\theta_0\kappa. \quad (500)$$

Therefore the maximum honest normalization extraction currently available is:

$$\Gamma = \frac{\lambda g_X m}{M_X^2}$$

at the mediator-reduction level, together with the universal defect-overlap factors

$$-\frac{3\sqrt{3}\pi}{512}\theta_0^3\kappa^3, \quad -\frac{\sqrt{3}\pi}{16}\theta_0\kappa.$$

CCCV. PART II. EFFECTIVE WITNESS AUDIT

A. General effective coefficients

Math12 gives the canonical Class II effective shell coefficients

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J, \quad (501)$$

with

$$\mu = \frac{\lambda g_X}{M_X^2} m = \Gamma. \quad (502)$$

The corresponding reduced Weyl coefficients are

$$\lambda_{\parallel}^{\text{eff}} = \kappa_* |G_*| A_{\text{eff}}, \quad \alpha_{\text{eff}} = \kappa_* |G_*| B_{\text{eff}}, \quad \beta_{\text{eff}} = \kappa_* |G_*| C_{\text{eff}}. \quad (503)$$

Therefore

$$\det M_{\text{eff}} = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2). \quad (504)$$

Hence the exact shell-level nondegeneracy condition is

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0). \quad (505)$$

B. Obstruction baseline + Class II correction

For the minimal locked obstruction baseline, the currently honest shell input is

$$A_0 = 0, \quad B_0 = \kappa_Z = \frac{4Z}{A}, \quad C_0 = 0, \quad (506)$$

because the locked minimal PDE supplies the identity-channel transverse witness but no longitudinal Pauli witness.

Substituting (506) into (501),

$$A_{\text{eff}} = -\Gamma c_{\parallel}, \quad B_{\text{eff}} = \frac{4Z}{A} - \Gamma c_I, \quad C_{\text{eff}} = -\Gamma c_J. \quad (507)$$

Therefore

$$\det M_{\text{eff}} = (\kappa_* |G_*|)^3 (-\Gamma c_{\parallel}) \left[\left(\frac{4Z}{A} - \Gamma c_I \right)^2 + (\Gamma c_J)^2 \right]. \quad (508)$$

C. Nondegeneracy region

From (508), the effective shell is Weyl iff

$$\boxed{\Gamma c_{\parallel} \neq 0, \quad \left(\frac{4Z}{A} - \Gamma c_I, \Gamma c_J \right) \neq (0, 0).} \quad (509)$$

Equivalently, assuming $\Gamma \neq 0$,

$$\boxed{c_{\parallel} \neq 0, \quad \left(\frac{4Z}{A\Gamma} - c_I, c_J \right) \neq (0, 0).} \quad (510)$$

So the bad set is the codimension-one tuned locus

$$c_{\parallel} = 0 \quad \text{or} \quad \left(c_J = 0, c_I = \frac{4Z}{A\Gamma} \right). \quad (511)$$

Away from that tuned locus, the shell is nondegenerate.

D. Interpretation

This gives two complementary viewpoints.

a. Viewpoint 1: obstruction baseline plus Class II repair. Because $B_0 = \kappa_Z \neq 0$ already survives from the identity channel, the Class II sector does not need to generate transverse witness from scratch. It is sufficient to generate a nonzero longitudinal coefficient:

$$c_{\parallel} \neq 0$$

already gives a Weyl shell generically, except on the tuned cancellation locus for the transverse factor.

b. Viewpoint 2: pure Class II local transport sector. If one works in the purely reduced local S2 transport normalization, one sets the baseline to the Class II transport coefficients themselves and obtains the stricter condition

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0),$$

equivalently

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0),$$

inside the concrete Class II truncation.

So the two descriptions are not contradictory: they are two different normalizations of the same shell problem.

E. What is now honestly closed

The combination of normalization extraction and witness audit closes the following chain:

$$\boxed{L_{\text{UV}}^{(II)} \implies \Gamma = \frac{\lambda g_X m}{M_X^2} \implies (\delta A, \delta B, \delta C) = (-\Gamma c_{\parallel}, -\Gamma c_I, -\Gamma c_J)} \quad (512)$$

and therefore

$$\boxed{\det M_{\text{eff}} = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2),} \quad (513)$$

with the explicit obstruction-baseline realization

$$\boxed{\det M_{\text{eff}} = (\kappa_* |G_*|)^3 (-\Gamma c_{\parallel}) \left[\left(\frac{4Z}{A} - \Gamma c_I \right)^2 + (\Gamma c_J)^2 \right].} \quad (514)$$

Hence the effective shell is Weyl on the open region

$$\boxed{c_{\parallel} \neq 0, \quad \left(\frac{4Z}{A\Gamma} - c_I, c_J \right) \neq (0, 0).} \quad (515)$$

F. Honest frontier

What still remains open is not the shell projection or the effective determinant formula. It is only:

1. whether the true complete TECT action actually realizes a nontrivial Class II carrier sector in the relevant defect background;
2. the final absolute values of the normalization constants $\nu_{\parallel}, \nu_{\perp}$ entering the exact transport coefficients.

CCCVI. FINAL SUMMARY OF NORMALIZATION EXTRACTION AND WITNESS AUDIT

a. 1. Strongest honest normalization extraction. From the shell-relevant quadratic Class II reduction,

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab},$$

the natural Class II scale is

$$\Gamma = \frac{\lambda g_X m}{M_X^2}.$$

For the minimal shell-adapted seed family,

$$\delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J.$$

Hence

$$g_c = \nu_{\parallel} \Gamma c_{\parallel}, \quad g_E = \nu_{\perp} \Gamma c_I, \quad g_O = \nu_{\perp} \Gamma c_J,$$

with remaining overlap-normalization constants $\nu_{\parallel}, \nu_{\perp}$.

b. 2. Effective witness audit. Using the canonical effective shell coefficients,

$$A_{\text{eff}} = A_0 - \Gamma c_{\parallel}, \quad B_{\text{eff}} = B_0 - \Gamma c_I, \quad C_{\text{eff}} = C_0 - \Gamma c_J,$$

one gets

$$\det M_{\text{eff}} = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2).$$

c. 3. Obstruction baseline. For the minimal locked obstruction baseline

$$A_0 = 0, \quad B_0 = \frac{4Z}{A}, \quad C_0 = 0,$$

this becomes

$$\det M_{\text{eff}} = (\kappa_* |G_*|)^3 (-\Gamma c_{\parallel}) \left[\left(\frac{4Z}{A} - \Gamma c_I \right)^2 + (\Gamma c_J)^2 \right].$$

d. 4. Weyl region. Therefore the effective shell is Weyl iff

$$c_{\parallel} \neq 0, \quad \left(\frac{4Z}{A\Gamma} - c_I, c_J \right) \neq (0, 0).$$

e. 5. Honest frontier. So both requested steps are now done at the strongest currently honest level:

1. the Class II normalization scale Γ and the induced witness directions are extracted;
2. the effective witness determinant and its nondegeneracy region are written explicitly.

What still remains open is only:

$$\text{whether the full complete TECT action truly realizes the Class II sector,}$$

and

$$\text{the final absolute overlap normalizations } \nu_{\parallel}, \nu_{\perp}.$$

**CCCVII. DIRAC STEP 2: OPPOSITE-VALLEY WITNESS MATCHING WITH OBSTRUCTION
BASELINE + CLASS II CORRECTION**

A. Goal

We now continue from the previously obtained effective shell coefficients

$$A_{\text{eff}} = A_0 - \Gamma c_{\parallel}, \quad B_{\text{eff}} = B_0 - \Gamma c_I, \quad C_{\text{eff}} = C_0 - \Gamma c_J,$$

and verify opposite-valley witness matching in the matched shell frame. We will do this for the obstruction baseline

$$A_0 = 0, \quad B_0 = \frac{4Z}{A}, \quad C_0 = 0,$$

together with the Class II correction.

B. Matched opposite-valley frames

Let

$$\hat{n}_+ := \frac{G_*}{|G_*|}, \quad \hat{n}_- := -\hat{n}_+. \quad (516)$$

Choose a positively oriented orthonormal frame at $+G_*$:

$$\{e_{1+}, e_{2+}, \hat{n}_+\}, \quad e_{1+} \times e_{2+} = \hat{n}_+. \quad (517)$$

Define the matched opposite-valley frame at $-G_*$ by

$$e_{1-} := e_{1+}, \quad e_{2-} := -e_{2+}, \quad \hat{n}_- = -\hat{n}_+. \quad (518)$$

Then

$$e_{1-} \times e_{2-} = e_{1+} \times (-e_{2+}) = -\hat{n}_+ = \hat{n}_-, \quad (519)$$

so $\{e_{1-}, e_{2-}, \hat{n}_-\}$ is again positively oriented.

This is the standard matched opposite-valley convention already used in the notes.

C. Effective shell tensor family at the two valleys

Write the effective shell-enriched tensor family at each valley in the canonical axial basis:

$$N_{ij}^3(\pm) = A_{\text{eff}} \hat{n}_{\pm i} \hat{n}_{\pm j}, \quad (520)$$

$$N_{ij}^1(\pm) = B_{\text{eff}} \hat{n}_{\pm i} e_{1\pm j} + C_{\text{eff}} \hat{n}_{\pm i} e_{2\pm j}, \quad (521)$$

$$N_{ij}^2(\pm) = -C_{\text{eff}} \hat{n}_{\pm i} e_{1\pm j} + B_{\text{eff}} \hat{n}_{\pm i} e_{2\pm j}. \quad (522)$$

The crucial point is that the scalar amplitudes

$$A_{\text{eff}}, \quad B_{\text{eff}}, \quad C_{\text{eff}}$$

are assumed to be even under the matched opposite-valley transformation. In the canonical Class II seed family this is exactly the statement that

$$c_{\parallel}^- = c_{\parallel}^+, \quad c_I^- = c_I^+, \quad c_J^- = c_J^+,$$

hence the same effective amplitudes are read at both valleys.

D. Witness functionals at the two valleys

Define the matched-frame witness functionals:

$$W_{\parallel}^{\pm}[N] := \hat{n}_{\pm i} \hat{n}_{\pm j} N_{ij}^3(\pm), \quad (523)$$

$$W_I^{\pm}[N] := \frac{1}{2} \left(e_{1\pm j} \hat{n}_{\pm i} N_{ij}^1(\pm) + e_{2\pm j} \hat{n}_{\pm i} N_{ij}^2(\pm) \right), \quad (524)$$

$$W_J^{\pm}[N] := \frac{1}{2} \left(e_{1\pm j} \hat{n}_{\pm i} N_{ij}^2(\pm) - e_{2\pm j} \hat{n}_{\pm i} N_{ij}^1(\pm) \right). \quad (525)$$

Substituting (520)–(522):

a. Longitudinal witness

$$\begin{aligned} W_{\parallel}^{\pm} &= \hat{n}_{\pm i} \hat{n}_{\pm j} (A_{\text{eff}} \hat{n}_{\pm i} \hat{n}_{\pm j}) \\ &= A_{\text{eff}} (\hat{n}_{\pm}^2)^2 = A_{\text{eff}}. \end{aligned} \quad (526)$$

b. Even transverse witness

$$\begin{aligned} W_I^{\pm} &= \frac{1}{2} \left[e_{1\pm j} \hat{n}_{\pm i} (B_{\text{eff}} \hat{n}_{\pm i} e_{1\pm j} + C_{\text{eff}} \hat{n}_{\pm i} e_{2\pm j}) \right. \\ &\quad \left. + e_{2\pm j} \hat{n}_{\pm i} (-C_{\text{eff}} \hat{n}_{\pm i} e_{1\pm j} + B_{\text{eff}} \hat{n}_{\pm i} e_{2\pm j}) \right] \\ &= \frac{1}{2} (B_{\text{eff}} + B_{\text{eff}}) = B_{\text{eff}}. \end{aligned} \quad (527)$$

c. Odd transverse witness

$$\begin{aligned} W_J^{\pm} &= \frac{1}{2} \left[e_{1\pm j} \hat{n}_{\pm i} (-C_{\text{eff}} \hat{n}_{\pm i} e_{1\pm j} + B_{\text{eff}} \hat{n}_{\pm i} e_{2\pm j}) \right. \\ &\quad \left. - e_{2\pm j} \hat{n}_{\pm i} (B_{\text{eff}} \hat{n}_{\pm i} e_{1\pm j} + C_{\text{eff}} \hat{n}_{\pm i} e_{2\pm j}) \right]. \end{aligned} \quad (528)$$

Depending on the sign convention used for the O_J -basis, this equals either $+C_{\text{eff}}$ or $-C_{\text{eff}}$. What matters physically is that the same convention is used at both valleys. Fixing the shell basis once and for all, we may write simply

$$W_J^{\pm} = C_{\text{eff}} \quad (529)$$

up to that global convention.

Thus the witness triples are

$$\boxed{(W_{\parallel}^{\pm}, W_I^{\pm}, W_J^{\pm}) = (A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}).} \quad (530)$$

E. Obstruction baseline + Class II correction

From the previous step,

$$A_{\text{eff}} = -\Gamma c_{\parallel}, \quad B_{\text{eff}} = \frac{4Z}{A} - \Gamma c_I, \quad C_{\text{eff}} = -\Gamma c_J. \quad (531)$$

Therefore

$$\boxed{W_{\parallel}^{\pm} = -\Gamma c_{\parallel}, \quad W_I^{\pm} = \frac{4Z}{A} - \Gamma c_I, \quad W_J^{\pm} = -\Gamma c_J,} \quad (532)$$

again up to the fixed sign convention for W_J .

Hence

$$\boxed{(W_{\parallel}^{-}, W_I^{-}, W_J^{-}) = (W_{\parallel}^{+}, W_I^{+}, W_J^{+}).} \quad (533)$$

So the obstruction baseline does not spoil opposite-valley matching. It merely shifts the common transverse scalar amplitude B_{eff} .

F. Consequence for opposite-valley pairing

By the previously established witness-pairing theorem, equality of the witness triples in the matched opposite-valley frames is equivalent to the standard reduced pairing relation

$$h_-(p) = -h_+(-p) \quad (534)$$

after subtraction of the common scalar part.

Therefore the obstruction-baseline + Class II corrected representative already satisfies the Step-2 pairing criterion inside the concrete truncation.

G. Combined Step 1 + Step 2

From the previous witness audit, the shell is Weyl iff

$$\Gamma c_{\parallel} \neq 0, \quad \left(\frac{4Z}{A} - \Gamma c_I, \Gamma c_J \right) \neq (0, 0). \quad (535)$$

From (533), Step 2 is automatically satisfied in the same concrete family.

Hence, inside the obstruction-baseline + Class II corrected truncation, we obtain:

Step 1 (Weyl nondegeneracy) + Step 2 (opposite-valley matching) are closed simultaneously.

(536)

H. Honest frontier

What remains open is stronger than the above statement:

the full complete TECT action must still be shown to realize the required opposite shell pair $\pm G_*$ as actual low-energy valleys

(537)

So the exact honest conclusion is:

inside the concrete Class II witness truncation, Step 2 is closed;

but

the full microscopic existence of the opposite shell-pair valleys remains to be verified.

CCCVIII. FINAL SUMMARY OF OPPOSITE-VALLEY MATCHING WITH OBSTRUCTION BASELINE + CLASS II CORRECTION

a. What was actually computed. Using the matched opposite-valley frames

$$\hat{n}_- = -\hat{n}_+, \quad e_{1-} = e_{1+}, \quad e_{2-} = -e_{2+},$$

and the effective axial shell tensor

$$N_{ij}^3(\pm) = A_{\text{eff}} \hat{n}_{\pm i} \hat{n}_{\pm j},$$

$$N_{ij}^1(\pm) = B_{\text{eff}} \hat{n}_{\pm i} e_{1\pm j} + C_{\text{eff}} \hat{n}_{\pm i} e_{2\pm j},$$

$$N_{ij}^2(\pm) = -C_{\text{eff}} \hat{n}_{\pm i} e_{1\pm j} + B_{\text{eff}} \hat{n}_{\pm i} e_{2\pm j},$$

one finds

$(W_{\parallel}^{\pm}, W_I^{\pm}, W_J^{\pm}) = (A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}).$

b. *With the obstruction baseline plus Class II correction.* Since

$$A_{\text{eff}} = -\Gamma c_{\parallel}, \quad B_{\text{eff}} = \frac{4Z}{A} - \Gamma c_I, \quad C_{\text{eff}} = -\Gamma c_J,$$

the witness triples at both valleys are

$$W_{\parallel}^{\pm} = -\Gamma c_{\parallel}, \quad W_I^{\pm} = \frac{4Z}{A} - \Gamma c_I, \quad W_J^{\pm} = -\Gamma c_J.$$

c. *Matching result.* Therefore

$$(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+).$$

So Step 2 is closed inside the same concrete truncation.

d. *Combined Step 1 + Step 2.* The shell is Weyl iff

$$\Gamma c_{\parallel} \neq 0, \quad \left(\frac{4Z}{A} - \Gamma c_I, \Gamma c_J \right) \neq (0, 0),$$

and in the same region opposite-valley witness matching already holds.

e. *Honest frontier.* Thus, inside the concrete obstruction-baseline + Class II corrected truncation,

$$\text{Step 1 and Step 2 are closed together.}$$

What remains open is only whether the full complete TECT action actually realizes the required opposite shell pair $\pm G_*$ as physical low-energy valleys.

CCCIX. DIRAC STEP 3: ZERO CONSTANT MIXING IN THE OBSTRUCTION-BASELINE + CLASS II TRUNCATION

A. Goal

After Step 1 (effective Weyl nondegeneracy) and Step 2 (opposite-valley witness matching), the next local Dirac closure condition is

$$V(0) = 0. \tag{538}$$

Here $V(p)$ is the off-diagonal mixing block between the local doubled Dirac carrier and the remote sector in

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^{\dagger} & H_R(p) \end{pmatrix}. \tag{539}$$

The symbol $V(0) = 0$ means absence of *constant inter-sector hybridization* at the shell crossing.

B. The actual local Dirac carrier

The correct shell-relevant local carrier is not an ordinary cubic 2D irrep. Rather, the actual local Dirac block is

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2, \tag{540}$$

where

- $\mathbb{C}_{\text{valley}}^2$ is the opposite shell pair $\pm G_*$,
- $\mathbb{C}_{\text{int}}^2$ is the shell-relevant enriched internal/transverse doublet.

This is crucial: the zero-mixing question must be asked relative to the actual carrier $\mathcal{H}_D$, not relative to ordinary little-group irreducibility.

C. The theorem-level zero-mixing statement

Let H_* denote the little group at the crossing $p = 0$, and assume:

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R \quad (541)$$

is H_* -invariant, and that the microscopic Hamiltonian is H_* -equivariant at $p = 0$. Then

$$\boxed{V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).} \quad (542)$$

Therefore

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0.} \quad (543)$$

This theorem-level statement is already closed.

D. Practical remote-sector decomposition

Decompose the remote sector into H_* -invariant pieces:

$$\mathcal{H}_R = \bigoplus_a K_a. \quad (544)$$

Then

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_a \text{Hom}_{H_*}(K_a, \mathcal{H}_D). \quad (545)$$

Hence it is enough to prove

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \quad \forall a. \quad (546)$$

E. Shell-relevant mismatch labels

A remote sector K_a is excluded from constant mixing if it differs from $\mathcal{H}_D$ in at least one of the following shell-relevant labels:

$$\boxed{\pm G_*, \quad \text{transverse/non-volumetric character}, \quad \text{presence of enriched internal doublet}, \quad \text{grading/parity/chirality}, \quad \text{extra conserv}} \quad (547)$$

Thus the microscopic $V(0) = 0$ problem is reduced to a finite label-audit problem.

F. Minimal dangerous truncation

In the minimal dangerous truncation, write

$$\boxed{\mathcal{H}_{\text{full}}^{\text{tr}} = \mathcal{H}_D \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}.} \quad (548)$$

Here:

- $\mathcal{H}_D$: the local doubled Dirac carrier,
- $\mathcal{H}_{\text{vol}}$: volumetric sector,
- $\mathcal{H}_{\text{high}}$: higher-shell sector,
- $\mathcal{H}_{\text{non-enr}}$: same shell but lacking the enriched internal doublet,
- $\mathcal{H}_{\text{wrong-int}}$: wrong internal representation.

G. Sectorwise mismatch table

We now check the shell-relevant labels sector by sector.

a. $\mathcal{H}_{\text{vol}}$. This sector fails the transverse/non-volumetric character condition. Hence

$$\text{Hom}_{H_*}(\mathcal{H}_{\text{vol}}, \mathcal{H}_D) = 0.$$

b. $\mathcal{H}_{\text{high}}$. This sector lies on a different shell orbit, so it fails the shell-pair label $\pm G_*$. Hence

$$\text{Hom}_{H_*}(\mathcal{H}_{\text{high}}, \mathcal{H}_D) = 0.$$

c. $\mathcal{H}_{\text{non-enr}}$. Even if it shares the same shell pair, it lacks the enriched internal doublet $\mathbb{C}_{\text{int}}^2$. Hence

$$\text{Hom}_{H_*}(\mathcal{H}_{\text{non-enr}}, \mathcal{H}_D) = 0.$$

d. $\mathcal{H}_{\text{wrong-int}}$. This sector carries the wrong internal representation content. Hence

$$\text{Hom}_{H_*}(\mathcal{H}_{\text{wrong-int}}, \mathcal{H}_D) = 0.$$

Therefore every dangerous remote sector in (548) mismatches $\mathcal{H}_D$ in at least one shell-relevant label.

H. Conclusion of the audit

By (545),

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.} \quad (549)$$

Then by (543),

$$\boxed{V(0) = 0.} \quad (550)$$

Thus, in the obstruction-baseline + Class II corrected truncation, zero constant mixing is established once the above finite mismatch table is accepted.

I. Compatibility with Class II

The canonical minimal Class II data

$$m^a(P_0) = m\delta^{a3}, \quad C_{ij}^{33} = c_{\parallel}\hat{n}_i\hat{n}_j, \quad C_{ij}^{31} = c_I\hat{n}_ie_{1j} + c_J\hat{n}_ie_{2j}, \quad C_{ij}^{32} = -c_J\hat{n}_ie_{1j} + c_I\hat{n}_ie_{2j} \quad (551)$$

renormalize the shell-level witness coefficients A, B, C , but do not change the representation content of the remote sector classification itself.

Therefore:

$$\boxed{\text{Class II is compatible with } V(0) = 0, \text{ but does not by itself prove it.}} \quad (552)$$

The actual proof of $V(0) = 0$ remains the sectorwise mismatch audit above.

J. Combined Step 1 + Step 2 + Step 3

From the previous steps, the concrete obstruction-baseline + Class II truncation gives:

$$\Gamma c_{\parallel} \neq 0, \quad \left(\frac{4Z}{A} - \Gamma c_I, \Gamma c_J \right) \neq (0, 0) \quad (553)$$

for Weyl nondegeneracy, and opposite-valley witness matching is already closed in the same truncation.

Now Step 3 adds:

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0.} \quad (554)$$

Thus the concrete truncation has now closed:

Step 1: Weyl nondegeneracy, Step 2: opposite-valley matching, Step 3: zero constant mixing
at the theorem-reduction level, with Step 3 requiring only the finite mismatch audit.

K. Sharp honest conclusion

The strongest honest conclusion is:

$$\boxed{\text{inside the concrete obstruction-baseline + Class II truncation, } V(0) = 0 \text{ is not automatic,}} \quad (555)$$

but

$$\boxed{\text{its verification is fully reduced to a finite shell-relevant remote-sector label audit.}} \quad (556)$$

So Step 3 is now closed at the reduction-theorem level in exactly the same sense as the current TECT notes.

CCCX. FINAL SUMMARY OF DIRAC STEP 3 IN THE OBSTRUCTION-BASELINE + CLASS II TRUNCATION

a. What was actually established. The local Dirac carrier is

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

and the constant mixing block satisfies

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Therefore

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0.}$$

b. Practical finite audit. Decompose the remote sector as

$$\mathcal{H}_R = \bigoplus_a K_a.$$

A sector K_a is excluded from constant mixing if it differs from $\mathcal{H}_D$ in at least one of:

$\pm G_*$, transverse/non-volumetric character, enriched internal doublet, grading/parity/chirality, extra conserved labels.

c. Minimal dangerous truncation. In the truncation

$$\mathcal{H}_{\text{full}}^{\text{tr}} = \mathcal{H}_D \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}},$$

every remote sector mismatches $\mathcal{H}_D$ in at least one shell-relevant label, hence

$$\boxed{V(0) = 0.}$$

d. Compatibility with Class II. The canonical Class II data renormalize the shell-level witness coefficients A, B, C , but do not by themselves prove $V(0) = 0$. So:

$$\boxed{\text{Class II is compatible with } V(0) = 0, \text{ and } V(0) = 0 \text{ is proved by the independent mismatch audit.}}$$

e. Combined status. Inside the concrete obstruction-baseline + Class II truncation:

Step 1 (Weyl nondegeneracy), Step 2 (opposite-valley matching), and Step 3 (zero constant mixing) are all closed at the reduction level.

f. Honest frontier. What remains open microscopically is only whether the full complete TECT action realizes the required remote-sector decomposition and mismatch table exactly as assumed in the finite audit.

CCCXI. DIRAC STEP 4: REMOTE SPECTRAL GAP IN THE OBSTRUCTION-BASELINE + CLASS II TRUNCATION

A. Goal

After Step 1 (local Weyl nondegeneracy), Step 2 (opposite-valley matching), and Step 3 ($V(0) = 0$), the next remaining local Dirac closure condition is

$$\boxed{\Delta > 0.} \quad (557)$$

This means that the remote block $H_R(p)$ is spectrally separated from zero near the selected shell crossing.

The correct theorem-level point is that $\Delta > 0$ is an *independent spectral input*. It is not implied by shell nondegeneracy.

B. Why $\det M \neq 0$ does not prove the remote gap

The shell-reduced local Dirac symbol has the form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (558)$$

equivalently

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad \det M = \lambda_{\parallel}(\alpha^2 + \beta^2). \quad (559)$$

This controls only the nondegeneracy of the local shell Dirac block. It says nothing by itself about the complementary spectrum of $H_R(p)$.

Hence

$$\boxed{\det M \neq 0 \not\Rightarrow \text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset.} \quad (560)$$

C. Remote block decomposition

Write the full microscopic Hamiltonian as

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^{\dagger} & H_R(p) \end{pmatrix}, \quad (561)$$

where

- $H_D(p)$ is the isolated local doubled Dirac block,
- $H_R(p)$ is the remote sector,
- $V(p)$ is the inter-sector mixing.

The remote-gap condition is

$$\boxed{\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset \quad (|p| \leq \rho)} \quad (562)$$

for some $\rho > 0$ and $\Delta > 0$.

D. Route A: direct finite spectral inequality

At the minimal dangerous truncation level, write

$$H_R(p) = D_R(p) + R_R(p), \quad (563)$$

where $D_R(p)$ is the diagonal/sector-center benchmark part and $R_R(p)$ is the residual remote-remote mixing.

Define the pointwise sector-center lower bound

$$\delta_0(p) := \min\{|\mu_{\text{vol}}(p)|, |\mu_{\text{high}}(p)|, |\mu_{\text{non-enr}}(p)|, |\mu_{\text{wrong-int}}(p)|\}, \quad (564)$$

and the remote residual norm

$$\eta_R(p) := \|R_R(p)\|. \quad (565)$$

Then the pointwise estimate is

$$\|H_R(p)\psi\| \geq (\delta_0(p) - \eta_R(p))\|\psi\|. \quad (566)$$

Therefore, if

$$\delta_0(p) > \eta_R(p), \quad (567)$$

then

$$\text{spec}(H_R(p)) \cap (-\Delta(p), \Delta(p)) = \emptyset, \quad \Delta(p) \geq \delta_0(p) - \eta_R(p) > 0. \quad (568)$$

On the neighborhood

$$N_\rho := \{p : |p| \leq \rho\},$$

define

$$\delta_{0,\rho} := \inf_{|p| \leq \rho} \delta_0(p), \quad \eta_{R,\rho} := \sup_{|p| \leq \rho} \eta_R(p). \quad (569)$$

Then

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \implies \text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset \quad (|p| \leq \rho),} \quad (570)$$

with

$$\boxed{\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.} \quad (571)$$

E. Explicit benchmark version for the minimal dangerous truncation

A sufficient explicit benchmark is obtained by sectorwise lower bounds:

$$\Delta_{\text{vol},\rho}^{(0)} > 0, \quad \Delta_{\text{high},\rho}^{(0)} > 0, \quad \Delta_{\text{non-enr},\rho}^{(0)} > 0, \quad \Delta_{\text{wrong-int},\rho}^{(0)} > 0. \quad (572)$$

Define

$$\boxed{\delta_{0,\rho}^{\text{bench}} := \min\left\{\Delta_{\text{vol},\rho}^{(0)}, \Delta_{\text{high},\rho}^{(0)}, \Delta_{\text{non-enr},\rho}^{(0)}, \Delta_{\text{wrong-int},\rho}^{(0)}\right\}.} \quad (573)$$

Then a sufficient finite gap condition is

$$\boxed{\delta_{0,\rho}^{\text{bench}} > \eta_{R,\rho}.} \quad (574)$$

This proves

$$\boxed{\Delta_\rho > 0.} \quad (575)$$

F. Route B: bounded perturbation from an already-gapped benchmark

Assume instead that there exists an unperturbed realization

$$H_R^{(0)}(p) \quad \text{with} \quad \text{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \emptyset \quad (\Delta_0 > 0), \quad (576)$$

for all sufficiently small p , and that the actual remote block is

$$H_R(p) = H_R^{(0)}(p) + \delta H_R(p), \quad \sup_{|p| \leq p_*} \|\delta H_R(p)\| \leq \varepsilon. \quad (577)$$

Then the bounded-perturbation theorem gives

$$\boxed{\varepsilon < \Delta_0 \implies \text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset, \quad \Delta := \Delta_0 - \varepsilon > 0.} \quad (578)$$

Under the stronger bound

$$\varepsilon < \frac{\Delta_0}{2},$$

one gets the convenient estimate

$$\Delta > \frac{\Delta_0}{2} > 0.$$

G. Class II compatibility statement

The canonical minimal Class II correction changes only the shell-relevant derivative tensor coefficients:

$$A_{\text{eff}} = A_0 - \Gamma c_{\parallel}, \quad B_{\text{eff}} = B_0 - \Gamma c_I, \quad C_{\text{eff}} = C_0 - \Gamma c_J, \quad (579)$$

with

$$\Gamma = \frac{\lambda g_X m}{M_X^2}. \quad (580)$$

It does *not* by itself prove a remote gap. Rather, its correct theorem-level status is:

$$\boxed{\text{Class II is gap-compatible, but } \Delta > 0 \text{ still requires an independent spectral check.}} \quad (581)$$

H. Schur-complement consequence

From Step 3 we already have

$$V(0) = 0, \quad \|V(p)\| \leq c|p|. \quad (582)$$

If $|E| < \Delta/2$, then

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta}. \quad (583)$$

Hence the Schur-complement self-energy

$$\Sigma(E, p) = V(p)(H_R(p) - E)^{-1}V(p)^\dagger \quad (584)$$

satisfies

$$\boxed{\|\Sigma(E, p)\| \leq \frac{2c^2}{\Delta}|p|^2.} \quad (585)$$

Therefore the effective low-energy Hamiltonian is

$$\boxed{H_{\text{eff}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right).} \quad (586)$$

So once $\Delta > 0$ is established, the remote sector changes only the subleading quadratic curvature and does not destroy the leading local Dirac cone.

I. Step 4 in the obstruction-baseline + Class II truncation

Combining the previous steps, the concrete truncation now has:

1. Step 1: local Weyl nondegeneracy on the region

$$\Gamma c_{\parallel} \neq 0, \quad \left(\frac{4Z}{A} - \Gamma c_I, \Gamma c_J \right) \neq (0, 0);$$

2. Step 2: opposite-valley witness matching;
3. Step 3: zero constant mixing $V(0) = 0$ reduced to the finite mismatch audit;
4. Step 4: remote-gap positivity reduced to either:

$$\delta_{0,\rho}^{\text{bench}} > \eta_{R,\rho},$$

or

$$\|\delta H_R\| < \Delta_0.$$

Thus Step 4 is closed at the same honest level as the notes:

the remote-gap problem is fully reduced to a finite spectral inequality / bounded-perturbation audit.

 (587)

J. Sharp honest conclusion

The strongest honest statement is:

inside the obstruction-baseline + Class II truncation, $\Delta > 0$ is not automatic,

 (588)

but

its verification is completely reduced to an explicit finite remote spectral audit.

 (589)

So Step 4 is now closed at the theorem-reduction level.

CCCXII. FINAL SUMMARY OF DIRAC STEP 4 IN THE OBSTRUCTION-BASELINE + CLASS II TRUNCATION

- a. *What was actually established.* The remote gap condition

$$\Delta > 0$$

is an independent spectral input. It is not implied by the local shell determinant

$$\det M = A(B^2 + C^2) \neq 0.$$

- b. *Two honest theorem-level routes.* Step 4 is reduced to one of the following finite checks:
- c. (A) *Direct finite spectral inequality* Define

$$\delta_{0,\rho} := \inf_{|p| \leq \rho} \delta_0(p), \quad \eta_{R,\rho} := \sup_{|p| \leq \rho} \|R_R(p)\|.$$

Then

$\delta_{0,\rho} > \eta_{R,\rho} \implies \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.$

d. (B) *Bounded perturbation from an already-gapped benchmark* If

$$\text{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \emptyset$$

and

$$\sup_{|p| \leq p_*} \|\delta H_R(p)\| < \Delta_0,$$

then

$$\Delta = \Delta_0 - \sup \|\delta H_R\| > 0.$$

e. *Class II status.* The canonical minimal Class II correction only renormalizes the shell-relevant coefficients

$$A_{\text{eff}}, \quad B_{\text{eff}}, \quad C_{\text{eff}},$$

so

$$\boxed{\text{Class II is gap-compatible, but does not by itself prove } \Delta > 0.}$$

f. *Schur-complement consequence.* Once

$$V(0) = 0, \quad \|V(p)\| \leq c|p|, \quad \Delta > 0,$$

one gets

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta} \quad (|E| < \Delta/2),$$

and

$$\|\Sigma(E, p)\| \leq \frac{2c^2}{\Delta} |p|^2, \quad H_{\text{eff}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right).$$

g. *Honest frontier.* So Step 4 is now closed at the reduction-theorem level. What remains open is only the actual finite remote spectral audit:

$$\boxed{\delta_{0,\rho}^{\text{bench}} - \eta_{R,\rho} \stackrel{?}{>} 0}$$

or, equivalently,

$$\boxed{\Delta_0 - \sup \|\delta H_R\| \stackrel{?}{>} 0.}$$

CCCXIII. DIRAC STEP 5: MASS PROTECTION IN THE OBSTRUCTION-BASELINE + CLASS II TRUNCATION

A. Goal

After Step 1 (Weyl nondegeneracy), Step 2 (opposite-valley matching), Step 3 ($V(0) = 0$), and Step 4 ($\Delta > 0$), the last remaining local Dirac closure condition is the absence of a symmetry-allowed constant Dirac mass.

The correct theorem-level question is:

$$\boxed{A_{\text{res}} \cap \{K \in \text{Herm}(4) : \{K, \Gamma_i\} = 0 \ \forall i\} = \{0\}}, \quad (590)$$

where A_{res} is the allowed constant operator algebra under the residual low-energy symmetry.

B. Doubled local Dirac block and the mass space

After Step 2, the local doubled shell block has the Dirac form

$$H_D(p) = \sum_{i=1}^3 v_i p_i (\tau_z \otimes \sigma_i). \quad (591)$$

The full little-group-scalar constant Hermitian operator space on the doubled block is

$$K = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2 + m_1 \tau_x \otimes \mathbf{1}_2 + m_2 \tau_y \otimes \mathbf{1}_2. \quad (592)$$

Among these, the genuine constant Dirac-mass space is exactly

$$\mathcal{M}_D = \text{span}_{\mathbb{R}} \{ \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2 \}. \quad (593)$$

Thus the actual microscopic mass-protection problem is:

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}. \quad (594)$$

C. Why opposite-valley matching alone is not enough

Opposite-valley matching fixes the relation

$$h_-(p) = -h_+(-p),$$

and therefore yields the doubled Dirac structure. But it does *not* by itself remove the allowed constant operators

$$\tau_x \otimes \mathbf{1}_2, \quad \tau_y \otimes \mathbf{1}_2.$$

Therefore:

$$\boxed{\text{opposite-valley matching is necessary for doubling, but not sufficient for mass protection.}} \quad (595)$$

D. Residual-symmetry audit table

At the theorem level, the practical classification is:

Residual structure	Action on $\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2$	Does it forbid all constant masses?
little-group scalarity only	allows both	No
opposite-valley matching only	fixes doubling but does not remove τ_x, τ_y	No
exact valley $U(1)_V$	(τ_x, τ_y) rotate as a charged doublet	Yes
exact valley $\mathbb{Z}_2^V$	$\tau_x, \tau_y \mapsto -\tau_x, -\tau_y$	Yes
Class II gradient-linear structure	does not insert a constant mass directly	Compatible only

E. Exact valley $U(1)_V$ as a sufficient protection mechanism

Define the valley-charge generator

$$Q_V := \tau_z \otimes \mathbf{1}_2, \quad U_\theta := e^{i\theta Q_V/2}. \quad (596)$$

Since

$$[Q_V, H_D(p)] = 0, \quad (597)$$

the doubled kinetic block is compatible with a residual valley $U(1)_V$.

Now

$$U_\theta(\tau_x \otimes \mathbf{1}_2)U_\theta^{-1} = \cos \theta (\tau_x \otimes \mathbf{1}_2) + \sin \theta (\tau_y \otimes \mathbf{1}_2), \quad (598)$$

$$U_\theta(\tau_y \otimes \mathbf{1}_2)U_\theta^{-1} = -\sin \theta (\tau_x \otimes \mathbf{1}_2) + \cos \theta (\tau_y \otimes \mathbf{1}_2). \quad (599)$$

Therefore no nonzero element of $\mathcal{M}_D$ is invariant under all θ , so

$$\boxed{A_{U(1)_V} \cap \mathcal{M}_D = \{0\}.} \quad (600)$$

Hence exact residual valley $U(1)_V$ is sufficient for full constant mass protection.

F. Exact valley $\mathbb{Z}_2^V$ is already enough

The full continuous $U(1)_V$ is stronger than necessary. The discrete remnant

$$S_V := \tau_z \otimes \mathbf{1}_2, \quad S_V^2 = \mathbf{1}_4 \quad (601)$$

already suffices, because

$$S_V(\tau_x \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_x \otimes \mathbf{1}_2), \quad S_V(\tau_y \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_y \otimes \mathbf{1}_2). \quad (602)$$

Thus

$$\boxed{A_{\mathbb{Z}_2^V} \cap \mathcal{M}_D = \{0\}.} \quad (603)$$

So exact residual valley $\mathbb{Z}_2^V$ is a weaker but still sufficient mechanism.

G. Compatibility with the obstruction-baseline + Class II truncation

The canonical minimal Class II representative is

$$m^a(P_0) = m\delta^{a3}, \quad C_{ij}^{33} = c_{\parallel}\hat{n}_i\hat{n}_j, \quad C_{ij}^{31} = c_I\hat{n}_ie_{1j} + c_J\hat{n}_ie_{2j}, \quad C_{ij}^{32} = -c_J\hat{n}_ie_{1j} + c_I\hat{n}_ie_{2j}. \quad (604)$$

Its role is to renormalize the derivative shell tensor coefficients A, B, C , equivalently the first-order local Weyl/Dirac witnesses. It does not by construction insert an independent constant 4×4 Dirac mass operator.

Therefore:

$$\boxed{\text{Class II is compatible with mass protection, but is not by itself a proof of mass protection.}} \quad (605)$$

H. Radiative stability under remote-sector elimination

Assume the full microscopic theory preserves the residual valley symmetry, meaning there exists a unitary action

$$U_\theta \oplus U_{R,\theta}$$

such that

$$(U_\theta \oplus U_{R,\theta})H_{\text{full}}(p)(U_\theta \oplus U_{R,\theta})^{-1} = H_{\text{full}}(p). \quad (606)$$

Then the Schur-complement shell Hamiltonian

$$H_{\text{eff}}(p, \omega) = H_D(p) + V(p)^\dagger (\omega - H_R(p))^{-1} V(p) \quad (607)$$

inherits the same symmetry. In particular, the zero-momentum self-energy

$$\Sigma(0, 0) := V(0)^\dagger (-H_R(0))^{-1} V(0) \quad (608)$$

must commute with Q_V :

$$[Q_V, \Sigma(0, 0)] = 0. \quad (609)$$

If we also impose little-group scalarity at the crossing, then

$$\Sigma(0, 0) = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2. \quad (610)$$

Hence no radiatively generated term proportional to

$$\tau_x \otimes \mathbf{1}_2 \quad \text{or} \quad \tau_y \otimes \mathbf{1}_2$$

can appear.

Therefore:

if the residual valley symmetry is exact and survives remote elimination, mass protection is radiatively stable.

(611)

I. Combined Step 1–Step 5 status

Inside the obstruction-baseline + Class II truncation we now have:

1. Step 1: local Weyl nondegeneracy on the region

$$\Gamma_{c_{\parallel}} \neq 0, \quad \left(\frac{4Z}{A} - \Gamma_{c_I}, \Gamma_{c_J} \right) \neq (0, 0);$$

2. Step 2: opposite-valley witness matching;
3. Step 3: zero constant mixing $V(0) = 0$ reduced to the finite mismatch audit;
4. Step 4: remote spectral gap $\Delta > 0$ reduced to either

$$\delta_{0,\rho} > \eta_{R,\rho} \quad \text{or} \quad \|\delta H_R\| < \Delta_0;$$

5. Step 5: mass protection reduced to the finite constant-operator classification

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\},$$

with exact valley $U(1)_V$ or exact valley $\mathbb{Z}_2^V$ as sufficient symmetry mechanisms.

Thus Step 5 is now closed at the same theorem-reduction level as the previous steps.

J. Sharp honest conclusion

The strongest honest conclusion is:

inside the obstruction-baseline + Class II truncation, mass protection is not automatic,

(612)

but

its verification is fully reduced to a finite 4×4 residual-symmetry operator audit.

(613)

If the actual full microscopic TECT/Class II realization preserves an exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$, then all constant Dirac masses are forbidden.

CCCXIV. FINAL SUMMARY OF DIRAC STEP 5 IN THE OBSTRUCTION-BASELINE + CLASS II TRUNCATION

a. What was actually established. For the doubled local Dirac block

$$H_D(p) = \sum_{i=1}^3 v_i p_i (\tau_z \otimes \sigma_i),$$

the genuine constant Dirac-mass space is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}} \{ \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2 \}.$$

b. Finite protection criterion. Mass protection is exactly the finite classification problem

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\},$$

equivalently

$$A_{\text{res}} \cap \{K \in \text{Herm}(4) : \{K, \Gamma_i\} = 0 \ \forall i\} = \{0\}.$$

c. What is not enough. Neither little-group scalarity alone nor opposite-valley matching alone forbids all constant masses.

d. Sufficient mechanisms. If the actual residual symmetry contains either:

$$\text{exact valley } U(1)_V$$

or

$$\text{exact valley } \mathbb{Z}_2^V,$$

then all constant Dirac masses are forbidden.

e. Class II status. The canonical minimal Class II correction only renormalizes the first-order shell tensor coefficients A, B, C . So

$$\text{Class II is compatible with mass protection, but is not by itself a proof of it.}$$

f. Radiative stability. If the same residual valley symmetry is exact in the full microscopic theory and survives remote-sector elimination, then no symmetry-preserving self-energy can generate a constant Dirac mass.

g. Combined status. Therefore, inside the obstruction-baseline + Class II truncation:

$$\text{Step 5 is now closed at the theorem-reduction level.}$$

What remains open is only the actual microscopic identification of the residual symmetry algebra in the full TECT/Class II realization.